

The long-range interactions of 3D Genome

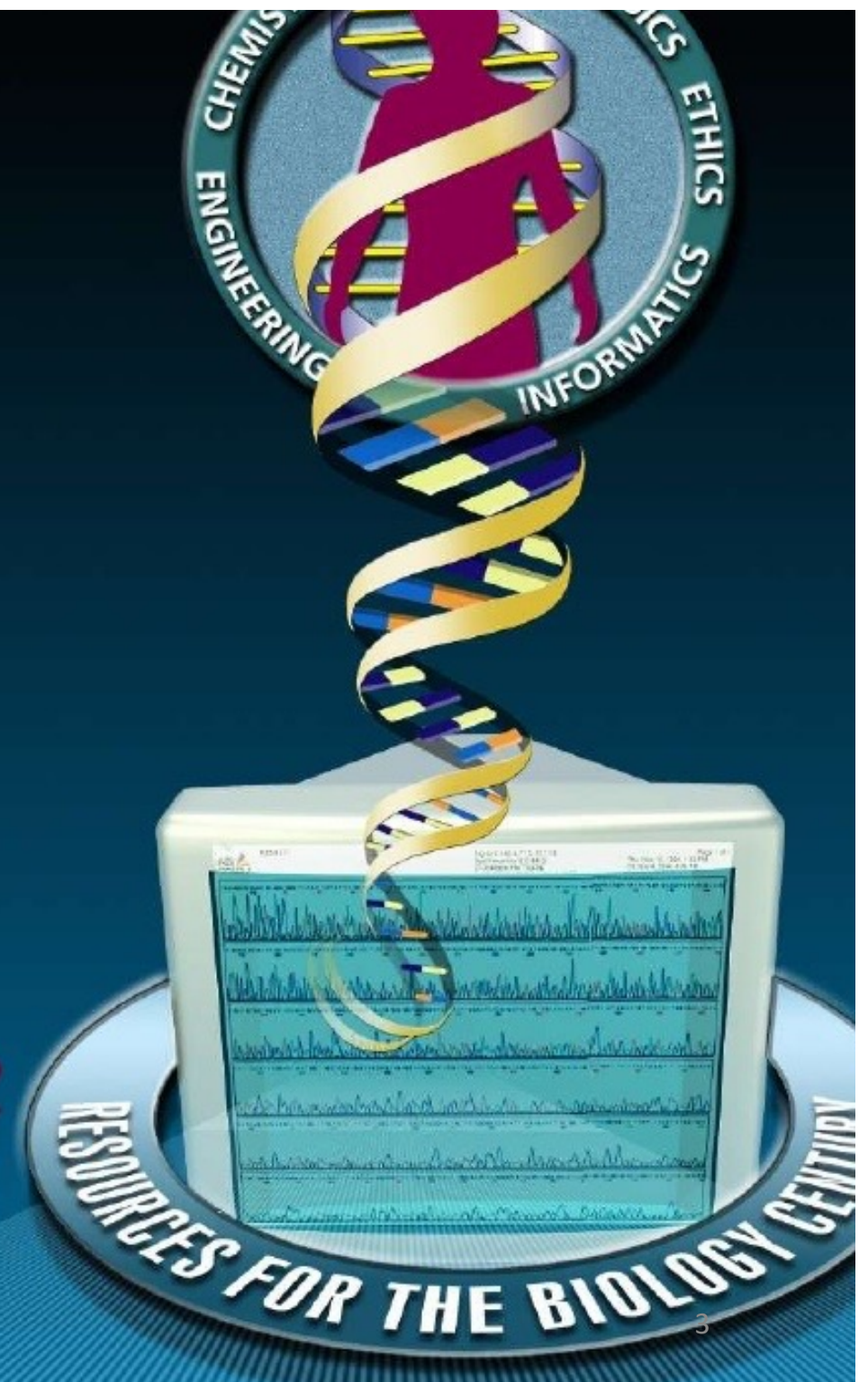
Chapter 12

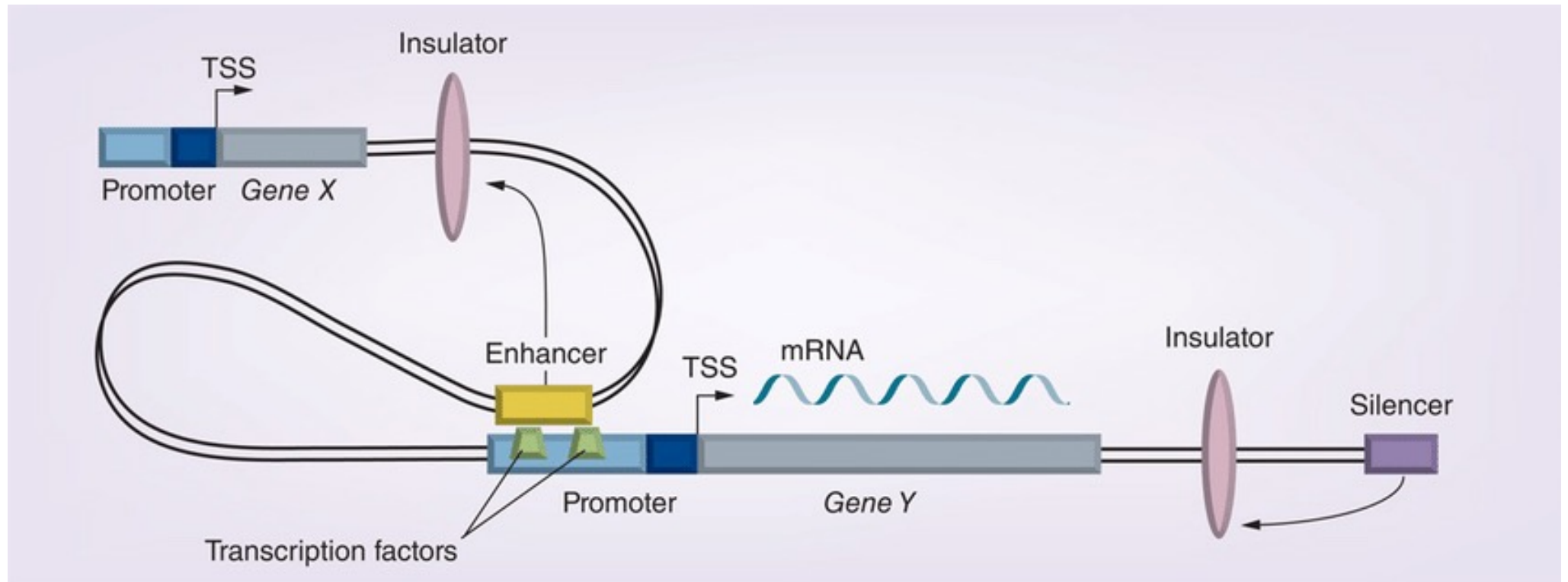
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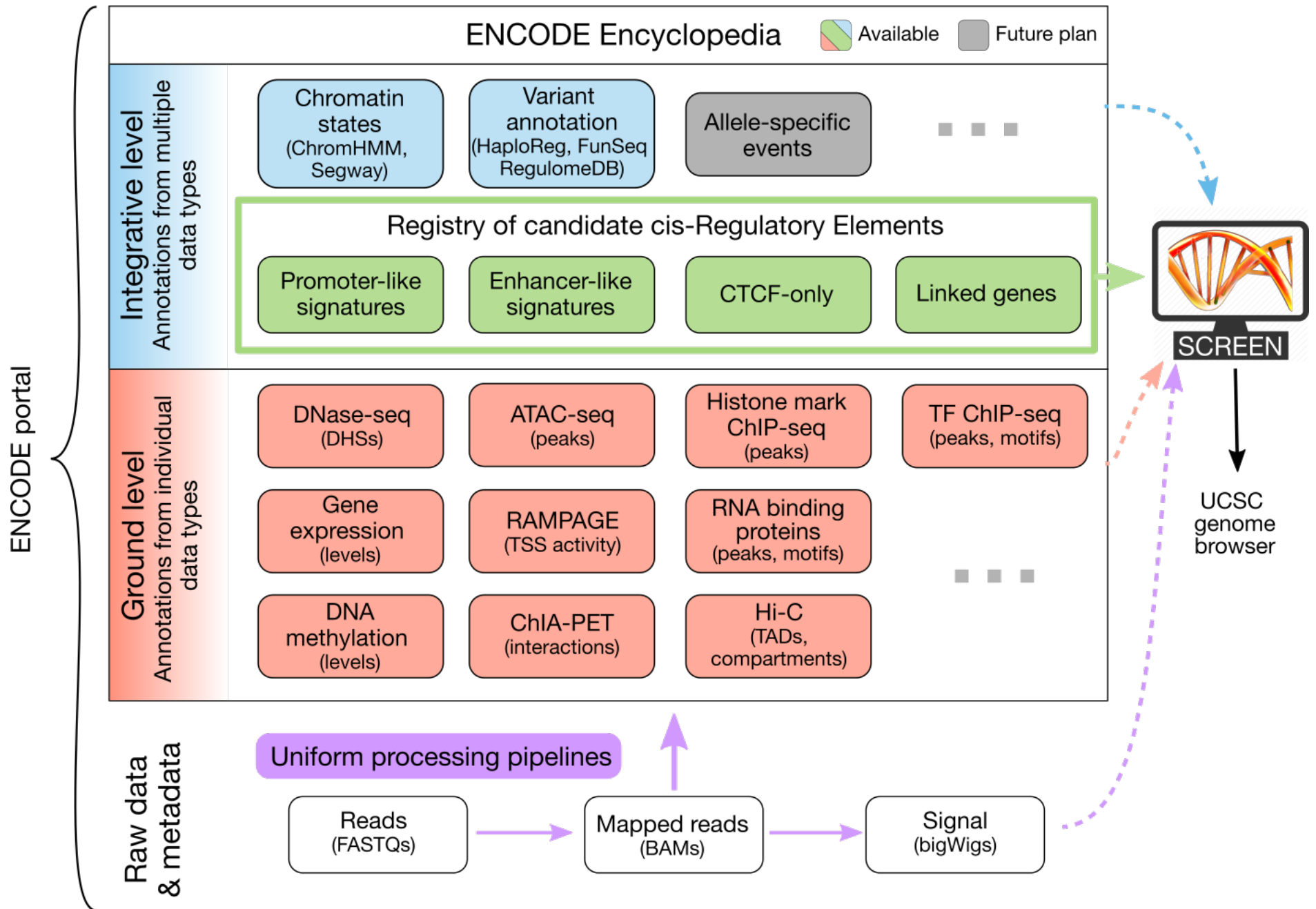
Human Genome Project

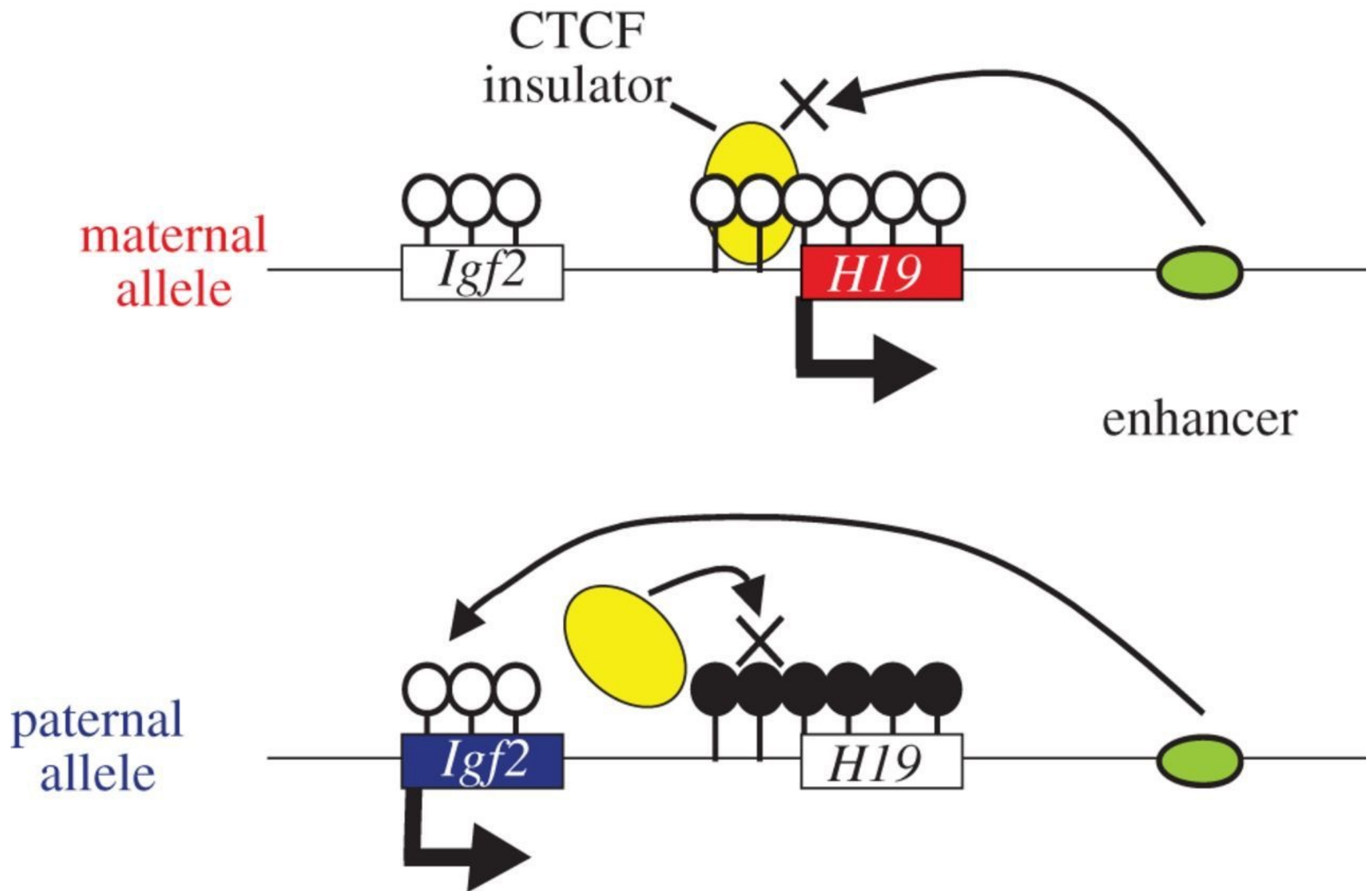
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Chapter 2

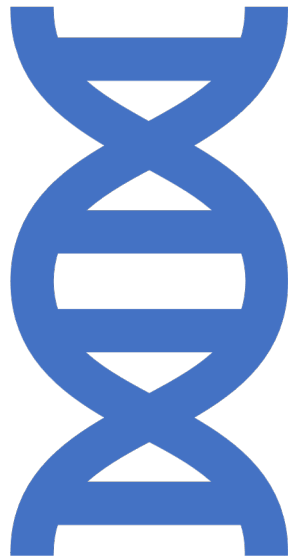




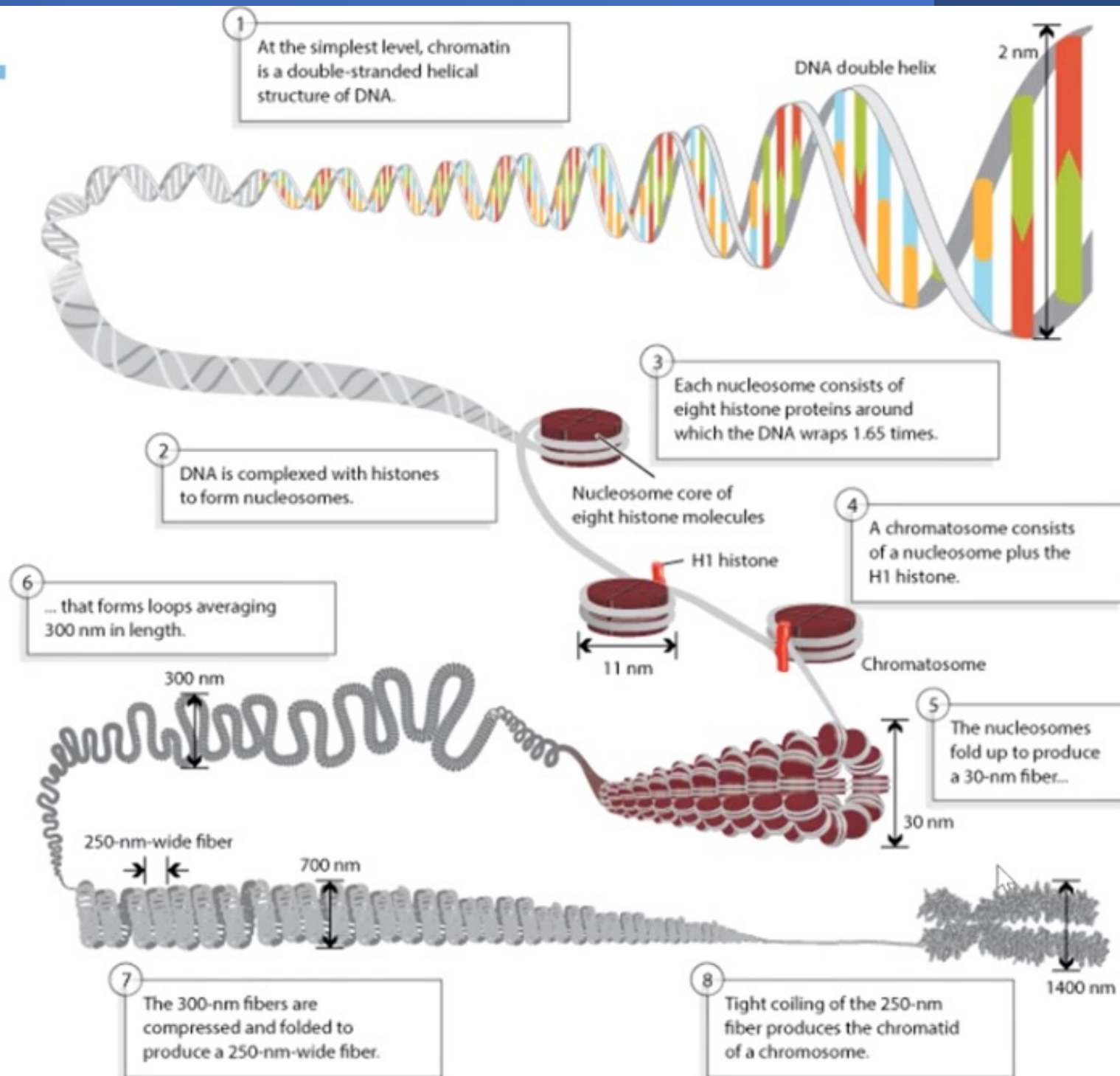


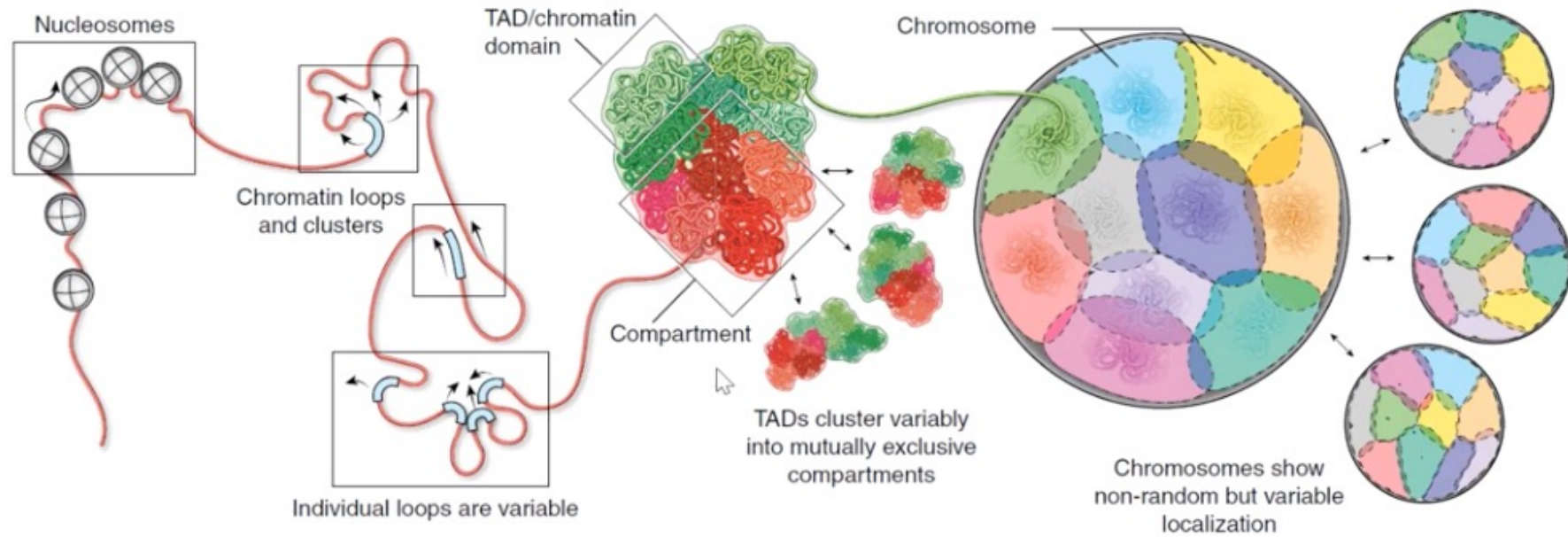


- How are the profound differences in gene activities established and maintained in a large number of cell types to ensure the development and functioning of a complex multicellular organism?



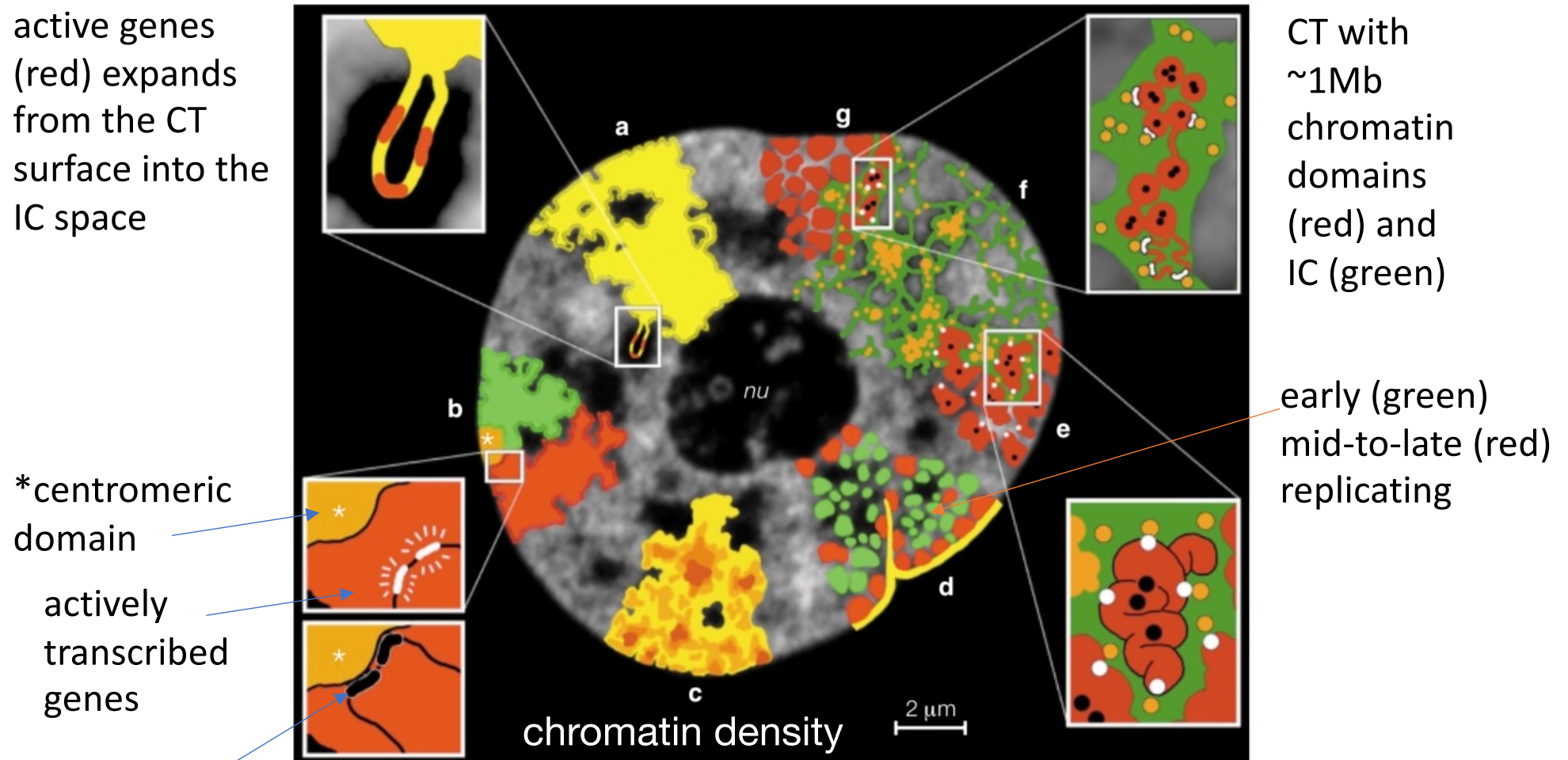
We need to understand how genomes are organized in the nucleus, the basic principles of nuclear architecture and the changes in nuclear organization that occur during cellular differentiation.





Evidence for a compartmentalized nuclear architecture

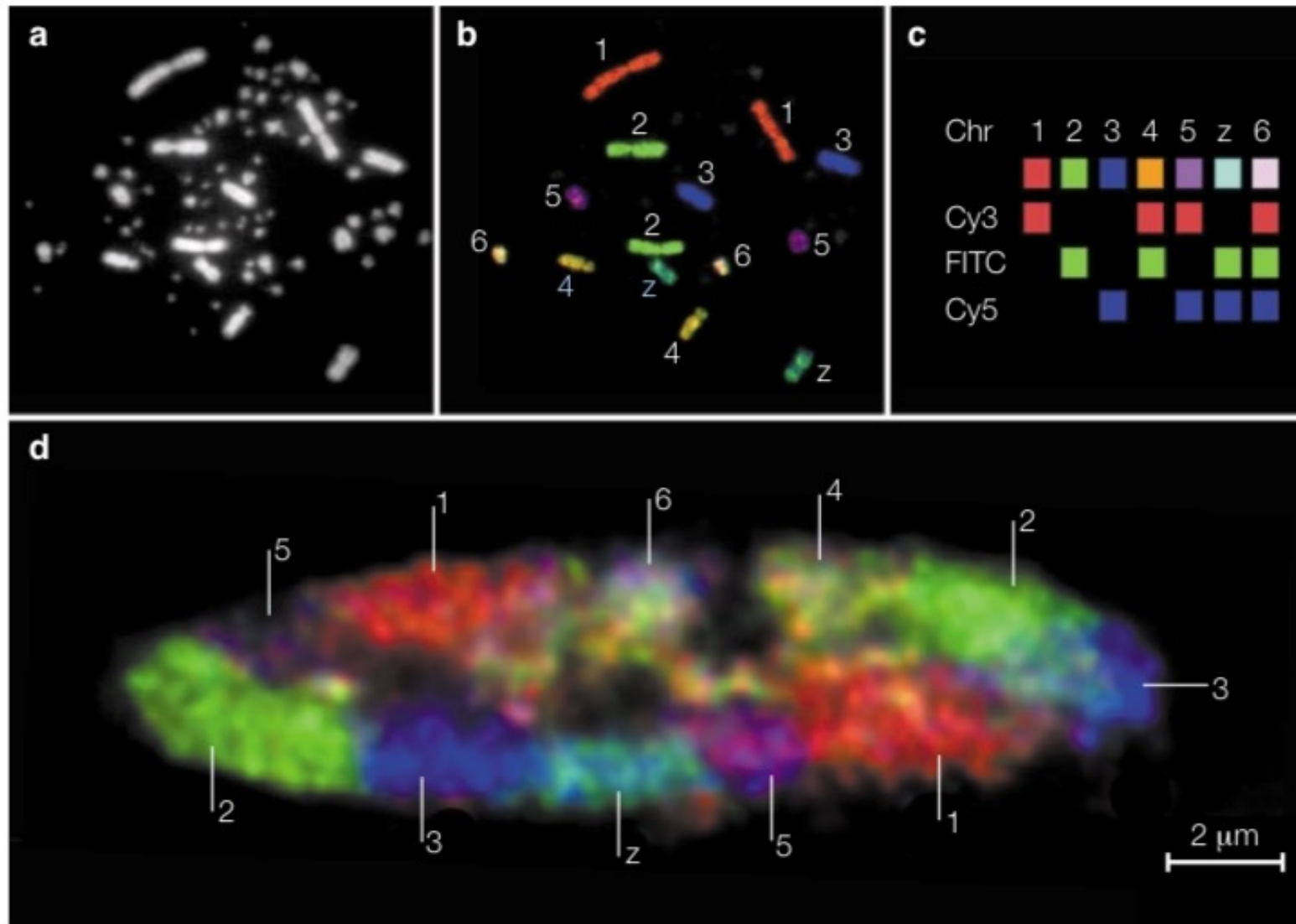
Model of functional nuclear architecture



silencing

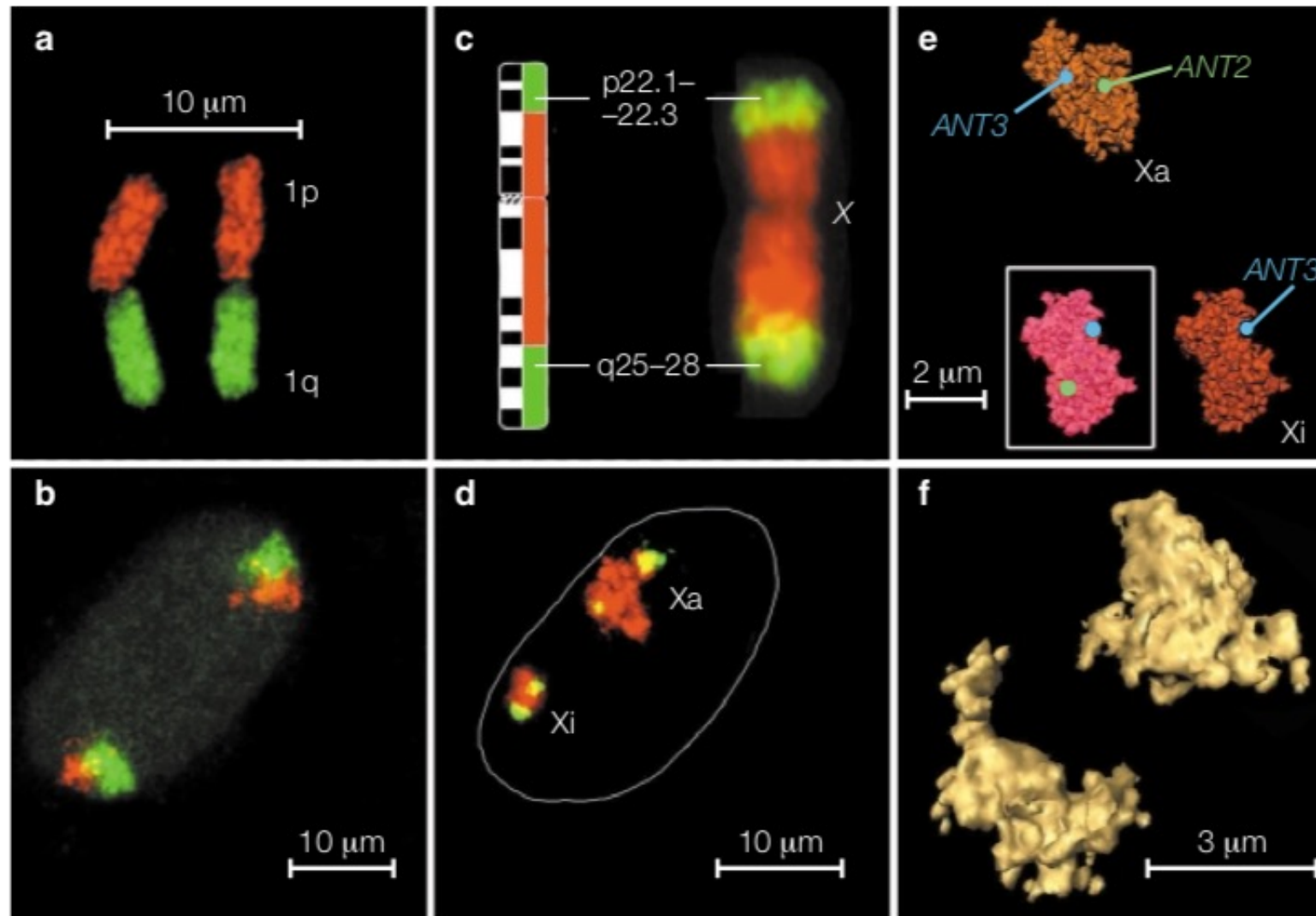
T. Cremer and C. Cremer, Nature Reviews Genetics, 2001

Chromosome territories in the chicken

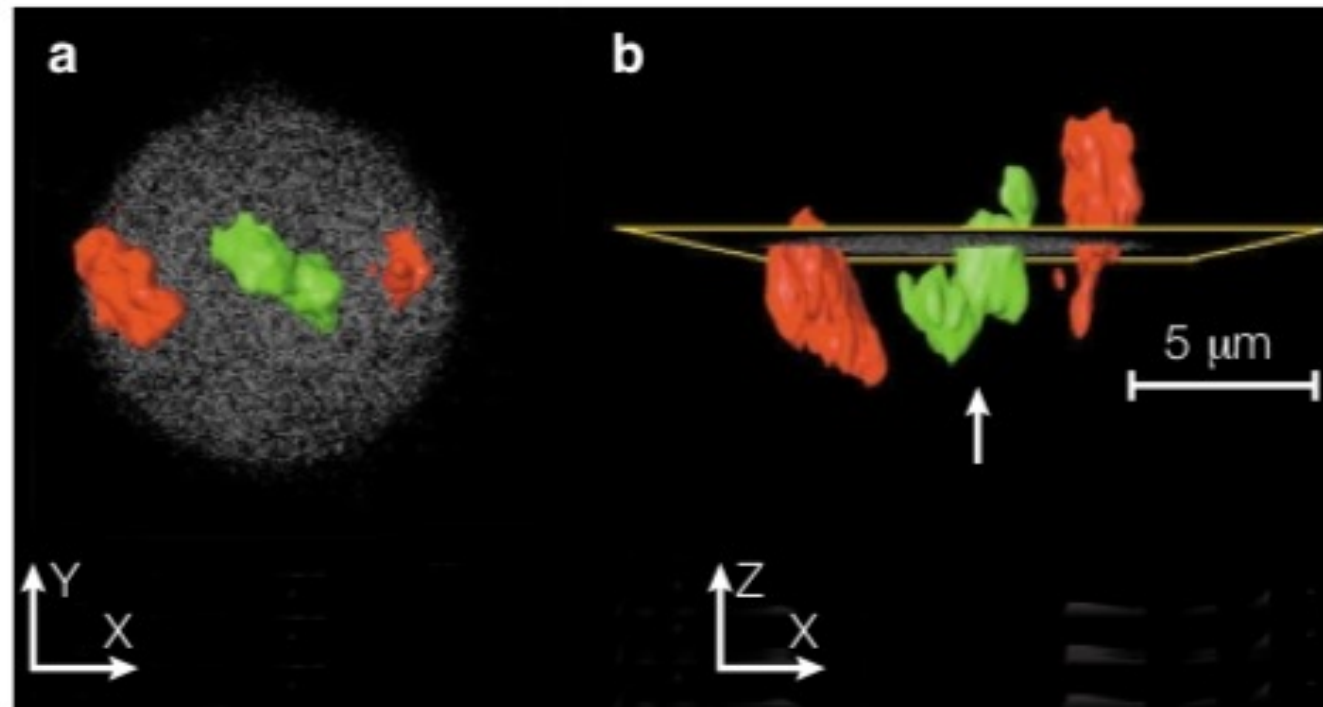


T. Cremer and C. Cremer, Nature Reviews 2001

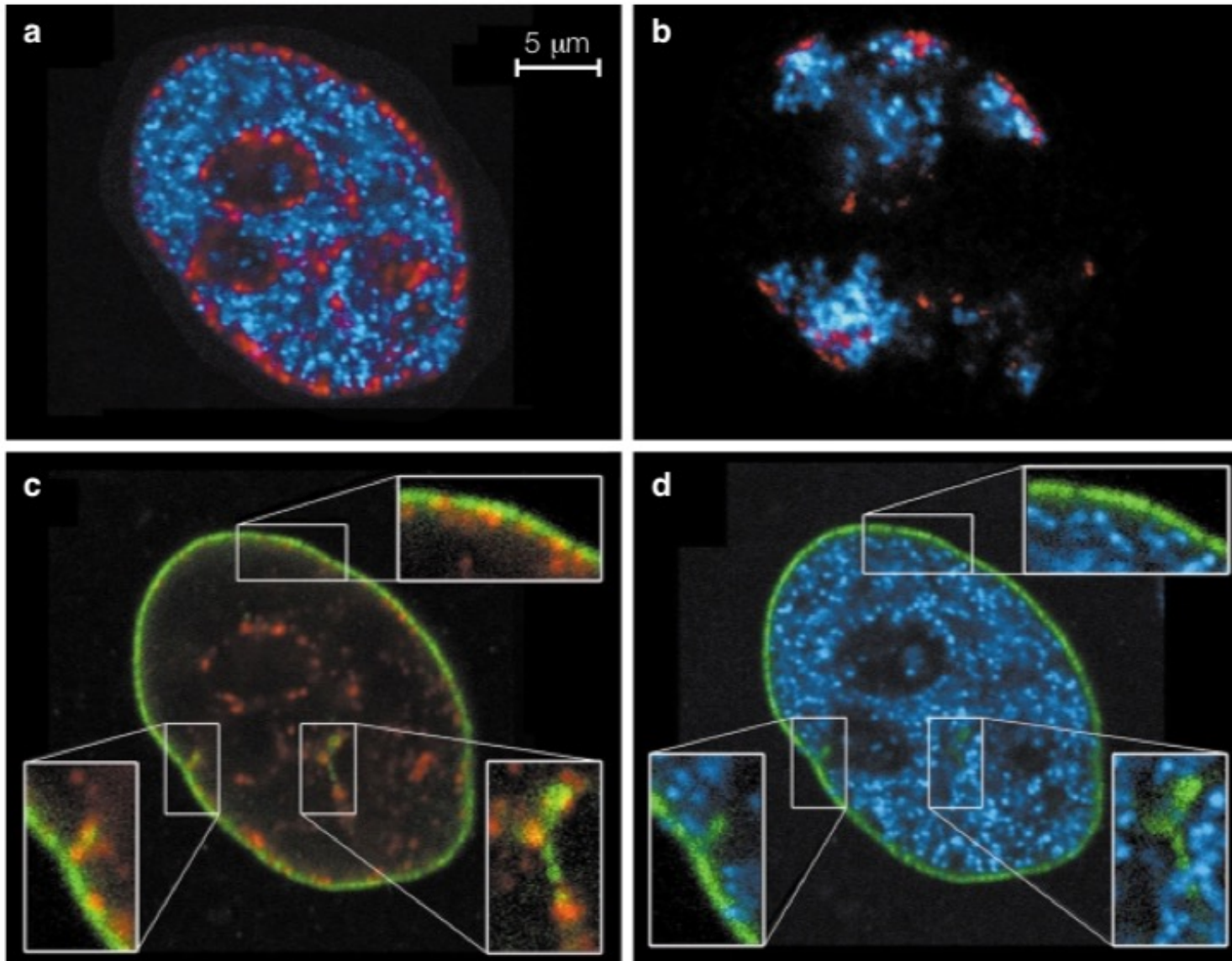
Features of human chromosome territories



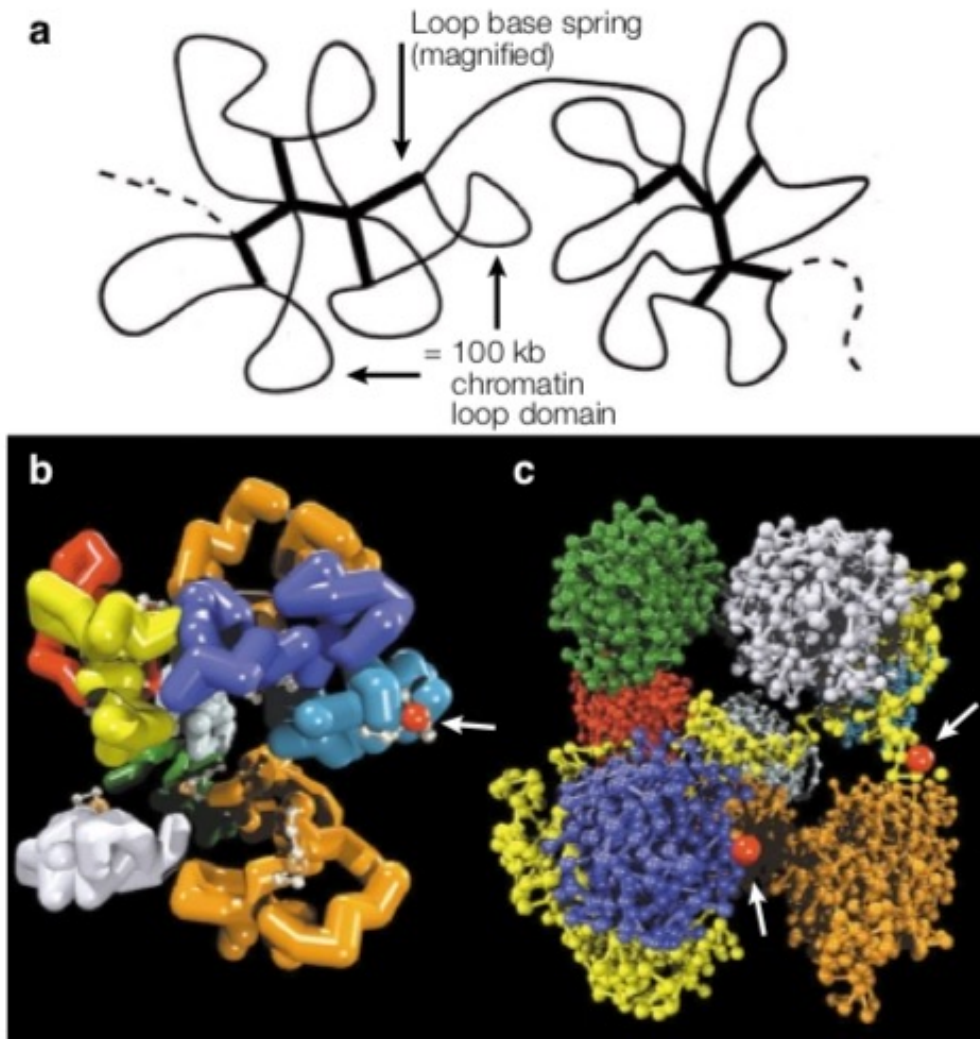
Gene-rich and gene-poor chromosome territories



Early- and mid-to-late-replicating chromatin domains



The multiloop subcompartment model



T. Cremer and C. Cremer, Nature Reviews Genetics, 2001

The following chronology indicates when specific techniques were first introduced. In subsequent years, the techniques have been developed and improved, and continue to be used in studies of nuclear organization and function. References to original publications in this box should be regarded as examples.

1950s: Electron microscopy allowed studies of chromatin and other nuclear structures at ultrahigh resolution^{55,58,94–96}.

1970s: Microbeam irradiation of chromatin in living-cell nuclei showed the presence of **chromosome territories (CTs)**, and allowed the first studies of their arrangements and of post-irradiation dynamics in living cells⁹⁷. Laser micro-irradiation was also used to manipulate chromosomes in living cells⁹⁸. Recently, fluorescence recovery after photobleaching (FRAP) and fluorescence loss in photobleaching (FLIP) have allowed the study of rapid movements of nuclear proteins tagged with green fluorescence protein (GFP) throughout the nuclear space^{60,99}.

1980s: In 1983, EPIFLUORESCENCE MICROSCOPY was combined with image reconstruction techniques to study the **three-dimensional chromosome topography** in polytene nuclei of *Drosophila melanogaster*¹⁰⁰. In the mid-1980s, the first fluorescence-laser scanning microscopes were ready to use¹⁰¹. Since then, important efforts have been made to develop advanced types of laser microscope with resolution beyond the **ABBE LIMIT** of conventional light microscopy^{102,103}. Submicrometre particle tracking has been used to directly measure the motion of interphase chromatin¹⁰⁴. Spectral-precision distance microscopy (SPDM)¹⁰⁵ allows precise three-dimensional distance measurements down to the nanometre scale. The development of fluorescence-resonance energy-transfer (FRET) microscopy has allowed studies of macromolecule interactions in living-cell nuclei¹⁰⁶. *In vivo* fluorescence correlation spectroscopy (FCS) showed that nuclear poly(A) RNA moves by a diffusion-like process in the interchromatin space¹⁰⁷. Fluorescence lifetime imaging (FLIM), based on the use of fluorochromes with different fluorescence lifetimes, has increased the number of targets that can be simultaneously discriminated in single-cell studies¹⁰⁸.

1980s: Fluorescence *in situ* hybridization (**FISH**)¹⁷ provided a tool for visualizing specific chromatin structures, from CTs, to chromosome arms and bands, down to the level of individual genes. Concomitantly, RNA *in situ* hybridization protocols were developed for studies of gene expression in single cells¹⁰⁹.

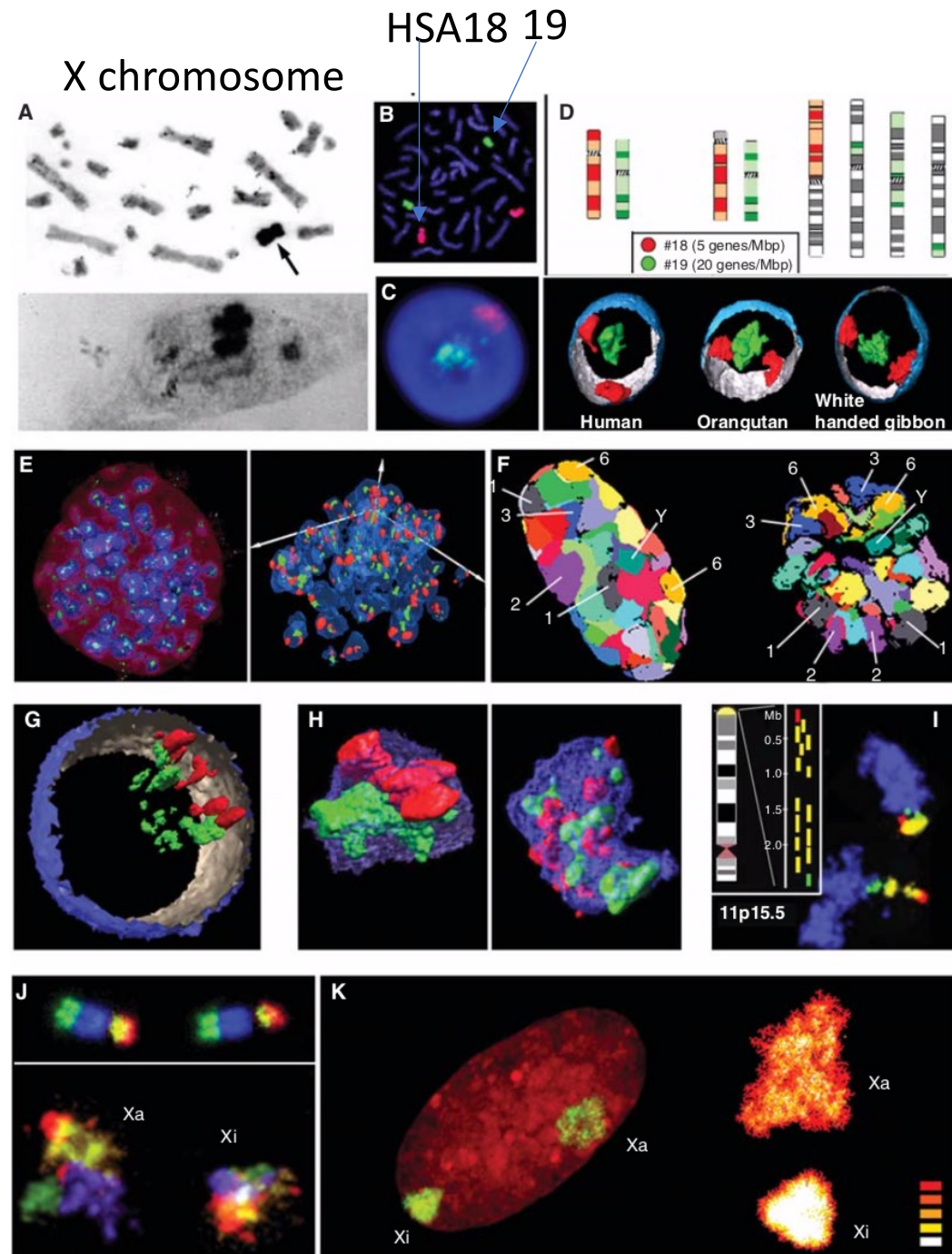
1980s: Labelling of DNA using base analogues (BrdU, CldU, IdU) allowed visualization of early- and mid-to-late-**replicating chromatin** in the nuclei of fixed cells^{110,111}. Replication labelling of DNA, followed by culture of cells for several additional cell cycles, provided another tool for visualizing individual CTs²⁵. DNA-labelling protocols with fluorochrome-linked nucleotides allowed studies of CTs and higher-order chromatin domains in living-cell nuclei^{30,31}.

1994: DNA fusion constructs for the expression of proteins tagged with GFP¹¹² allowed the visualization of specific proteins in living cells. Artificial chromatin structures that contain repeated *lac* operons were studied in living cells using GFP-tagged lac repressor protein⁸⁴. Recently, GFP colour variants (for example, yellow and red fluorescent protein)¹¹³ have opened the door to multicolour, living-cell studies of nuclear proteins. In combination with the visualization of chromatin, these advances will lead to rapid advances in our understanding of the topology, movements and dynamic interactions of chromatin and non-chromatin domains within living-cell nuclei.

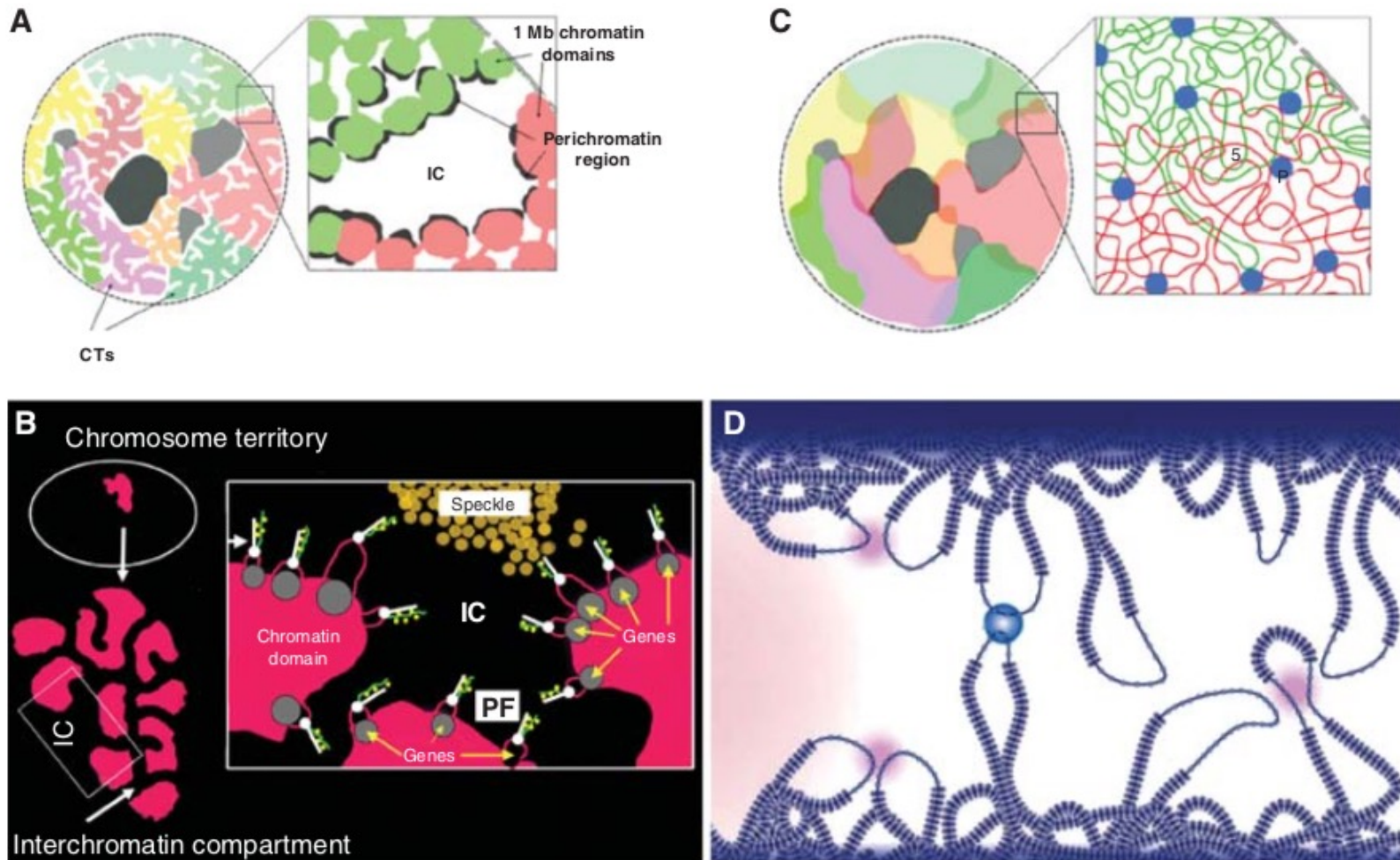
1995: A first quantitative simulation of CTs was published⁹⁰. Since then, several other mathematical and computer models have been developed to simulate higher-order chromatin structure and arrangements^{91,114}. Computer models that allow the simulation of dynamic aspects of nuclear architecture are now being developed.

T. Cremer and C. Cremer, Nature Reviews Genetics, 2001

Direct evidence for chromosome territories (CTs) by in situ hybridization experiments.



Different models of nuclear architecture



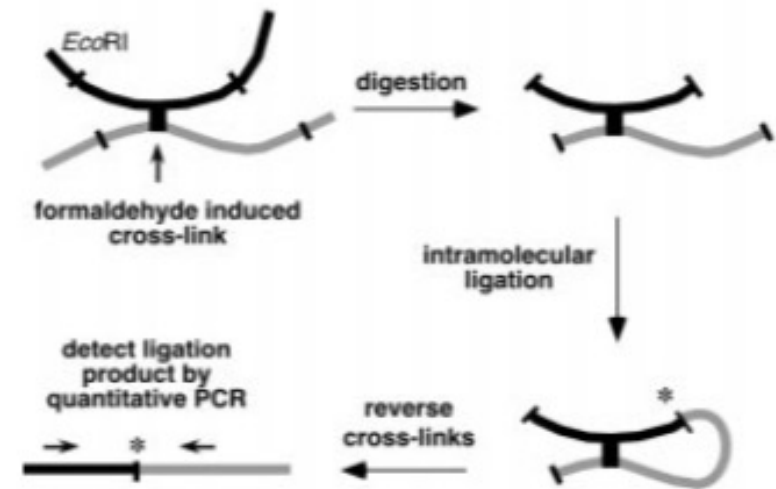
Capturing Chromosome Conformation

Science 2002

Job Dekker,^{1*} Karsten Rippe,² Martijn Dekker,³ Nancy Kleckner¹

We describe an approach to detect the frequency of interaction between any two genomic loci. Generation of a matrix of interaction frequencies between sites on the same or different chromosomes reveals their relative spatial disposition and provides information about the physical properties of the chromatin fiber. This methodology can be applied to the spatial organization of entire genomes in organisms from bacteria to human. Using the yeast *Saccharomyces cerevisiae*, we could confirm known qualitative features of chromosome organization within the nucleus and dynamic changes in that organization during meiosis. We also analyzed yeast chromosome III at the G₁ stage of the cell cycle. We found that chromatin is highly flexible throughout. Furthermore, functionally distinct AT- and GC-rich domains were found to exhibit different conformations, and a population-average 3D model of chromosome III could be determined. Chromosome III emerges as a contorted ring.

A



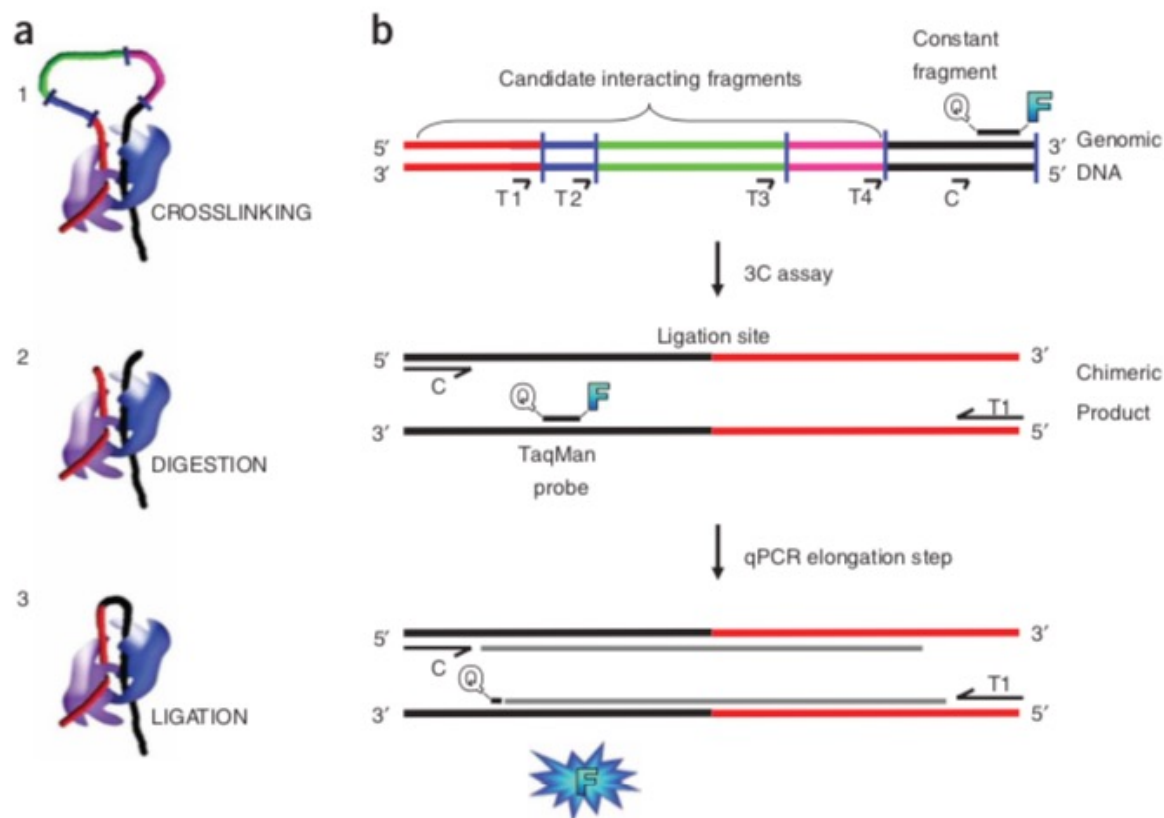
Quantitative analysis of chromosome conformation capture assays (3C-qPCR)

Hélène Hagège¹, Petra Klous², Caroline Braem¹, Erik Splinter², Job Dekker³, Guy Cathala¹, Wouter de Laat² & Thierry Forné¹

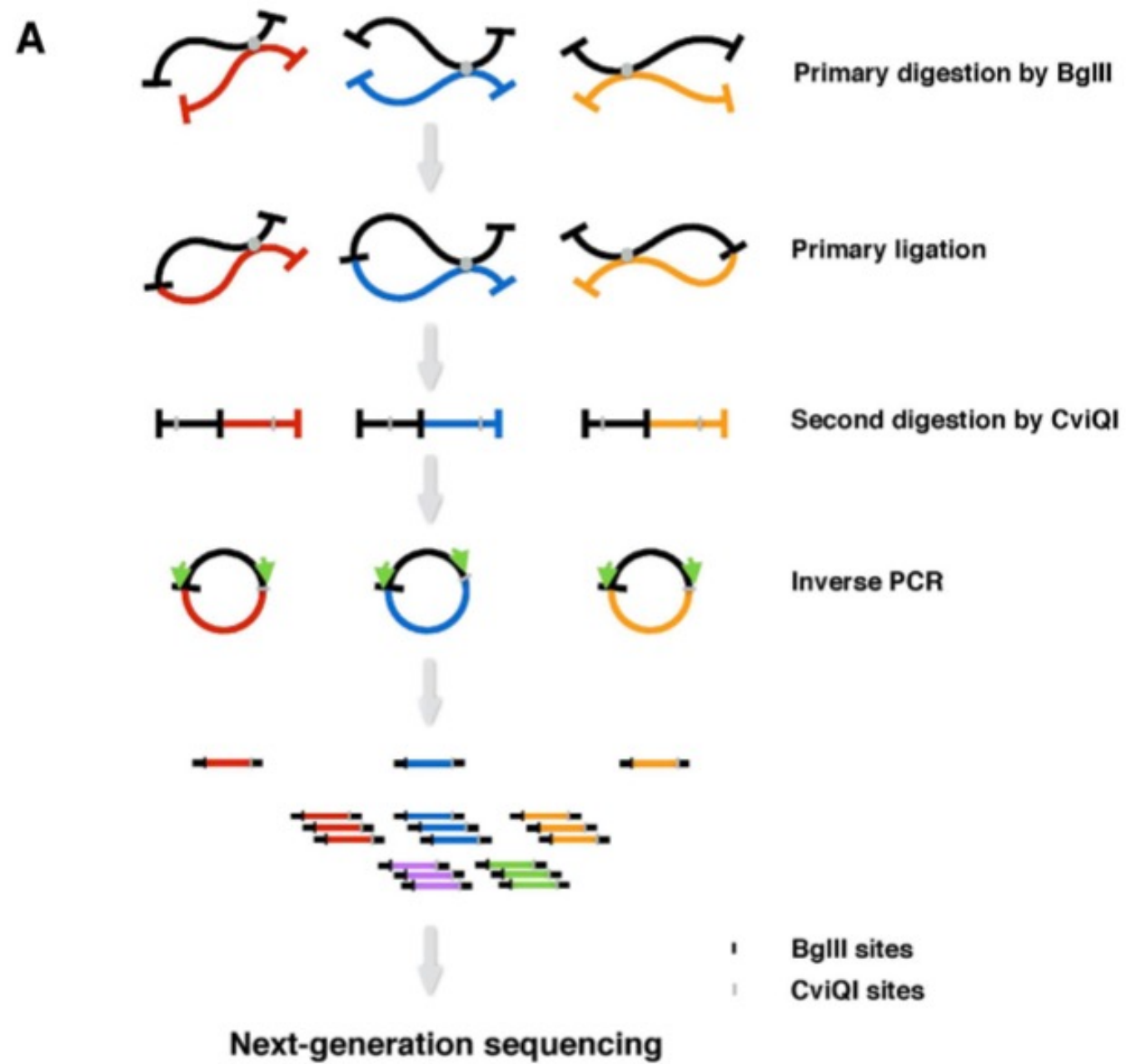
¹UMR5535 CNRS-UMI; IFR122; Institut de Génétique Moléculaire de Montpellier; IGMM; 1919, Route de Mende, 34293 Montpellier Cedex 5, France. ²Department of Cell Biology and Genetics, Erasmus MC, PO Box 2040, 3000 CA Rotterdam, The Netherlands. ³Program in Gene Function and Expression and Department of Biochemistry and Molecular Pharmacology, University of Massachusetts Medical School, Worcester, Massachusetts 01605-0103, USA. Correspondence should be addressed to W.d.L. (w.delaat@erasmusmc.nl) or T.F. (forne@igmm.cnrs.fr).

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Chromosome conformation capture (3C) technology is a pioneering method explored at a scale encompassing a few tens to a few hundred kilobase particularly important in mammals where dispersed regulatory elements. 3C technology involves formaldehyde fixation of cells, followed by a procedure with which pairs of selected DNA fragments are crosslinked in the population. Frequencies require the best quantification techniques. We recently adapted 3C assays, resulting in a method that more accurately determines crosslinks that rely on measuring the intensity of ethidium bromide-stained PCR products. We have named this method 3C-qPCR. Once the whole procedure (3C assays and quantitative analyses) can be completed.

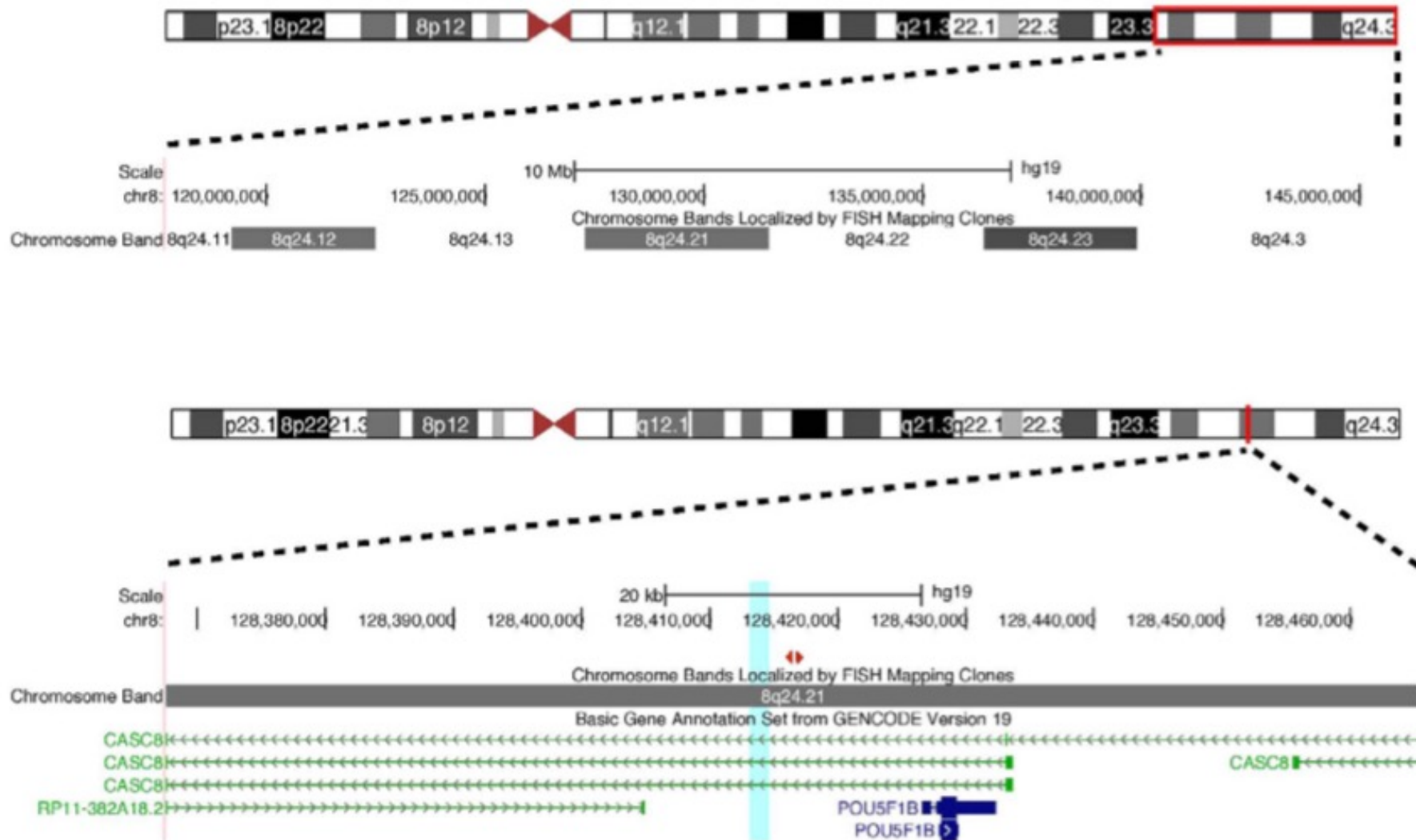


4C-seq

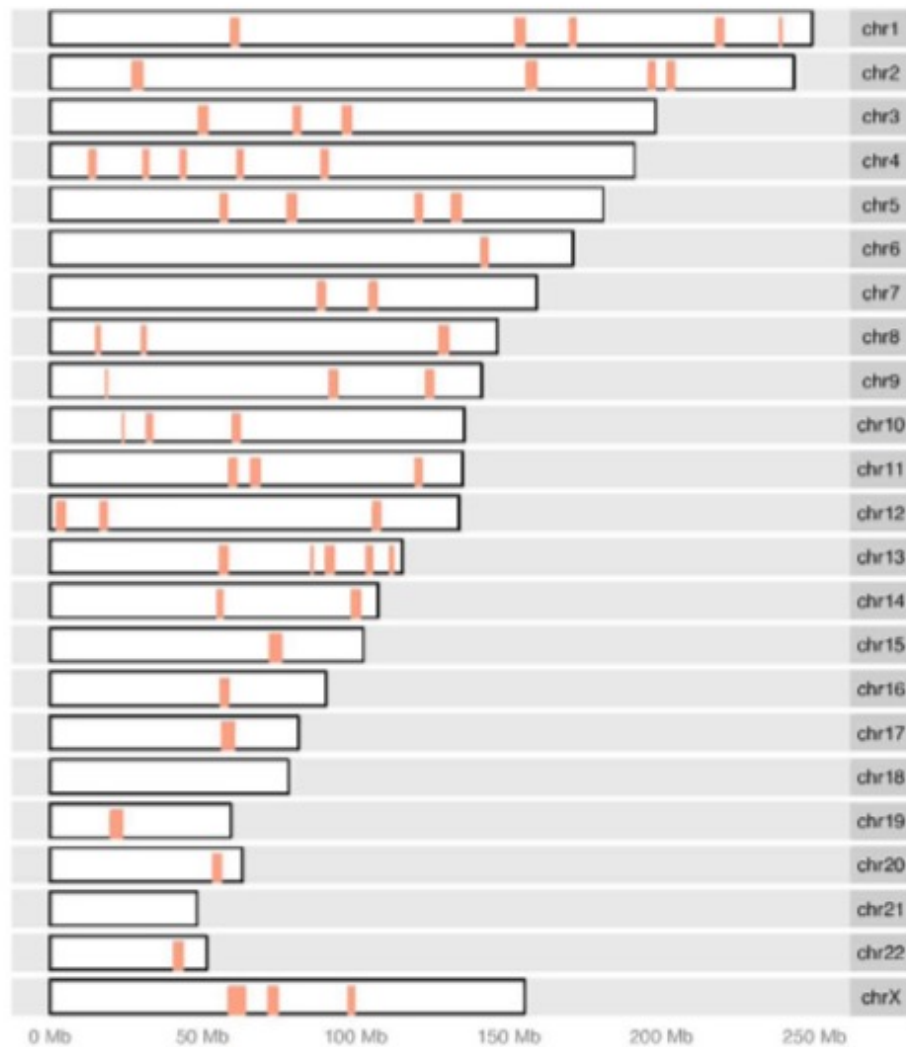


4C-seq

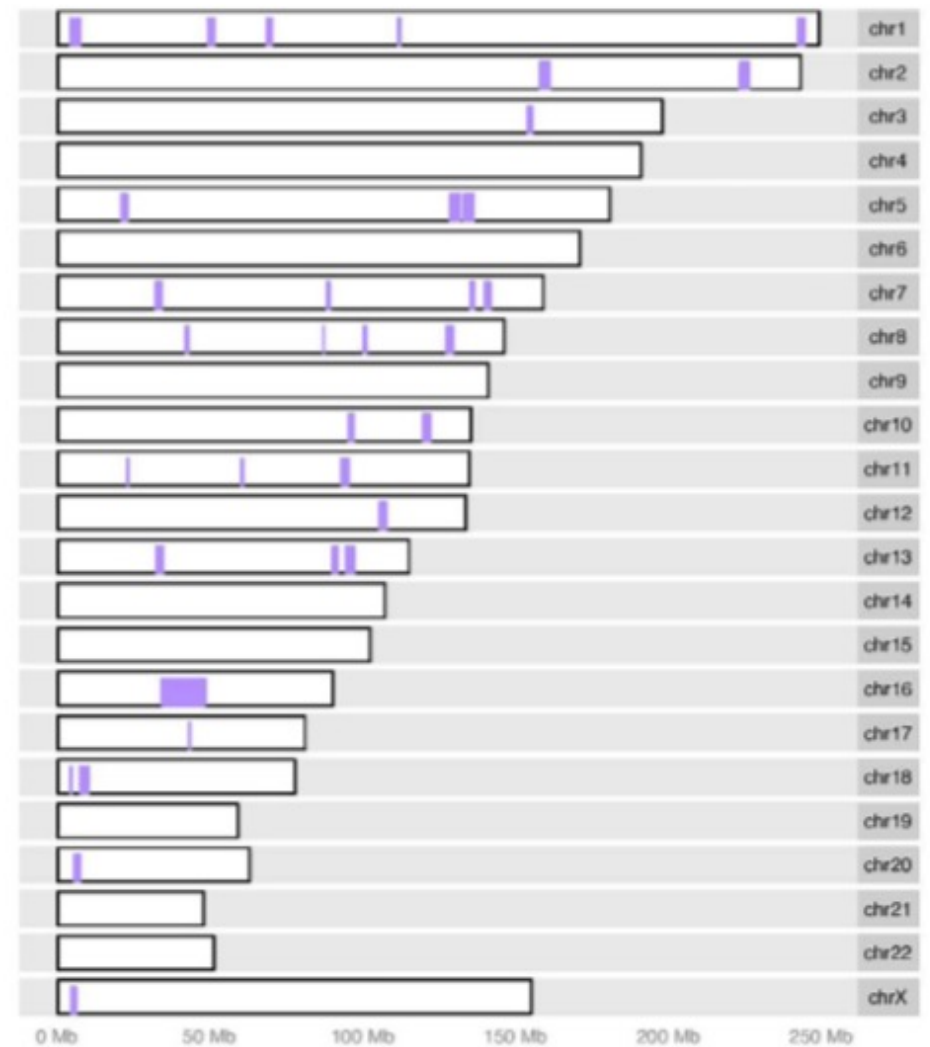
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4C-seq

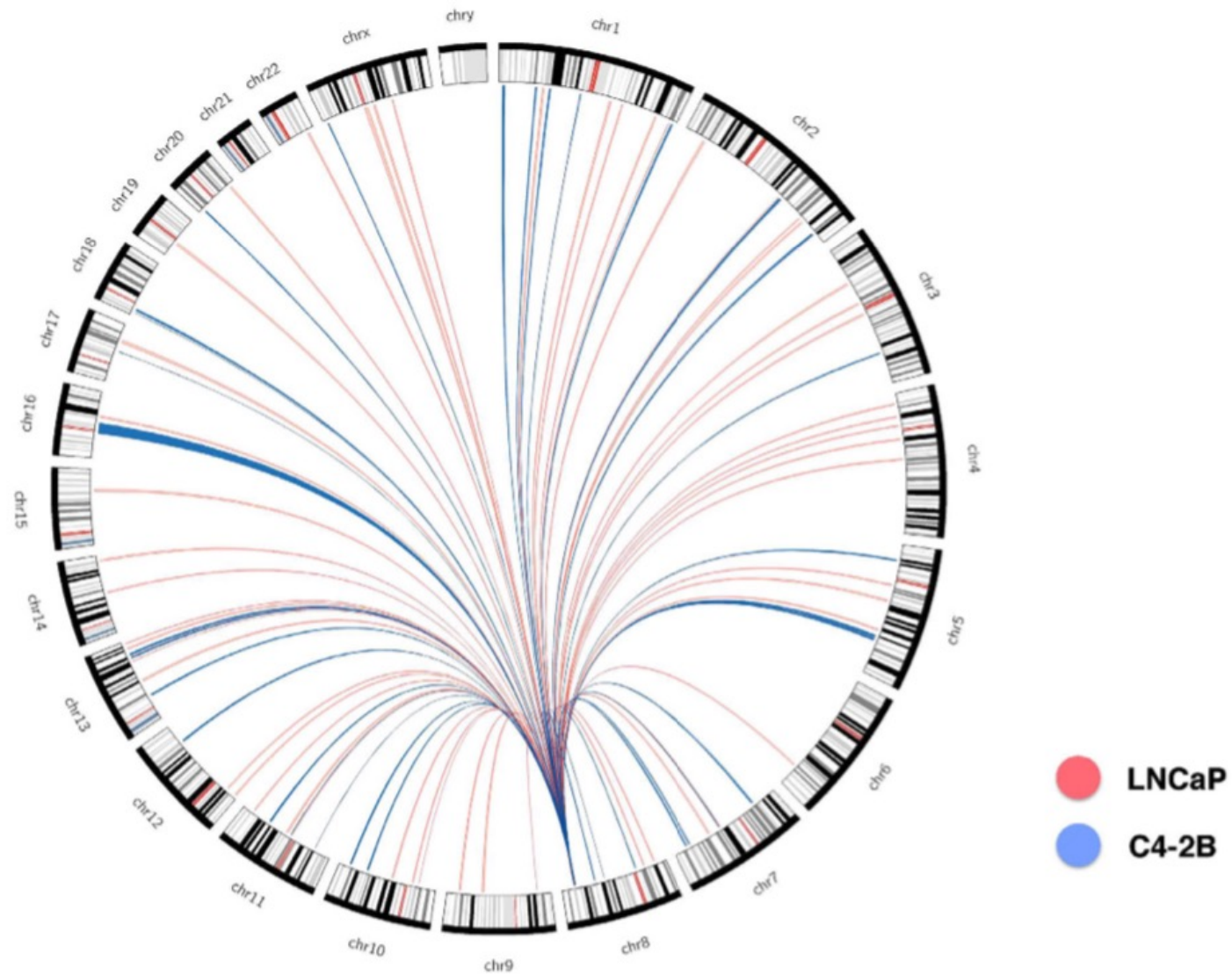


LNCaP



C4-2B

4C-seq

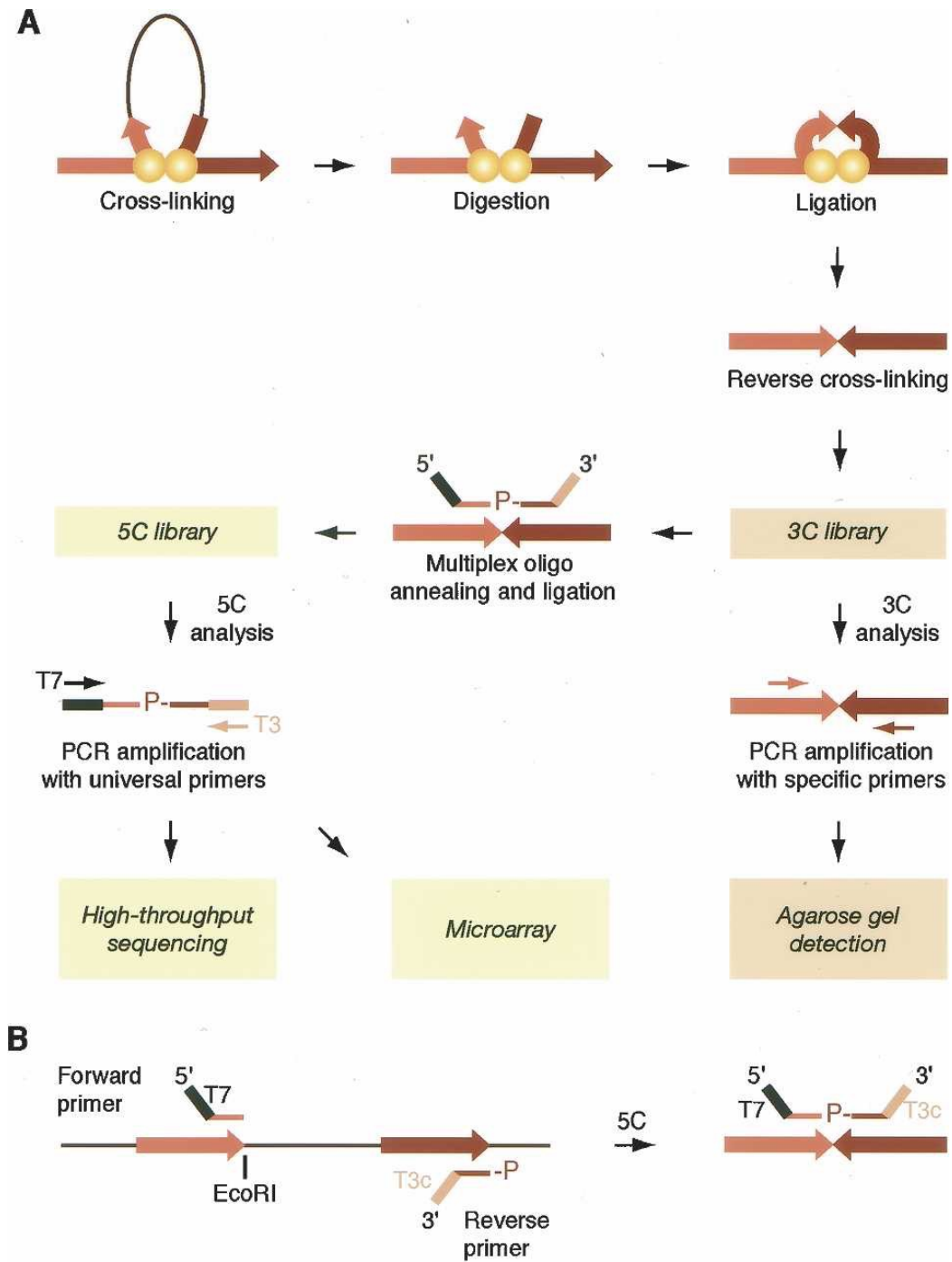


Chromosome Conformation Capture Carbon Copy (5C): A massively parallel solution for mapping interactions between genomic elements

Josée Dostie,¹ Todd A. Richmond,² Ramy A. Arnaout,^{3,4,5} Rebecca R. Selzer,² William L. Lee,³ Tracey A. Honan,³ Eric D. Rubio,⁶ Anton Krumm,⁶ Justin Lamb,³ Chad Nusbaum,³ Roland D. Green,² and Job Dekker^{1,7}

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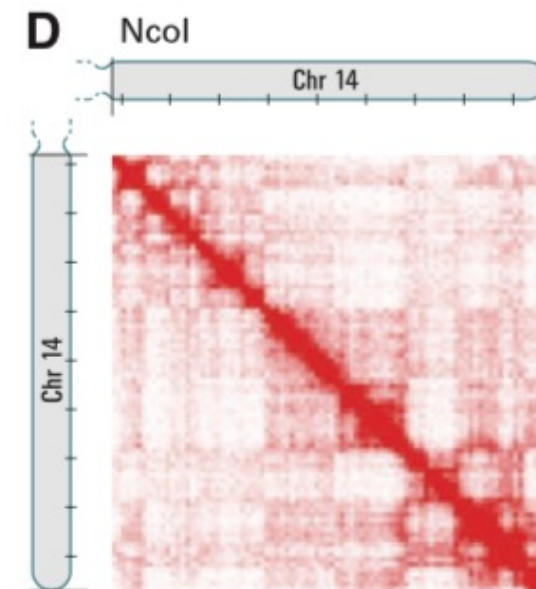
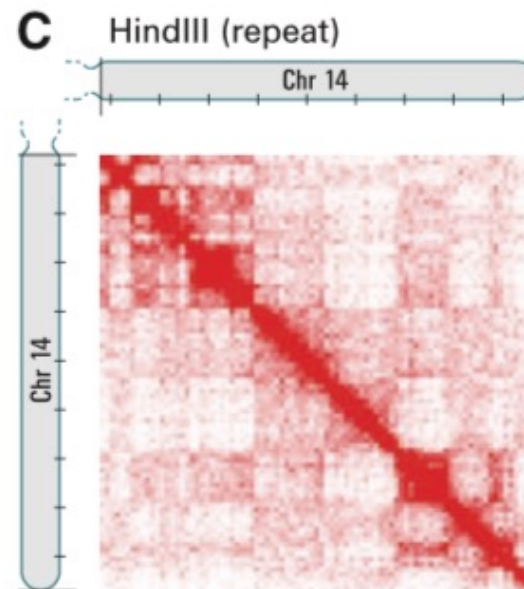
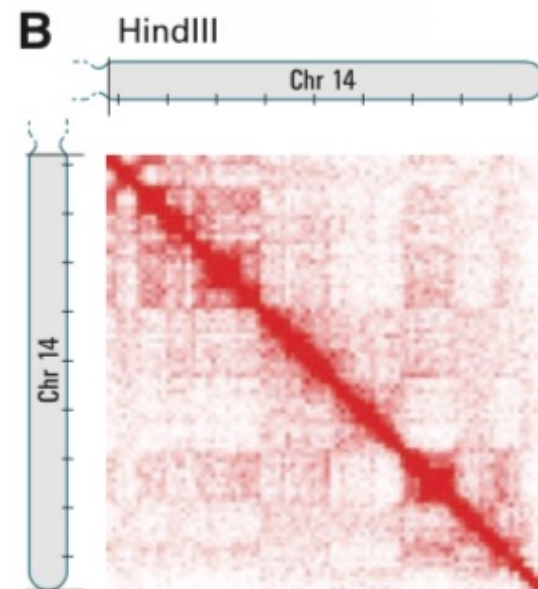
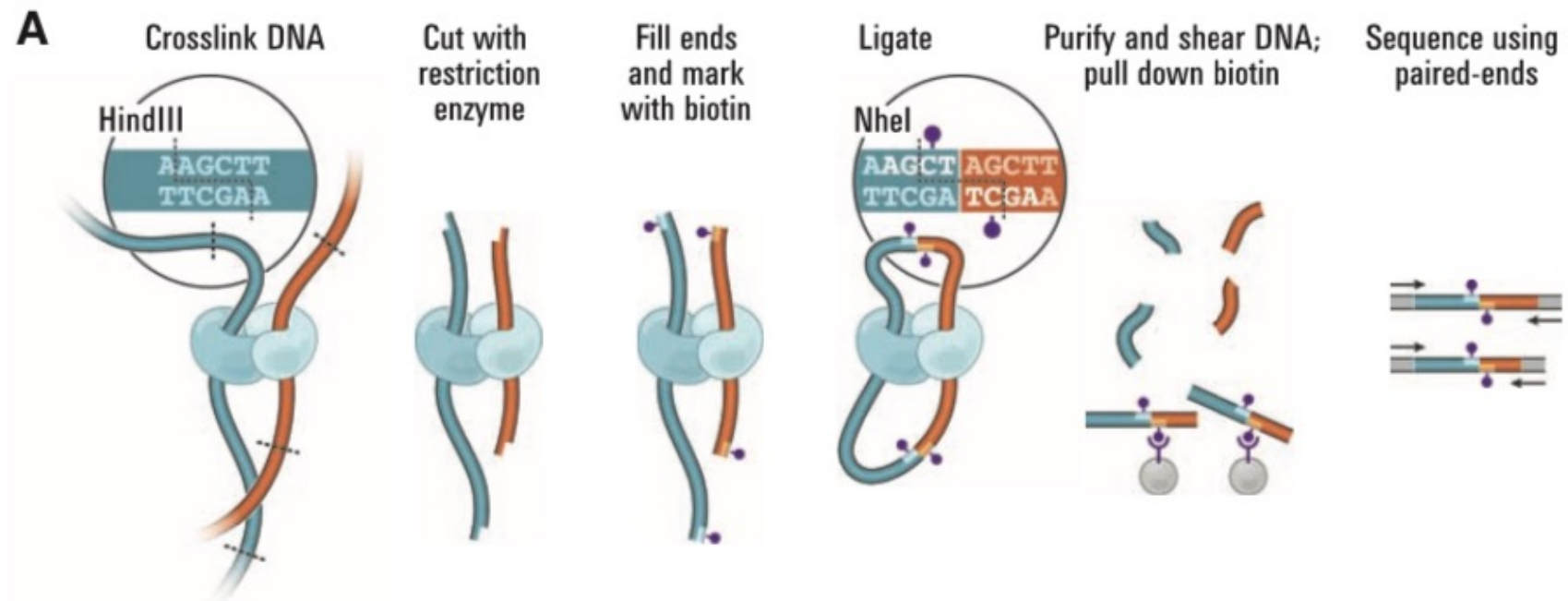
Physical interactions between genetic elements located throughout the genome play important roles in gene regulation and can be identified with the Chromosome Conformation Capture (3C) methodology. 3C converts physical chromatin interactions into specific ligation products, which are quantified individually by PCR. Here we present a **high-throughput 3C approach, 3C-Carbon Copy (5C)**, that employs microarrays or quantitative DNA sequencing using 454-technology as detection methods. We applied 5C to analyze a 400-kb region containing the human β -globin locus and a 100-kb conserved gene desert region. We validated 5C by detection of several previously identified looping interactions in the β -globin locus. We also identified a new looping interaction in K562 cells between the β -globin Locus Control Region and the γ - δ -globin intergenic region. Interestingly, this region has been implicated in the control of developmental globin gene switching. 5C should be widely applicable for large-scale mapping of *cis*- and *trans*- interaction networks of genomic elements and for the study of higher-order chromosome structure.



Comprehensive Mapping of Long-Range Interactions Reveals Folding Principles of the Human Genome

Erez Lieberman-Aiden,^{1,2,3,4*} Nynke L. van Berkum,^{5*} Louise Williams,¹ Maxim Imakaev,² Tobias Ragoczy,^{6,7} Agnes Telling,^{6,7} Ido Amit,¹ Bryan R. Lajoie,⁵ Peter J. Sabo,⁸ Michael O. Dorschner,⁸ Richard Sandstrom,⁸ Bradley Bernstein,^{1,9} M. A. Bender,¹⁰ Mark Groudine,^{6,7} Andreas Gnirke,¹ John Stamatoyannopoulos,⁸ Leonid A. Mirny,^{2,11} Eric S. Lander,^{1,12,13†} Job Dekker^{5†}

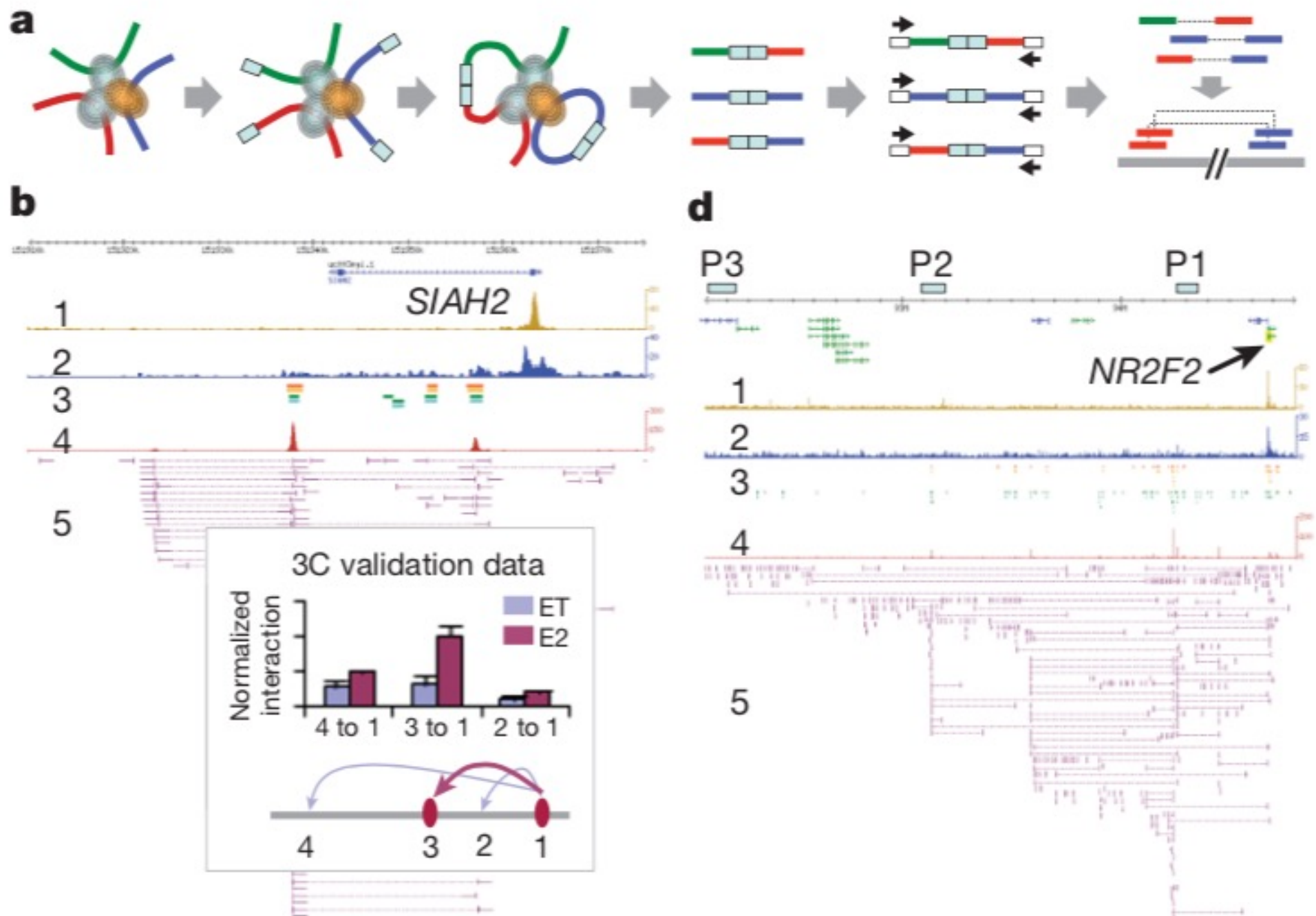
We describe Hi-C, a method that probes the three-dimensional architecture of whole genomes by coupling proximity-based ligation with massively parallel sequencing. We constructed spatial proximity maps of the human genome with **Hi-C** at a resolution of 1 megabase. These maps confirm the presence of chromosome territories and the spatial proximity of small, gene-rich chromosomes. We identified an additional level of genome organization that is characterized by the spatial segregation of open and closed chromatin to form two genome-wide compartments. At the megabase scale, the chromatin conformation is consistent with a fractal globule, a knot-free, polymer conformation that enables maximally dense packing while preserving the ability to easily fold and unfold any genomic locus. The fractal globule is distinct from the more commonly used globular equilibrium model. Our results demonstrate the power of Hi-C to map the dynamic conformations of whole genomes.

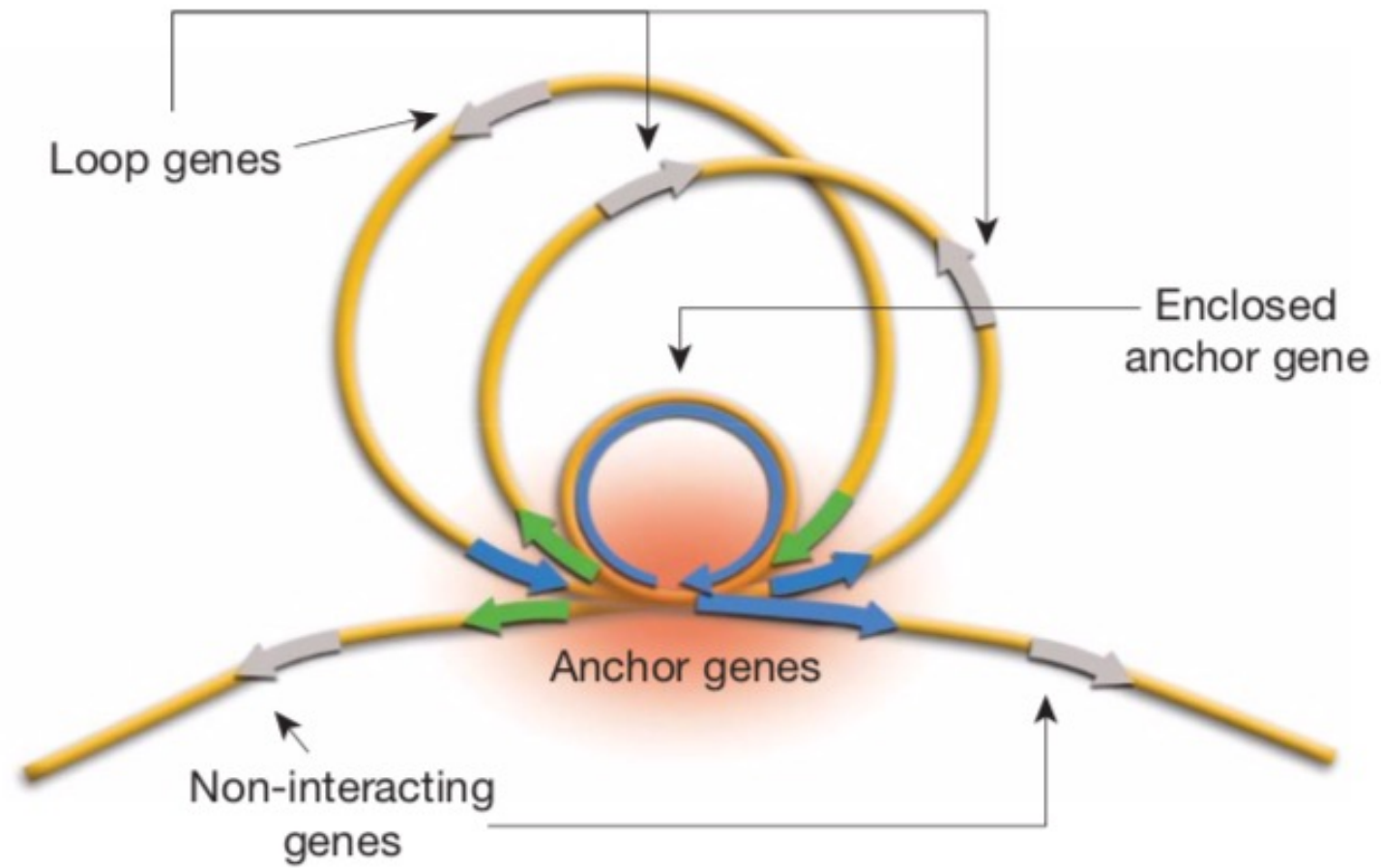


An oestrogen-receptor- α -bound human chromatin interactome

Melissa J. Fullwood¹, Mei Hui Liu¹, You Fu Pan¹, Jun Liu¹, Han Xu¹, Yusoff Bin Mohamed¹, Yuriy L. Orlov¹, Stoyan Velkov¹, Andrea Ho¹, Poh Huay Mei¹, Elaine G. Y. Chew¹, Phillips Yao Hui Huang¹, Willem-Jan Welboren², Yuyuan Han¹, Hong Sain Ooi¹, Pramila N. Ariyaratne¹, Vinsensius B. Vega¹, Yanquan Luo¹, Peck Yean Tan¹, Pei Ye Choy¹, K. D. Senali Abayratna Wansa¹, Bing Zhao¹, Kar Sian Lim¹, Shi Chi Leow¹, Jit Sin Yow¹, Roy Joseph¹, Haixia Li¹, Kartiki V. Desai¹, Jane S. Thomsen¹, Yew Kok Lee¹, R. Krishna Murthy Karuturi¹, Thoreau Herve¹, Guillaume Bourque¹, Hendrik G. Stunnenberg², Xiaoan Ruan¹, Valere Cacheux-Rataboul¹, Wing-Kin Sung^{1,3}, Edison T. Liu¹, Chia-Lin Wei¹, Edwin Cheung^{1,4,5} & Yijun Ruan^{1,4}

Genomes are organized into high-level three-dimensional structures, and DNA elements separated by long genomic distances can in principle interact functionally. Many transcription factors bind to regulatory DNA elements distant from gene promoters. Although distal binding sites have been shown to regulate transcription by long-range chromatin interactions at a few loci, chromatin interactions and their impact on transcription regulation have not been investigated in a genome-wide manner. Here we describe the development of a new strategy, chromatin interaction analysis by paired-end tag sequencing (**ChIA-PET**) for the *de novo* detection of global chromatin interactions, with which we have comprehensively mapped the chromatin interaction network bound by oestrogen receptor α (ER- α) in the human genome. We found that most high-confidence remote ER- α -binding sites are anchored at gene promoters through long-range chromatin interactions, suggesting that ER- α functions by extensive chromatin looping to bring genes together for coordinated transcriptional regulation. We propose that chromatin interactions constitute a primary mechanism for regulating transcription in mammalian genomes.

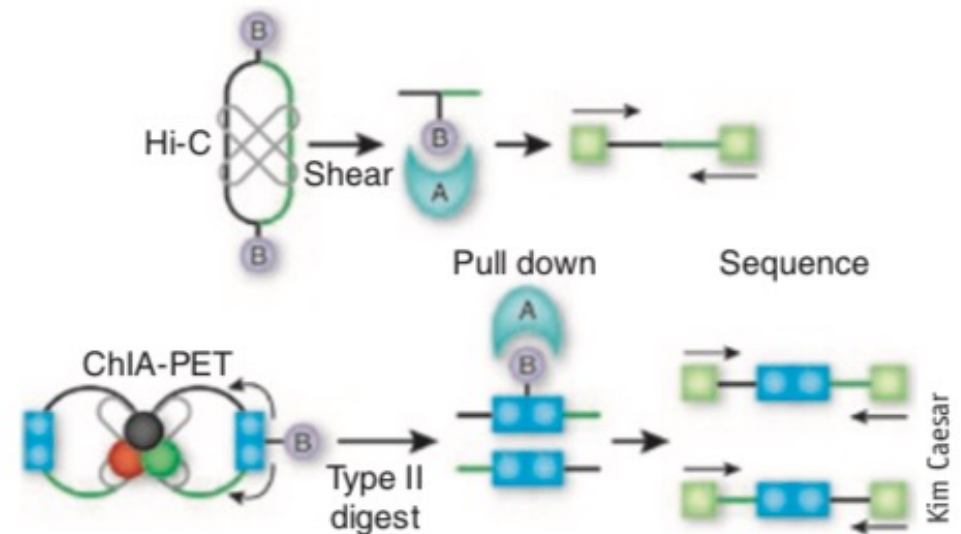




When ChIA PETs meet Hi-C

Two methods enable the drawing of genome-wide chromatin interaction maps: one looks at protein-independent folding principles, the other at protein-mediated functional interactions.

In its stretched out form, human chromatin adds up to a roughly two meter long polymer that needs to fit into a nucleus approximately ten micrometers in diameter. In addition, genes and their regulatory elements need to communicate, and chromatin must open to allow transcription and replication. Important questions are 'how does the linear order of genes relate to their spatial arrangement?' and 'how are genes regulated by distant elements?'. Two independent groups recently tackled these questions from opposite directions.



Schematic of Hi-C and ChIA-PETs.

one-kilobase resolution, hundreds of millions of reads will be needed in addition to a new way to model the interaction data.

But in the meantime, the scientists found surprising results with the number of reads

and replication. Important questions are ‘how does the linear order of genes relate to their spatial arrangement?’ and ‘how are genes regulated by distant elements?’. Two independent groups recently tackled these questions from opposite directions.

Job Dekker from the University of Massachusetts and Eric Lander from the Broad Institute headed an effort to gain an unbiased view of chromatin organization in the nucleus. “We wanted to have a technique to look at the structure of an entire chromosome at kilobase resolution,” recalls Dekker. Unlike previous chromatin conformation

It is Ruan’s goal to now generate a more generic map of all interactions involved in transcription. “With Hi-C you hope to capture all interactions,” Ruan says, “but you do not have their function. With ChIA PETs, we can experimentally annotate each of the interaction sites.”

Yijun Ruan and Edwin Cheung from the Genome Institute of Singapore were interested in these functional relationships between a subset of loci. They asked specific questions, namely, ‘how does a single transcription factor affect the chromatin interactome?’ and ‘why are so many transcription factor binding sites a long distance away from the gene promoters?’. For answers, they elegantly combined the existing methods of chromatin immunoprecipitation (ChIP) and paired-end ditag (PET) sequencing. The researchers ligated immunoprecipitated DNA fragments to a barcoded linker with a type II restriction enzyme

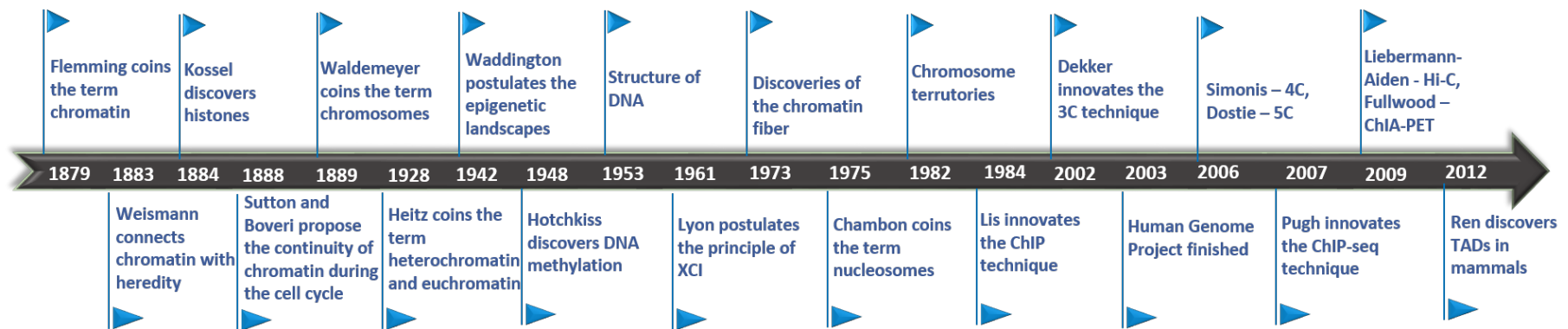


Nature 2009

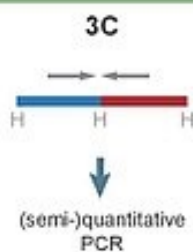
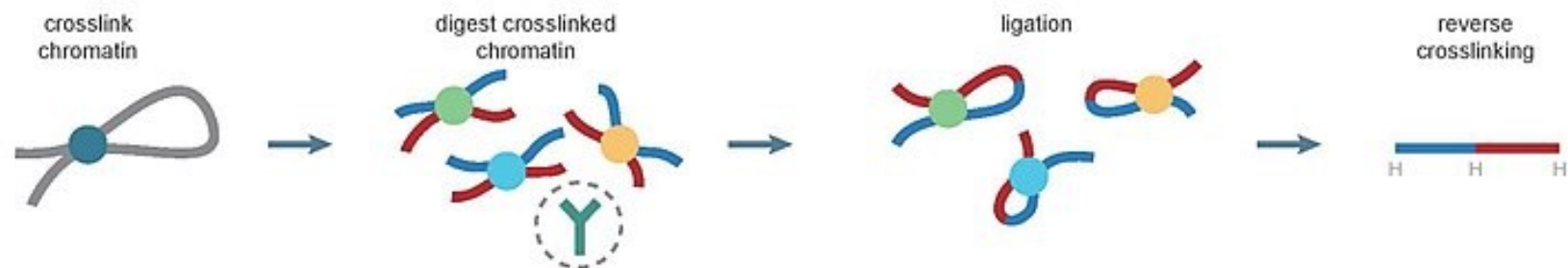
An oestrogen-receptor- α -bound human chromatin interactome

Melissa J. Fullwood¹, Mei Hui Liu¹, You Fu Pan¹, Jun Liu¹, Han Xu¹, Yusoff Bin Mohamed¹, Yuriy L. Orlov¹, Stoyan Velkov¹, Andrea Ho¹, Poh Huay Mei¹, Elaine G. Y. Chew¹, Phillips Yao Hui Huang¹, Willem-Jan Welboren², Yuyuan Han¹, Hong Sain Ooi¹, Pramila N. Ariyaratne¹, Vinsensius B. Vega¹, Yanquan Luo¹, Peck Yean Tan¹, Pei Ye Choy¹, K. D. Senali Abayratna Wansa¹, Bing Zhao¹, Kar Sian Lim¹, Shi Chi Leow¹, Jit Sin Yow¹, Roy Joseph¹, Haixia Li¹, Kartiki V. Desai¹, Jane S. Thomsen¹, Yew Kok Lee¹, R. Krishna Murthy Karuturi¹, Thoreau Herve¹, Guillaume Bourque¹, Hendrik G. Stunnenberg², Xiaolan Ruan¹, Valere Cacheux-Rataboul¹, Wing-Kin Sung^{1,3}, Edison T. Liu¹, Chia-Lin Wei¹, Edwin Cheung^{1,4,5} & Yijun Ruan^{1,4}

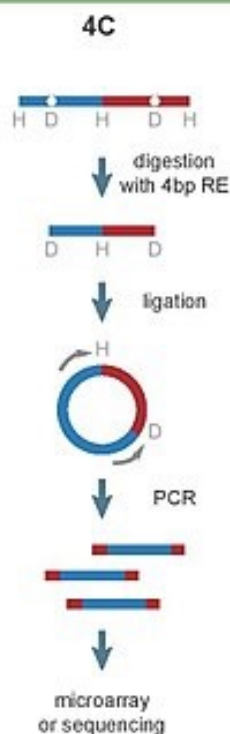
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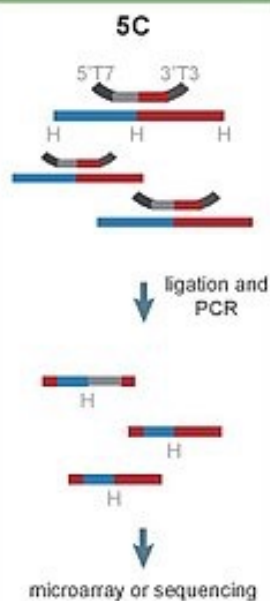
Chromosome Conformation Technologies



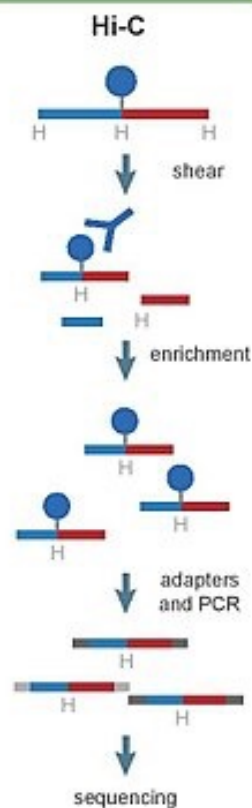
one vs one



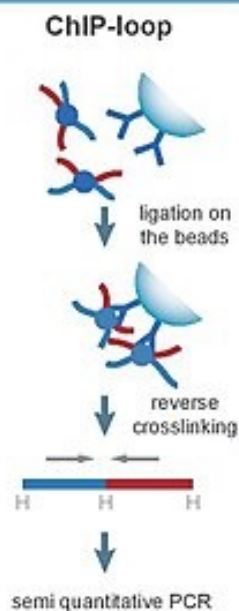
one vs all



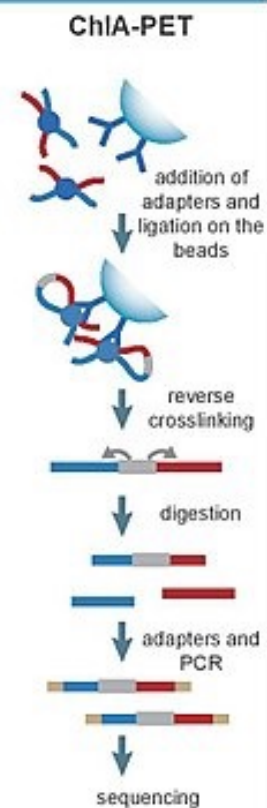
many vs many



all vs all

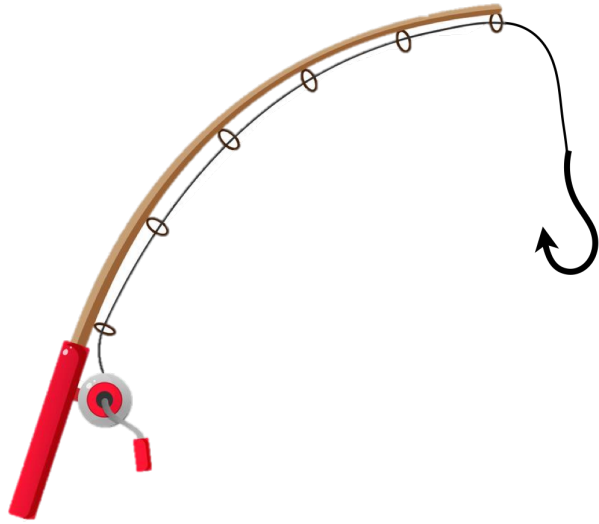


one vs one

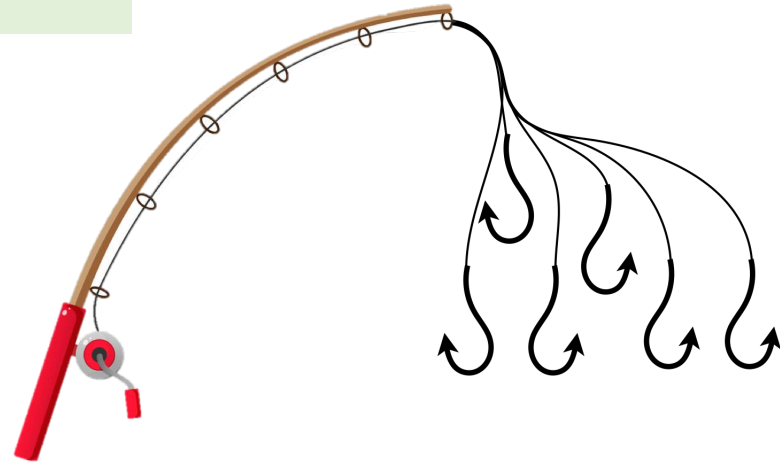


all vs all

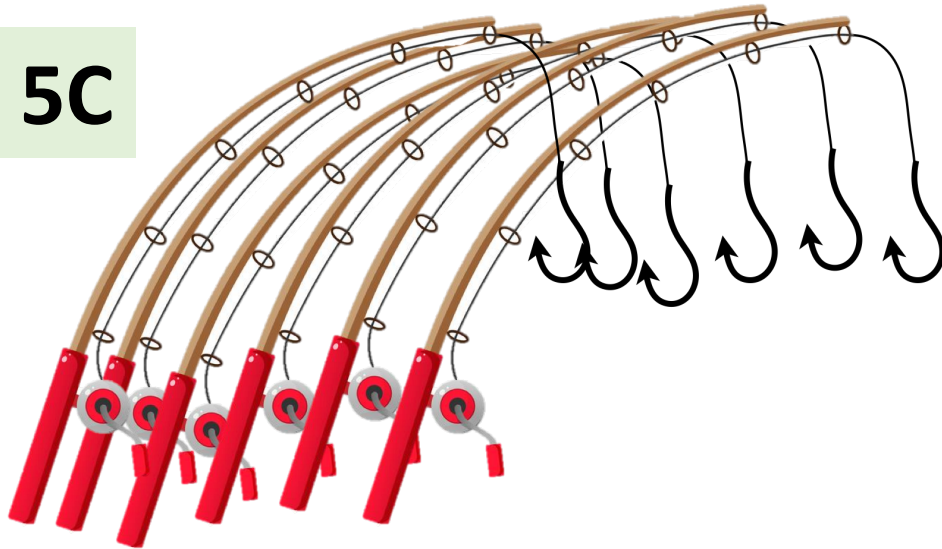
3C



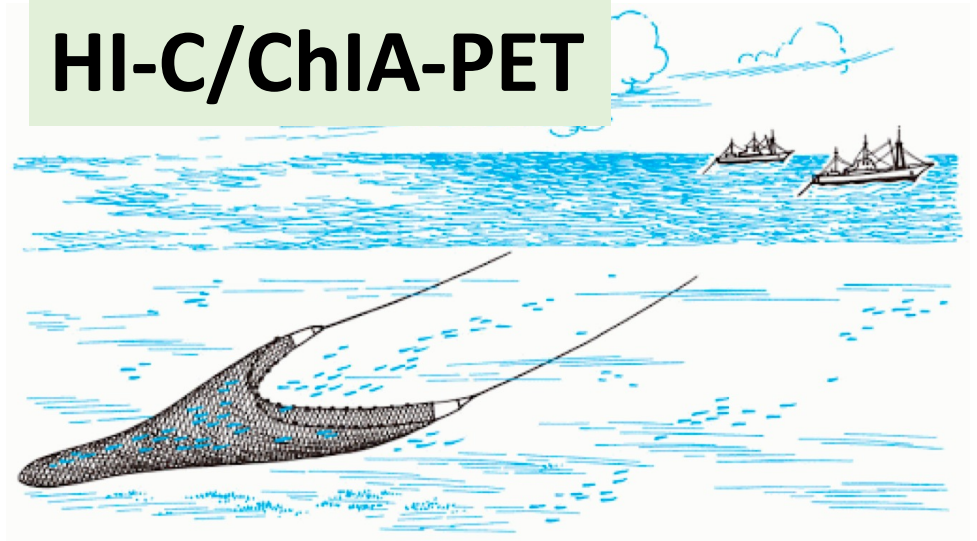
4C



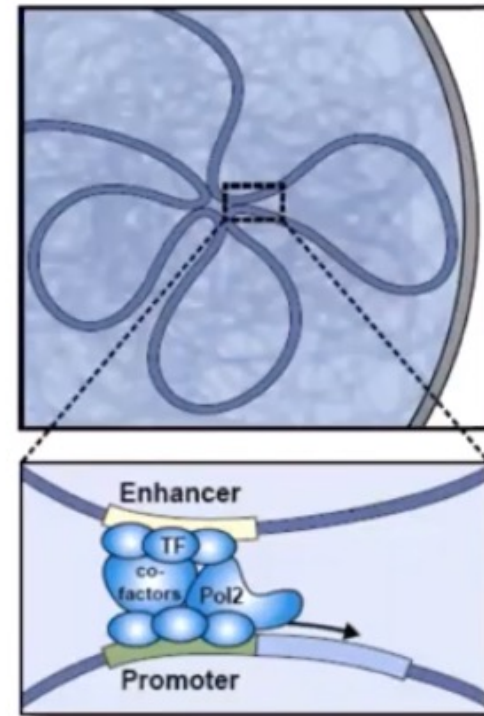
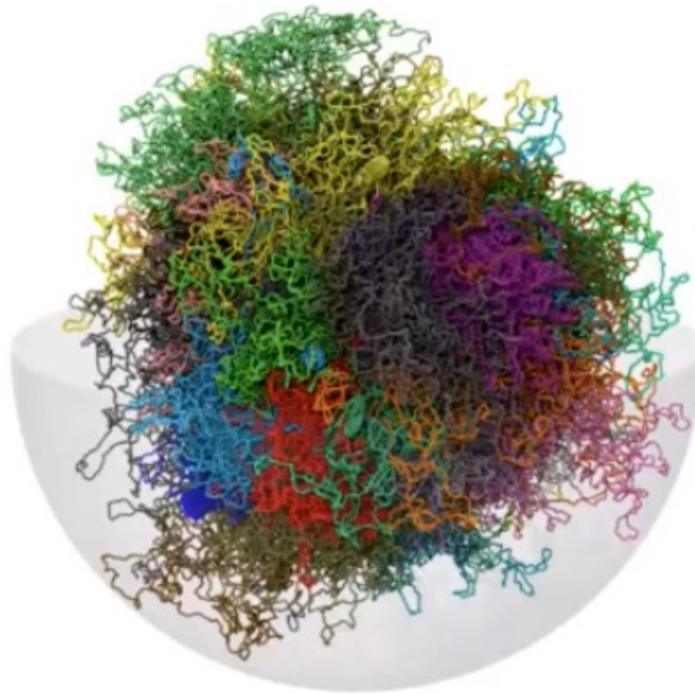
5C



HI-C/ChIA-PET



- Fill the human genome into the nucleus is similar to fill a string of 44miles long into a basketball



From Sissa Medialab and David Gorkin

STAT115

4



nature research



39

4DN

The 4D nucleome project

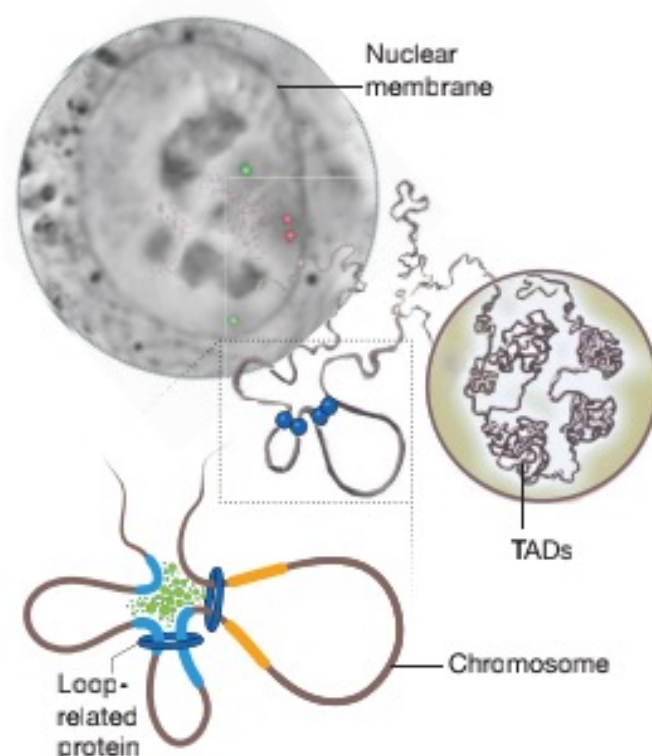


<https://www.4dnucleome.org/>

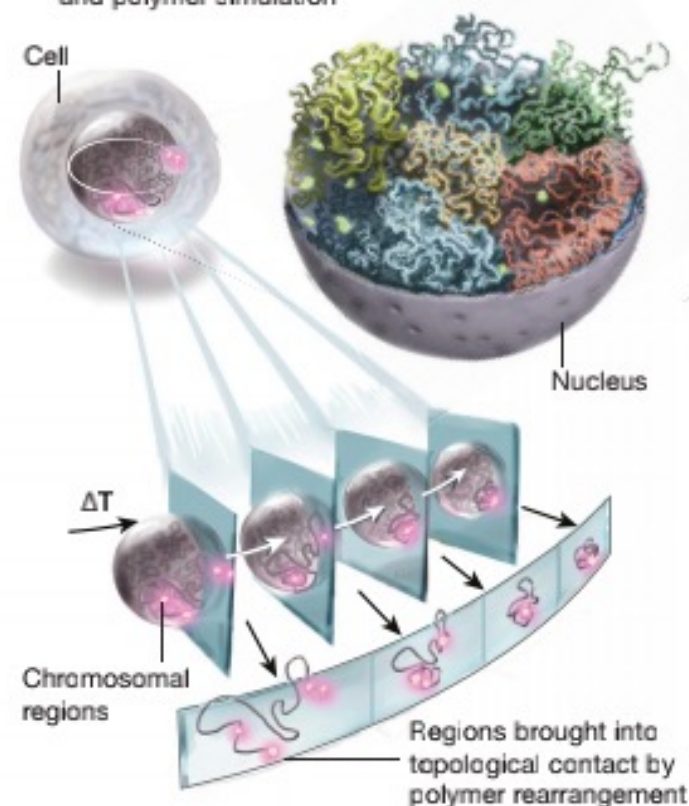
The 4D nucleome project

Job Dekker¹, Andrew S. Belmont², Mitchell Guttman³, Victor O. Leshyk⁴, John T. Lis⁵, Stavros Lomvardas⁶, Leonid A. Mirny⁷, Clodagh C. O'Shea⁸, Peter J. Park⁹, Bing Ren¹⁰, Joan C. Ritland Politz¹¹, Jay Shendure¹², Sheng Zhong⁴ & the 4D Nucleome Network*

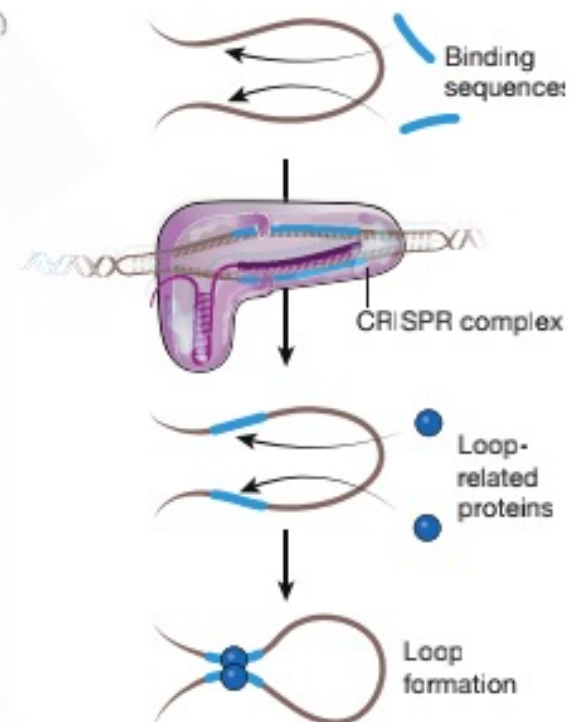
a Mapping – molecular genomic mapping of contacts, imaging of contacts and dynamics



b Model building – by constraint modelling and polymer simulation

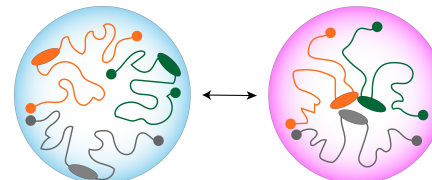


c Functional validation – by genetic and biophysical perturbation experiments



About 4DN

- The NIH 4D Nucleome Program (<https://commonfund.nih.gov/4Dnucleome>), funded as part of the **NIH Common Fund**.
- Aims to understand the principles underlying the **three-dimensional organization of the nucleus**.
- The role this organization plays in **regulating gene expression and cellular function** in the context of normal development and differentiation, and ultimately how disrupting this organization may contribute to disease.
- Researchers funded as part of this program **are developing novel technologies, resources, and data** intended to catalyze research in this field.



Nucleomics Tools (U01) RFA-RM-14-007

Nucleomics Tools (U01) RFA-RM-14-007		
PI Name	Institution Name	Title
FRASER, PETER J (contact) TANAY, AMOS	THE BABRAHAM INSTITUTE	New automated experimental and computational pipeline for high coverage single-cell Hi-C and its integration with single cell RNA-seq: enabling 4D Nucleomics at single cell resolution
GUTTMAN, MITCHELL (contact) ISMAGILOV, RUSTEM F	CALIFORNIA INSTITUTE OF TECHNOLOGY	Novel tools to comprehensively map dynamic organization of RNA and DNA in higher-order nuclear structures within single cells
LIEBERMAN-AIDEN, EREZ	BAYLOR COLLEGE OF MEDICINE	Beyond pairwise DNA contacts: exploring higher-order genome structure using proximity ligation and split pool barcoding
LIS, JOHN T (contact) LIN, HENING ZIPFEL, WARREN R	CORNELL UNIVERSITY	Distance-Hi-C: Creating Photo Activated X-linkers To Define Nuclear Architecture
RAJ, ARJUN (contact) BLOBEL, GERD A PHILLIPS-CREMINS, JENNIFER	UNIVERSITY OF PENNSYLVANIA	Engineering and visualizing genome folding at high spatiotemporal resolution

Study of Nuclear Bodies and Compartments (U01) RFA-RM-14-008

Study of Nuclear Bodies and Compartments (U01) RFA-RM-14-008		
PI Name	Institution Name	Title
BRANGWYNNE, CLIFFORD P	PRINCETON UNIVERSITY	Optogenetic Droplets: Using Light to Control Nucleoplasmic Phase Separation
GERACE, LARRY R.	SCRIPPS RESEARCH INSTITUTE	Mapping the organization of the peripheral nuclear compartment
GROUDINE, MARK T	FRED HUTCHINSON CANCER RES CENTER	A toolkit to reversibly disrupt nuclear bodies and move genes among compartments
GUTTMAN, MITCHELL (contact) HESS, SONJA RINN, JOHN LOUIS	CALIFORNIA INSTITUTE OF TECHNOLOGY	Deciphering the function and mechanisms of lncRNA-mediated organization of nuclear compartments
KAUFMAN, PAUL D. (contact) DEKKER, JOB PEDERSON, THORU	UNIV OF MASSACHUSETTS MED SCH WORCESTER	Nucleolar Genomics During Early Mammalian Development
LOMVARDAS, STAVROS (contact) BROWN, LEWIS M.	COLUMBIA UNIVERSITY HEALTH SCIENCES	Deciphering nuclear bodies and compartments that govern singular olfactory receptor expression

4D Nucleome Network Organizational Hub (U01) RFA-RM-14-010

4D Nucleome Network Organizational Hub (U01) RFA-RM-14-010		
PI Name	Institution Name	Title
ZHONG, SHENG (contact) CHIEN, SHU REN, BING SUBRAMANIAM, SHANKAR	UNIVERSITY OF CALIFORNIA SAN DIEGO	The Organizational Hub and Web Portal for the 4D Nucleome Network

4D Nucleome Network Data Coordination and Integration Center (U01) RFA-RM-14-011

4D Nucleome Network Data Coordination and Integration Center (U01) RFA-RM-14-011		
PI Name	Institution Name	Title
PARK, PETER J	HARVARD MEDICAL SCHOOL	4D Nucleome Network Data Coordination and Integration Center
WANG, TING	WASHINGTON UNIVERSITY	The WashU 4DN Network Data Coordination and Integration Center

4D Nucleome Imaging Tools (U01) RFA-RM-14-009

4D Nucleome Imaging Tools (U01) RFA-RM-14-009		
PI Name	Institution Name	Title
BEWERSDORF, JOERG (contact) BADDELEY, DAVID	YALE UNIVERSITY	An Integrated Imaging System for High-throughput Nanoscopy of the 4D Nucleome
CAI, LONG (contact) QI, LEI STANLEY	CALIFORNIA INSTITUTE OF TECHNOLOGY	Hi-resolution dynamic imaging of chromosomes in single cells by combined CRISPR imaging and sequential FISH
ELLENBERG, JAN	EUROPEAN MOLECULAR BIOLOGY LABORATORY	Reconstructing the dynamic 3D architecture of the human genome by superresolution microscopy and DNA sequence modelling
GREGOR, THOMAS (contact) LEVINE, MICHAEL S.	PRINCETON UNIVERSITY	Imaging chromosome dynamics and measuring its impact on transcriptional activity
GRUNWALD, DAVID	UNIV OF MASSACHUSETTS MED SCH WORCESTER	Visualizing local and global chromatin architecture, and gene expression in the individual cell by structural (SUSHI) and temporal (3D-SMRT) single molecule imaging
LAM, KIT S (contact) CHENG, R.HOLLAND IZUMIYA, YOSHIHIRO	UNIVERSITY OF CALIFORNIA AT DAVIS	Genetically Encoded Small Illuminants for 4D nucleome imaging
LIPHARDT, JAN T.	STANFORD UNIVERSITY	Deep Super-localization Microscopy and Effectively Unbleachable Labeling for 4D Nucleomics
O'SHEA, CLODAGH (contact) ELLISMAN, MARK H	SALK INSTITUTE FOR BIOLOGICAL STUDIES	Imaging local and global chromatin structure as a 3D continuum within the nucleus
SINGER, ROBERT H. (contact) DARZACQ, XAVIER	ALBERT EINSTEIN COLLEGE OF MEDICINE	Tools For Imaging the Functional Genome in Living Cells and Tissues

Nuclear Organization and Function Interdisciplinary Consortium (NOFIC) (U54) RFA-RM-14-030

Nuclear Organization and Function Interdisciplinary Consortium (NOFIC) (U54) RFA-RM-14-030		
PI Name	Institution Name	Title
ALBER, FRANK (contact) CHEN, LIN	UNIVERSITY OF SOUTHERN CALIFORNIA	Mapping the 3D Genome Landscape
BELMONT, ANDREW STEVEN	UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN	Combined Cytological, Genomic, and Functional Mapping of Nuclear Genome Organization
DEKKER, JOB (contact) MIRNY, LEONID A	UNIV OF MASSACHUSETTS MED SCH WORCESTER	Center for 3D Structure and Physics of the Genome
REN, BING (contact) MURRE, CORNELIUS	LUDWIG INSTITUTE FOR CANCER RESEARCH LTD	San Diego Center for 4D Nucleome Research
RUAN, YIJUN	JACKSON LABORATORY	Nucleome Positioning System for Spatiotemporal Genome Organization and Regulation
SHENDURE, JAY ASHOK (contact) NOBLE, WILLIAM STAFFORD	UNIVERSITY OF WASHINGTON	University of Washington Center for Nuclear Organization and Function

New Investigator Projects on 4DN Organization and Function in Human Health and Disease (U01 Clinical Trial Not Allowed) RFA-RM-20-006		
PI Name	Institution Name	Title
DIAO, YARUI	DUKE UNIVERSITY	The 4D nucleome of muscle regeneration in ischemia-induced tissue damage and repair
KALHOR, REZA	JOHNS HOPKINS UNIVERSITY	Dynamics of lineage-specific genome reorganization in gastrulation and their response to disease-associated epigenetic perturbations
LI, WENBO	UNIVERSITY OF TEXAS HLTH SCI CTR HOUSTON	Beyond Gene Dosage: Understanding Down Syndrome via 4D Genome Organization
SATPATHY, ANSUMAN	STANFORD UNIVERSITY	4DN Interrogation of T Cell Exhaustion in Cancer
YAMADA, TOMOKO	NORTHWESTERN UNIVERSITY	Mechanisms of Genome Organization in Brain Development and Behavior
ZHANG, HUAIYING	CARNEGIE-MELLON UNIVERSITY	Phase separation-induced nuclear organization in ALT Cancer

Limited Competition: 4DN Organizational Hub (U01 Clinical Trial Not Allowed) RFA-RM-20-007		
PI Name	Institution Name	Title
ZHONG, SHENG	UNIVERSITY OF CALIFORNIA, SAN DIEGO	The second phase of NIH Common Fund 4D Nucleome Network Organizational Hub

Limited Competition 4DN Data Coordination & Integration Center (U01 Clinical Trial Not Allowed) RFA-RM-20-008		
PI Name	Institution Name	Title
PARK, PETER J	HARVARD MEDICAL SCHOOL	4D Nucleome Network Data Coordination and Integration Center

4DN Organization and Function in Human Health and Disease (U01 Clinical Trial Not Allowed) RFA-RM-20-005

4DN Organization and Function in Human Health and Disease (U01 Clinical Trial Not Allowed) RFA-RM-20-005		
PI Name	Institution Name	Title
BRUNEAU, BENOIT GAETAN (contact) POLLARD, KATHERINE S.	J. DAVID GLADSTONE INSTITUTES	Genetic determinants of 4D genome folding in human cardiac development
DIXON, JESSE R	SALK INSTITUTE FOR BIOLOGICAL STUDIES	Investigating the effects of structural variants on 3D genome organization and gene regulation in cancer genomes
HAWKINS, RAYMOND DAVID	UNIVERSITY OF WASHINGTON	Three-dimensional conformation changes associated with T cell memory and autoimmunity
HUANGFU, DANWEI (contact) APOSTOLOU, EFFIE LESLIE, CHRISTINA S	SLOAN-KETTERING INST CAN RESEARCH	Discovery of diabetes-relevant B cell enhancers through 4D enhancer mapping, integrative analysis, and large-scale CRISPRi perturbation screens
KAUFMAN, PAUL D. (contact) FOLTZ, DANIEL RICHARD HUANG, SUI	UNIV OF MASSACHUSETTS MED SCH WORCESTER	The role of the nucleolus in human genome organization in normal and disease states
LOMVARDAS, STAVROS	COLUMBIA UNIVERSITY HEALTH SCIENCES	Interrogating genome folding trajectories in health and disease
PHILLIPS-CREMINS, JENNIFER ELIZABETH (contact) JAIN, RAJAN JOYCE, ERIC F. LAKADAMYALI, MELIKE VAHEDI, GOLNAZ	UNIVERSITY OF PENNSYLVANIA	Single-cell dissection of chromatin architecture mechanisms connecting pathologic instability and transcriptional silencing
SCHATZ, DAVID G. (contact) WANG, SIYUAN	YALE UNIVERSITY	Genome Architecture in Human Germinal Center B Cell Development, Malignancy, and Somatic Hypermutation
SHEN, YIN (contact) KRIEGSTEIN, ARNOLD	UNIVERSITY OF CALIFORNIA, SAN FRANCISCO	Charting the 3D epigenome in human brain development and diseases
XU, XIANGMIN (contact) COTMAN, CARL WAYNE REN, BING	UNIVERSITY OF CALIFORNIA-IRVINE	Single-Cell Analysis of Aging-Associated 4D Nucleome in the Human Hippocampus

4DN Centers for Data Integration, Modeling and Visualization (UM1 Clinical Trial Not Allowed) RFA-RM-20-004

4DN Centers for Data Integration, Modeling and Visualization (UM1 Clinical Trial Not Allowed) RFA-RM-20-004		
PI Name	Institution Name	Title
DEKKER, JOB (contact) MIRNY, LEONID A	UNIV OF MASSACHUSETTS MED SCH WORCESTER	Center for 3D Structure and Physics of the Genome
MA, JIAN (contact) ALBER, FRANK BELMONT, ANDREW STEVEN WU, CHAO-TING	CARNEGIE-MELLON UNIVERSITY	Multiscale Analyses of 4D Nucleome Structure and Function by Comprehensive Multimodal Data Integration
NOBLE, WILLIAM STAFFORD (contact) DISTECHE, CHRISTINE M. SHENDURE, JAY ASHOK	CALIFORNIA INSTITUTE OF TECHNOLOGY	UW 4-Dimensional Genomic Organization of Mammalian Embryogenesis Center
REN, BING (contact) DULAC, CATHERINE G LEIN, ED NICODEMI, MARIO POMBO, ANA WANG, TING ZHUANG, XIAOWEI	UNIVERSITY OF CALIFORNIA, SAN DIEGO	Center for Integrated Multi-modal and Multi-scale Nucleome Research

Real-Time Chromatin Dynamics and Function (U01 Clinical Trial Not Allowed) RFA-RM-20-003

Real-Time Chromatin Dynamics and Function (U01 Clinical Trial Not Allowed) RFA-RM-20-003		
PI Name	Institution Name	Title
BELMONT, ANDREW STEVEN (contact) HAN, KYU YOUNG	UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN	Identification of the active nuclear niche(s) using novel proteomic, genomic, transgenic, and live-cell microscopy technologies
BINTU, LACRAMIOARA (contact) BOETTIGER, ALISTAIR	STANFORD UNIVERSITY	Live-cell multiplex super-resolution imaging of chromatin state transitions
CAI, LONG (contact) ELOWITZ, MICHAEL B GUTTMAN, MITCHELL	CALIFORNIA INSTITUTE OF TECHNOLOGY	Dynamics of chromosome organization and chromatin states in single cells
GREGOR, THOMAS (contact) LEVINE, MICHAEL STEVEN TOETTCHER, JARED E	PRINCETON UNIVERSITY	Control of the 4D chromatin landscape underlying gene activity during development
HA, TAEKJIP (contact) JOHNSON, MARGARET ELLEN WU, CARL	JOHNS HOPKINS UNIVERSITY	Chromatin Function During Transcription and DNA Repair at Single Molecule Resolution in Living Cells
HUANG, BO (contact) NARLIKAR, GEETA J SHU, XIAOKUN	UNIVERSITY OF CALIFORNIA, SAN FRANCISCO	Modulation and functional characterization of protein condensation in chromatin organization
PHILLIPS-CREMIN, JENNIFER ELIZABETH (contact) BLOBEL, GERD A QI, LEI STANLEY RAJ, ARJUN	UNIVERSITY OF PENNSYLVANIA	Engineering and Imaging 3D genome structure-function dynamics across time scales
TORRES PADILLA, MARIA ELENA	HELMHOLTZ ZENTRUM MUENCHEN - GMBH	Establishment and visualization of nuclear architecture at the onset of mammalian development

4DN Phase 2 Onboarding

The Goal and Aims of 4DN

- The **goal** of the Common Fund's 4D Nucleome (4DN) program is to study the **three-dimensional organization of the nucleus in space and time** (the 4th dimension).

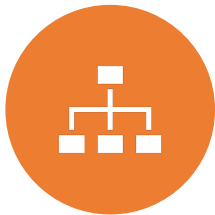
The Goal and Aims of 4DN

- The Common Funds's 4D Nucleome program **aims** to :
 - Understand the **principles** underlying nuclear organization in space and time
 - The **role** nuclear organization plays in gene expression and cellular function
 - **How changes** in nuclear organization affect normal development as well as various disease

The Goal and Aims of 4DN

- This program is **developing technologies, resources and data** to enable the study of the 4D Nucleome
- **Novel tools** to explore the dynamic nuclear architecture and its role in gene expression
- **Programs, models** to examine the relationship between nuclear organization and function in both normal development and disease
- **Reference maps** of nuclear architecture in a variety of cells and tissues

4DN Phase 2 Initiatives



Organizational Hub



Data Coordination and
Integration Center



Real Time Chromatin
Dynamics and Function



4DN Centers for Data
Integration, Modeling, and
Visualization

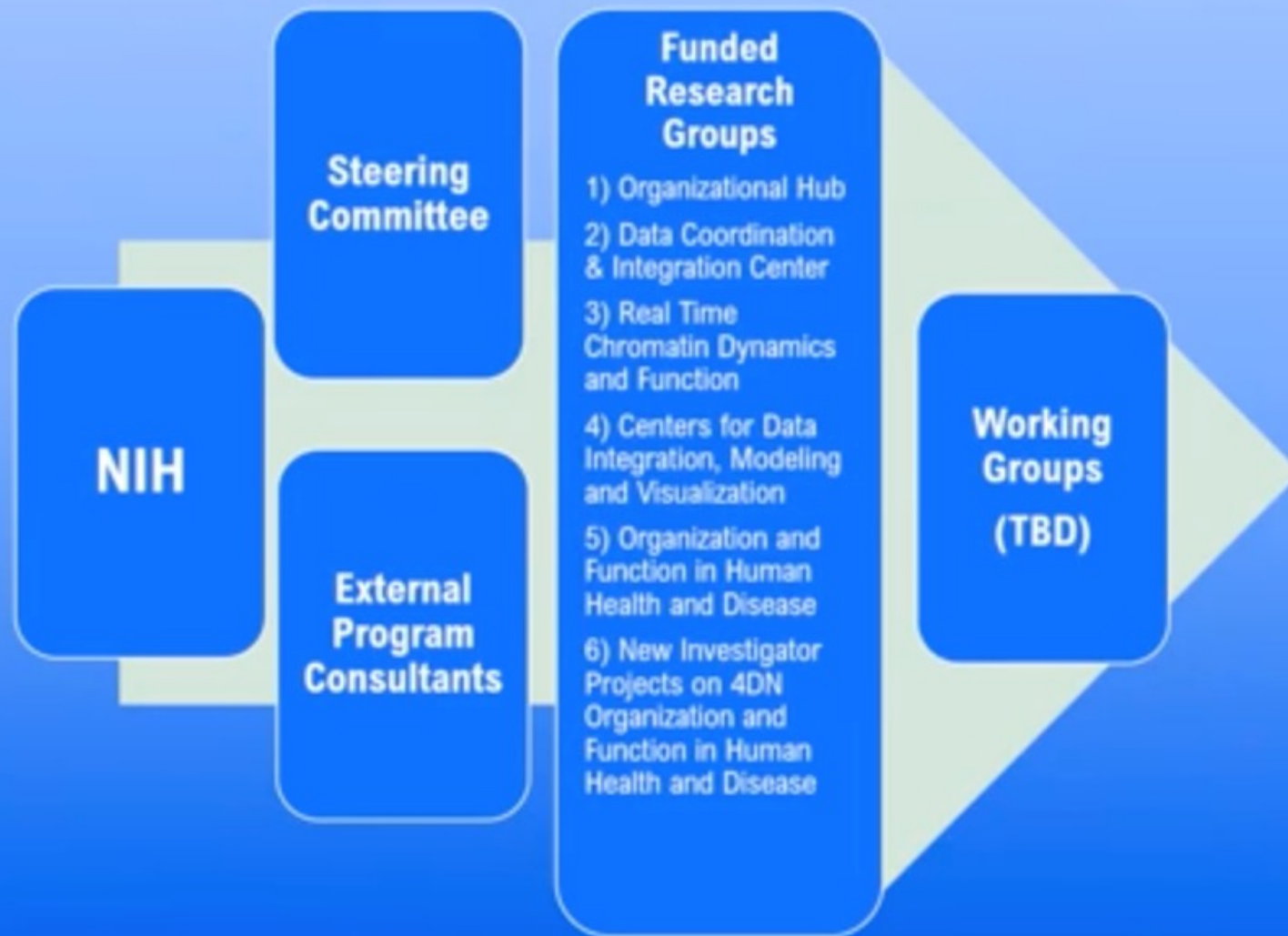


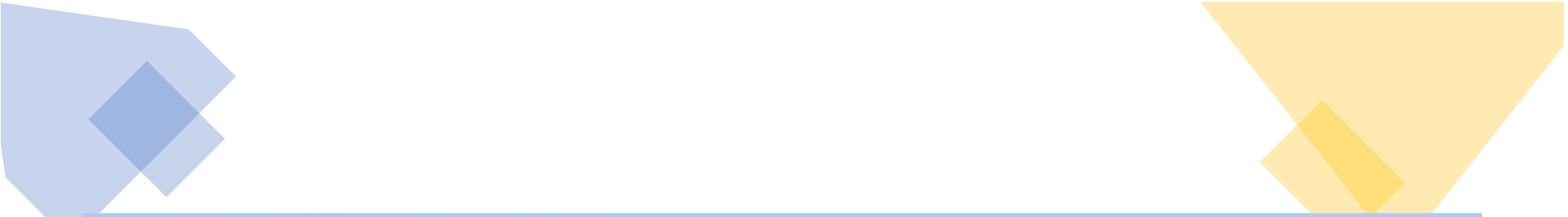
4DN Organization and
Function in Human Health
and Disease



New Investigator Projects
on 4DN Organization and
Function in Human Health
and Disease

4DN Communication Chart





4DN Member Access

- **4DN Web Portal**

To find information about the data, protocols, reagents, resources, and methods that are currently being developed by 4DN Network investigators.

- **4DN Data Portal**

The 4D Nucleome Data Portal hosts data generated by the 4DN Network and other reference nucleomics data sets, and an expanding tool set for open data processing and visualization.

- **4DN Wiki**

The 4D Nucleome Network Wiki is an internal collaboration site, accessible to 4DN investigators and members only.

- **4DN Directory**

Provides a one-stop directory for all current 4DN members and allows logged in 4DN members to manage their and their team members' profiles, participate in the interval working groups, and more.



4DN Web Portal

Launch Data Portal | Contact Us

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What is the 4D Nucleome Program?

Watch this video on Youtube to learn how 4DN aims to improve public health!

• • • • •

Tweets by @4DNucleome

4DNucleome Retweeted



Yun Li
@yunliunc

I'm very pleased to present our new 3D chromatin interaction caller: HiC-ACT, published with AJHG cell.com/ajhg/fulltext/... @AJHGNews. HiC-ACT is simple yet powerful. Great collaboration with @afhumung. Looking forward to applying to HiC-ACT to 4DN Phase2 data @4DNucleome



HiC-ACT: improved detection ...
Genome-wide chromatin confor...
cell.com



Feb 5, 2021

Embed

View on Twitter

Latest 4DN Publications



SPIN reveals genome-wide landscape of nuclear compartment...



Yuchuan Wang, Yang Zhang, Ruochi Zhang, Tom van Schaik, Liguang Zhang, Takayo Sasaki, Daniel Peric-Hupkes, Yu Chen, David M. Gilbert, Bas van Steensel, Andrew S. Belmont, Jian Ma

We report SPIN, an integrative computational method to reveal genome-wide intranuclear chromosome positioning and nuclear compartmentalization relative to multiple nuclear structures, which are pivotal for modulating genome function. As a proof-of-principle, we use SPIN to integrate nuclear compartment mapping (TSA-seq and DamID) and chromatin interaction data (Hi-C) from K562 cells to identify 10 spatial compartmentalization states genome-wide relative to nuclear speckles, lamina, and putative associations with nucleoli. These SPIN states show novel patterns of genome spatial organization and their relation to other 3D genome features and genome function (transcription and replication timing). SPIN provides critical insights into nuclear spatial and functional compartmentalization.

Jan 14, 2021

Launch Data Portal

The 4D Nucleome Data Portal hosts data generated by the 4DN Network and other reference nucleomics data sets, and an expanding tool set for open data processing and visualization.

Nucleome Browser

Nucleome Browser is a multimodal, interactive data visualization and exploration platform to integrate genomic, imaging, and 3D structure data.

Internal Wiki

The 4D Nucleome Network Wiki is an internal collaboration site, accessible to 4DN investigators and members only.

people.4dnucleome

people.4dnucleome provides a one-stop directory for all



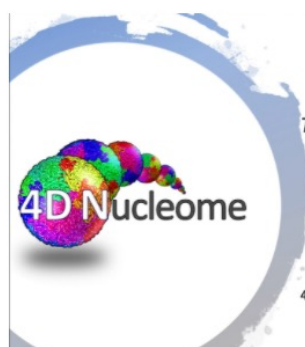
4DN Web Portal

[Launch Data Portal](#) | [Contact Us](#)

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Tutorial for 4DN New Members:

Tools and Services Provided by 4DN OH

4D Nucleome Network Organizational Hub
Contact: oh@4dnucleome.org

Tutorial for 4DN new members

Welcome to 4DN! Watch this video on YouTube to learn how to use 4DN tools and services!

• • • • •

Tweets by @4DNucleome

4DNucleome Retweeted



Yun Li
@yunliunc

I'm very pleased to present our new 3D chromatin interaction caller: HiC-ACT, published with AJHG cell.com/ajhg/fulltext/... @AJHGNews. HiC-ACT is simple yet powerful. Great collaboration with @afhumung. Looking forward to applying to HiC-ACT to 4DN Phase2 data @4DNucleome



HiC-ACT: improved detection o...
Genome-wide chromatin conform...
cell.com



Feb 5, 2021

[Embed](#)

[View on Twitter](#)

Latest 4DN Publications



TSA-seq reveals a largely conserved genome organization...



Liguo Zhang, Yang Zhang, Yu Chen, Omid Gholamalamdari, Yuchuan Wang, Jian Ma, Andrew S. Belmont

TSA-seq mapping suggests that gene distance to nuclear speckles is more deterministic and predictive of gene expression levels than gene radial positioning. Gene expression correlates inversely with distance to nuclear speckles, with chromosome regions of unusually high expression located at the apex of chromosome loops protruding from the nuclear periphery into the interior. Genomic distances to the nearest lamina-associated domain are larger for loop apexes mapping closest to nuclear speckles, suggesting the possibility of conservation of speckle-associated regions. To facilitate comparison of genome organization by TSA-seq, we reduced required cell numbers 10-20-fold for TSA-seq by deliberately saturating protein-labeling while preserving distance mapping by the still unsaturated DNA-labeling. [...]

Dec 17, 2020

[Close to the edge: Heterochromatin at the](#)

Launch Data Portal

The 4D Nucleome Data Portal hosts data generated by the 4DN Network and other reference nucleomics data sets, and an expanding tool set for open data processing and visualization.

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Nucleome Browser is a multimodal, interactive data visualization and exploration platform to integrate genomic, imaging, and 3D structure data.

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COVID-19

- [Get the latest public health information from CDC](#) »
- [Get the latest research information from NIH](#) » | [Español](#) »
- [NIH staff guidance on coronavirus \(NIH Only\)](#) »

4D Nucleome



Common Fund » Common Fund Programs » 4D Nucleome (4DN)

4D Nucleome (4DN)

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[Health Relevance](#)

[Highlights](#)

For Researchers

[Funding Opportunities](#)

[Funded Research](#)

[TCPA Awards](#)

[NIH Working Group](#)

[Program Publications](#)

What is the 4D Nucleome Program?

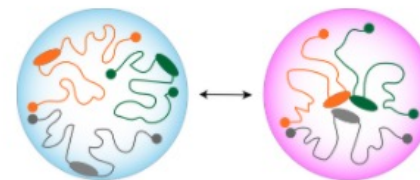
Watch this video to learn how 4DN aims to improve public health!

[Watch](#)



Program Snapshot

The goal of the Common Fund's **4D Nucleome (4DN)** program is to study the three-dimensional organization of the nucleus in space and time (the 4th dimension). The nucleus of a cell contains DNA, the genetic "blueprint" that encodes all of the genes a living organism uses to produce proteins needed to carry out life-sustaining



Under different conditions, such as disease, regions of our DNA (different colored ovals) may change shape or location inside the nucleus of a

Article Features 4DN Program

Read [here](#) about how NIH-supported research is advancing what we know about genome organization.

4D Nucleome Portal

Visit the [4D Nucleome Portal](#) to find information about the data, protocols, reagents, resources, and methods that are currently being developed by 4DN Network investigators.

4DN Program Announces Associate Membership

Learn more about becoming an [associate member](#) [pdf](#).



1749 Experiment Sets

4238 Experiments

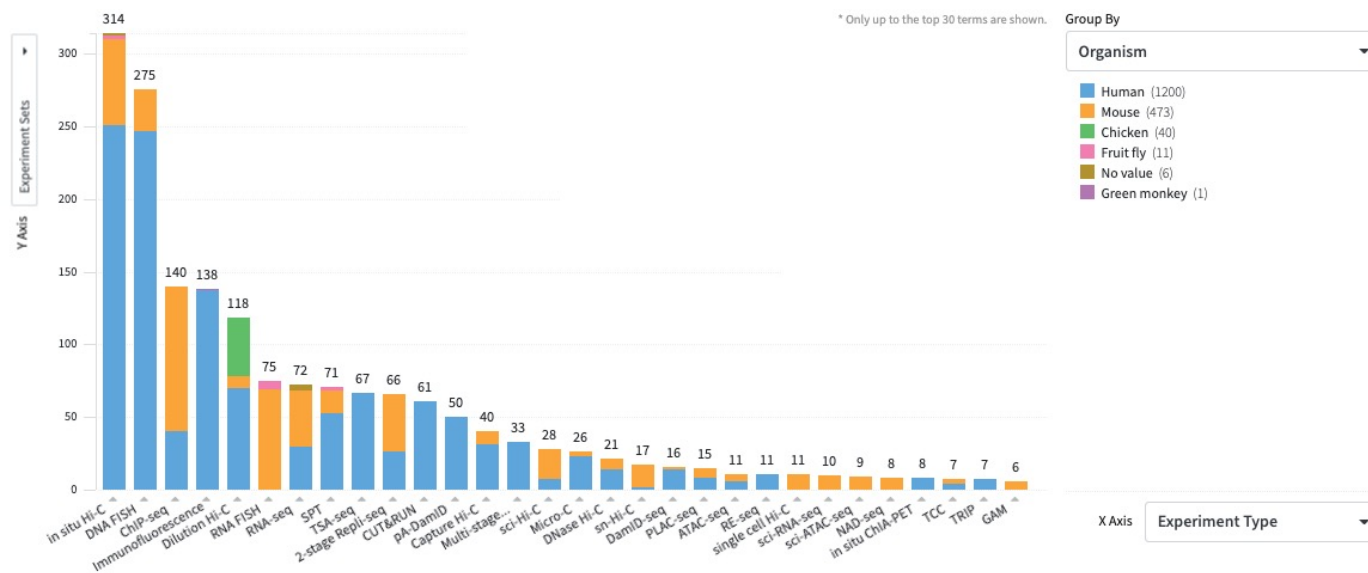
31591 Files



Include External Data

4D Nucleome Data Portal

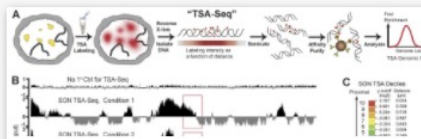
A platform to search, visualize, and download nucleomics data.



Share Your Feedback

We are listening!

In less than 60 seconds, you can provide feedback on your data portal experience.



Data Highlight

Chen et al. introduce TSA-seq and use it to estimate distances of loci genome-wide relative to nuclear compartments



Data Highlight

Finn et. al. utilize high throughput imaging to study heterogeneity in 3D genome structure.



Browse All >

Search All Experiment Sets in the 4D Nucleome Database



Browse Sequencing Data

Search Sequencing Experiment Sets in the 4D Nucleome Database



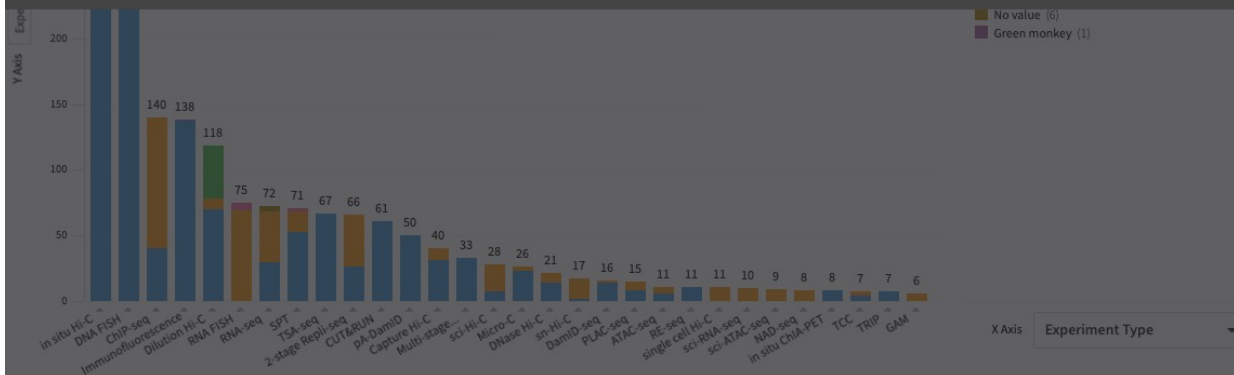
View Microscopy Data

View Microscopy Datasets in the 4D Nucleome Database



Browse by Publication

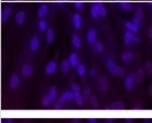
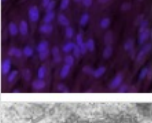
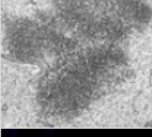
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4DN Microscopy Data Overview

Microscopy Datasets

Below is a summary table of the microscopy datasets that are publicly available on the 4DN data portal. Many of these datasets don't include the raw image files; this is because the submitters in many cases have deemed the processed data to be much more useful to collaborators and outside users than raw images. If you are interested in obtaining raw images or any other type of file that isn't available on the data portal, please contact the data generating lab.

Sample Image	Dataset	Experiment Type	Publication	Biosource	Lab	Data File Availability
	DNA FISH on Chr1 52 Experiment Sets (5 Experiment Sets with Raw Files)	DNA FISH	Finn et al (2019) Cell	HFF-hTERT	Tom Misteli NCI	Raw: 600 Processed: 272
	DNA FISH on Chr1 and Chr17 14 Experiment Sets	DNA FISH	Finn et al (2019) Cell	HFF-hTERT	Tom Misteli NCI	Raw: 0 Processed: 52
	DNA FISH on Chr17 71 Experiment Sets	DNA FISH	Finn et al (2019) Cell	HFF-hTERT	Tom Misteli NCI	Raw: 0 Processed: 256
	DNA FISH on Chr18 38 Experiment Sets	DNA FISH	Finn et al (2019) Cell	HFF-hTERT	Tom Misteli NCI	Raw: 0 Processed: 137
	DNA FISH on Chr4 70 Experiment Sets	DNA FISH	Finn et al (2019) Cell	HFF-hTERT	Tom Misteli NCI	Raw: 0 Processed: 247
	ChromEMT 3 Experiment Sets	Electron Tomography	Ou et al (2017) Science	Multiple	Clodagh O'Shea SALK	Raw: 6 Processed: 0



Data Portal

 Data ▼

Tools ▼

Resources ▼

Help ▼



Visualization & Analysis Tools

Visualization and Analysis Tools

4DN Visualization Workspace

4DN JupyterHub - BETA

MicroMeta - ALPHA

4DN Visualization Workspace

GRCh38

GRCm38

dm6

Please contact the [DCIC](#) with any questions, suggestions, or requests. Please note that the 4DN Visualization Workspace is a BETA release with active development. Features may change without notice.



4DN JupyterHub - BETA



Users should not need to download large files to explore 4DN Data. The 4DN Data Portal provides an accompanying [Jupyter Notebook Service](#) to all users for immediate access to the data on the portal. In addition to files released to public, users are automatically authenticated to access restricted data, shared with the 4DN Network or within their labs.

Please note that you need a Data Portal account to access the 4DN JupyterHub. You can register by clicking the **Log In/Register** button in the top bar.

The Jupyter Notebooks run Python 3 and come pre-installed with a number of tools to access the 4DN Data Portal and the data within. Existing ipython notebook (ipynb) files in the examples directory demonstrate sample use cases.

Please contact the [DCIC](#) with any questions, suggestions, or requests. Please note that the 4DN JupyterHub service is a BETA release with active development. Features may change without notice.

```
In [7]: # The jupyterhub utils package from dcic is used for programmatic access to the 4DN Data Portal.
        from dcicutils import jh_utils

In [8]: # Searching for Hi-C experiments on the H1-Tier1 cell line
search_hicexps = jh_utils.search_metadata('/search/?type=ExperimentSet'
                                         '&experiments_in_set.experiment_type=in+situ+Hi-C'
                                         '&experiments_in_set.biosample.biosource_summary=H1-hE')
print(len(search_hicexps), "Experiment Set matched the query.")

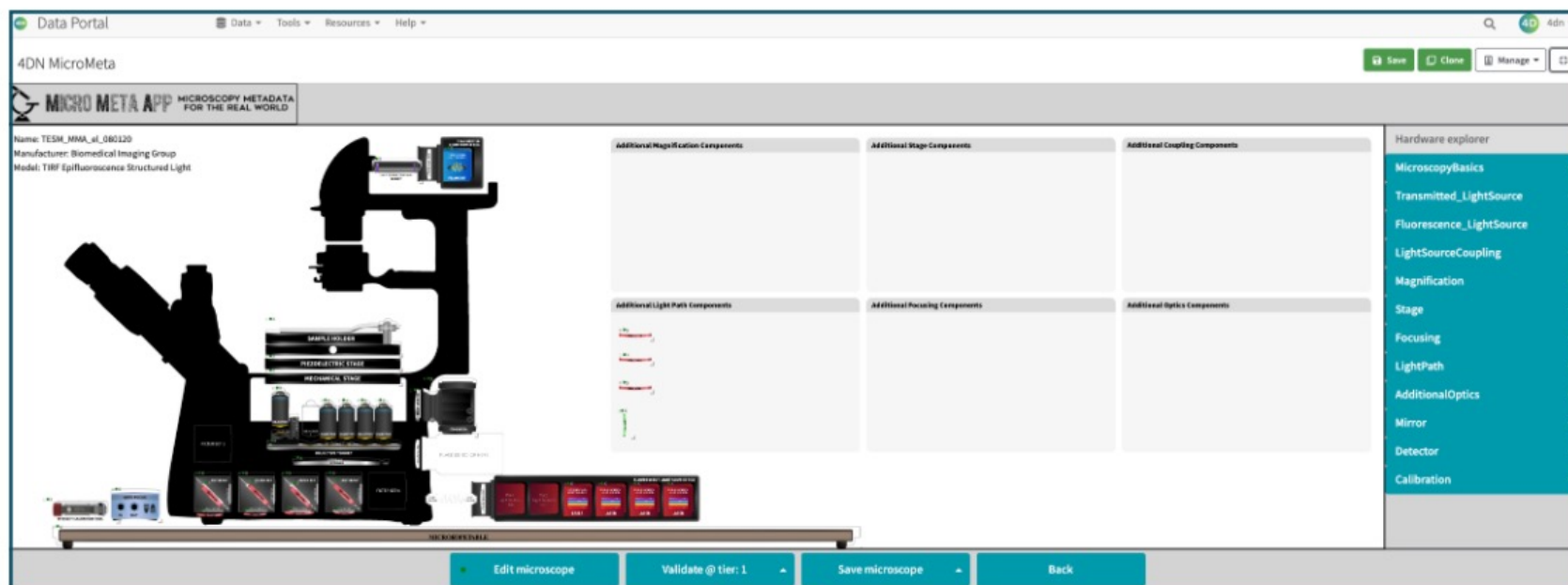
1 Experiment Set matched the query.
```

[Home](#) > [Visualization & Analysis Tools](#) >

MicroMeta - ALPHA

MicroMeta provides a user interface to collect metadata about microscopes and, soon, their settings based on the 4DN-OME standards. Please see the [MicroMeta documentation](#) for more information.

The MicroMeta App is integrated to the 4DN Data Portal allowing users to create, edit, save, and clone microscopes. Click "Manage Microscopes" to start.

[Manage Microscopes](#)



Resources

Tools and Information about Materials & Methods

Experimental Resources

- 4DN Experimental Protocols
- 4DN Cell Lines
- Experiment Types - Genomics and Imaging
- 4DN file formats - imaging

Data Analysis

- Reproducible Data Analysis
- Reference Files
- Hi-C Processing Pipeline
- Repli-seq Processing Pipeline
- ChIP-seq Processing Pipeline
- ATAC-seq Processing Pipeline
- RNA-seq Processing Pipeline
- Domain Calling Pipelines
- Quality Control Workflows

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- 4DN Joint Analysis Portal
- NOFIC AICS Collaborative Project
- Hi-C Data Sets and Studies on the 4DN Portal
- High-resolution Hi-C datasets

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4DN Experimental Protocols

Approved 4DN Experimental Protocols

The Protocols in the Table Below have been approved as 4DN Standard Protocols by the 4DN Genomics Working Group and Steering Committee. The utilization of these protocols whenever feasible is encouraged.

Title ▾	Experiment Type ▾	Approval Date ▲
+ 4DN Standard in situ Hi-C proto...	in situ Hi-C	February 21st, 2017
+ 4DN Standard 2-stage Repli-seq...	2-stage Repli-seq	June 20th, 2017
+ 4DN Standard Long Read ChIA-...	ChIA-PET	October 17th, 2017
+ 4DN Standard DamID-seq proto...	DamID-seq	December 12th, 2017
+ 4DN Standard DNase Hi-C proto...	DNase Hi-C	January 16th, 2018
+ 4DN Standard single cell Hi-C pr...	single cell Hi-C	April 17th, 2018
+ 4DN Standard PLAC-seq Protocol	PLAC-seq	August 18th, 2020
+ 4DN Standard HiChIP Protocol	HiChIP	August 18th, 2020
+ 4DN Standard in situ ChIA-PET	in situ ChIA-PET	August 18th, 2020

Protocol

4DN Standard in situ ChIA-PET Protocol

● Released

 [View JSON](#)

4DN standard protocol for in situ ChIA-PET to detect and quantify chromatin contacts anchored at genomic regions associated with specific DNA binding proteins or histone modifications - approved by SC 08/18/2020

 August 25th, 2020 at 8:00am

 Overview

 Attribution

 Details

Overview

Protocol Type 

Experimental protocol

Document Classification 

None

Attached File 

 [View 20200820_4DN_in_situ_chiaped_protocol.pdf ...](#)

MD5 Checksum  : 8077b53c402e744fa1137a9267fda1e2

Media Type : application/pdf

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4DN Cell Lines

— TIER-1

Common cell lines used by all 4DN labs, with controlled batches

SOP	Name	Source	Type	4DN Datasets
	GM12878	Human - B-lymphocyte	Immortalized cell line	Search
	HFFc6	Human - Skin fibroblast	Immortalized cell line	Search
	WTC-11	Human - Stem cell	Induced pluripotent stem cell	Search
	H1-hESC	Human - Stem cell	Stem cell derived cell line	Search
	IMR-90	Human - Fibroblast of lung	Primary cell line	Search

— TIER-2

Common cell lines used by all 4DN labs

SOP	Name	Source	Type	4DN Datasets
	HAP-1	Human - Blood cell	Immortalized cell line	Search
	K562	Human - Blood cell	Immortalized cell line	Search
	U2OS	Human - Bone	Immortalized cell line	Search
	HCT116	Human - Colon	Immortalized cell line	Search
	HEK293	Human - Kidney	Immortalized cell line	Search

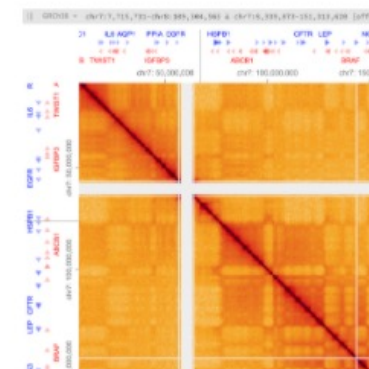
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Experiment Types - Genomics and Imaging

— Genomics Experiments

The largest category of genomics experiments are **Chromatin Conformation Capture (3C)** assays. In general these assays detect regions of the genome that are in close spatial proximity in the nucleus even though they may be separated by many kilobases in the linear chromosome DNA sequence or even reside on separate chromosomes. Many variations of this general approach have been developed :

- many recent ligation-based approaches perform the fixation, digestion and ligation in intact permeabilized nuclei
- different classes of enzymes have been used in the digestion step
- several ligation-free methodologies have been developed
- while many methods detect pairwise interactions, several assays have been developed that can detect multiplexes of interacting DNA regions
- some methods are designed to detect RNA-DNA contacts
- several methods use immunoprecipitation (IP) of specific chromatin associated proteins or pull downs with targeted specific labeled nucleotide probes to enrich specific contactx.



The types of 3C experiments for which data is available on the portal is summarized in the table below and includes links to the Datasets as well as the Experiment Type page, which in some cases includes additional description and information on uniform processing of that data type at the 4DN-DCIC. 3C datasets are usually visualized as 2-dimensional contact matrices - the majority of pages for 3C datasets contain HiGlass views of these matrices.

Chromatin Conformation Capture (3C) Experiment Types

Chromatin Conformation Capture (3C) Experiment Types

Experiment Type	Contacts	Ligation-free	Multi-contact	Enrichment	Single-cell	Public Experiment Sets	Reference	SOP
Dilution Hi-C	DNA					119 Public Experiment Sets	PMID:19815776	
in situ Hi-C	DNA					219 Public Experiment Sets	PMID:25497547	
single cell Hi-C	DNA				X	11 Public Experiment Sets	PMID:24067610	
sci-Hi-C	DNA				X	5 Public Experiment Sets	PMID:28135255	
sn-Hi-C	DNA				X	15 Public Experiment Sets	PMID:28355183	
DNase Hi-C	DNA					21 Public Experiment Sets	PMID:27685100	
Micro-C	DNA					7 Public Experiment Sets	PMID:26119342	
TCC	DNA					7 Public Experiment Sets	PMID:22198700	
Capture Hi-C	DNA			biotinylated oligonucleotides		29 Public Experiment Sets	PMID:25938943	
methyl Hi-C	DNA					Not yet ready for public release	PMID:31384045	
single cell methyl Hi-C	DNA				X	Not yet ready for public release	PMID:31384045	
MC Hi-C	DNA		X			Not yet ready for public release		
MC-3C	DNA		X			Not yet ready for public release		
ChIA-PET	DNA			protein-IP		4 Public Experiment Sets	PMID:28358394	
PLAC-seq	DNA			protein-IP		8 Public Experiment Sets	PMID:27886167	
DNA SPRITE	DNA	X	X			2 Public Experiment Sets	PMID:29887377	
RNA-DNA SPRITE	RNA-DNA	X	X			1 Public Experiment Set	PMID:29887377	
MARGI	RNA-DNA					2 Public Experiment Sets	PMID:28132817	
GAM	DNA	X	X			Not yet ready for public release	PMID:28273065	
TrAC-loop	DNA	X				Not yet ready for public release	PMID:30150754	



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4DN file formats - imaging

These are proposed file format specifications for processed files derived from imaging data.

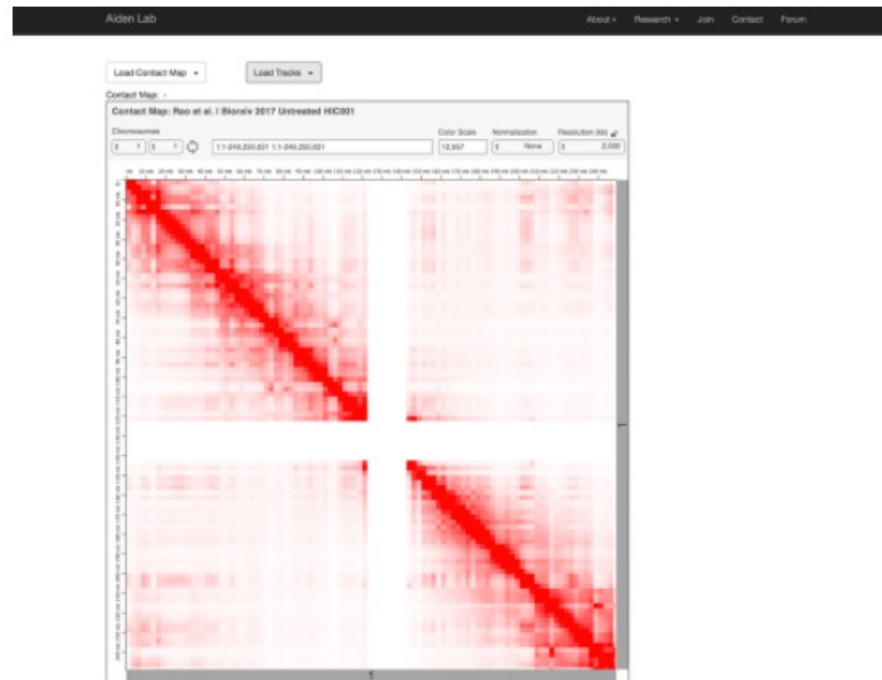
- [4DN SPT v1.0](#) (approved by the 4DN Imaging Working Group)
- [4DN DNA FISH v1.0](#) (DRAFT)
- [4DN segmentation v1.0](#) (DRAFT)



4DN Web Portal

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4DN Software



Juicebox.js is cloud-based visualization software for Hi-C data created by Jim Robinson and Douglass Turner of the IGV team, in collaboration with Neva C. Durand and the Aiden Lab. You can also download a full-fledged desktop version of Juicebox.

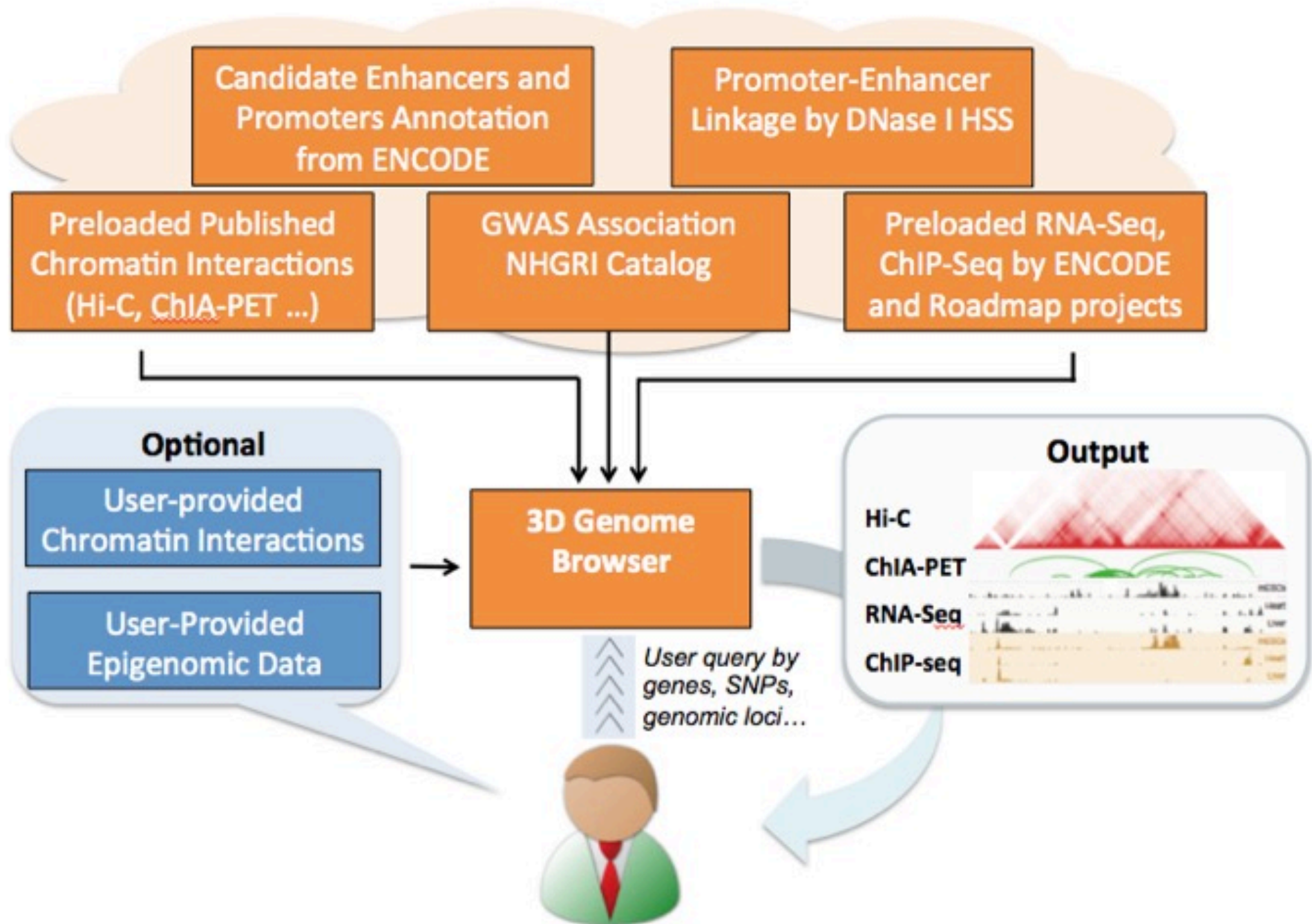
Juicebox is a cloud-based software for visualizing data from Hi-C and other proximity mapping experiments. Juicebox allows users to zoom in and out of Hi-C maps interactively, just as a user of Google Earth might zoom in and out of a geographic map. Maps can be compared to one another, or to 1D tracks or 2D feature sets.

• • • • •

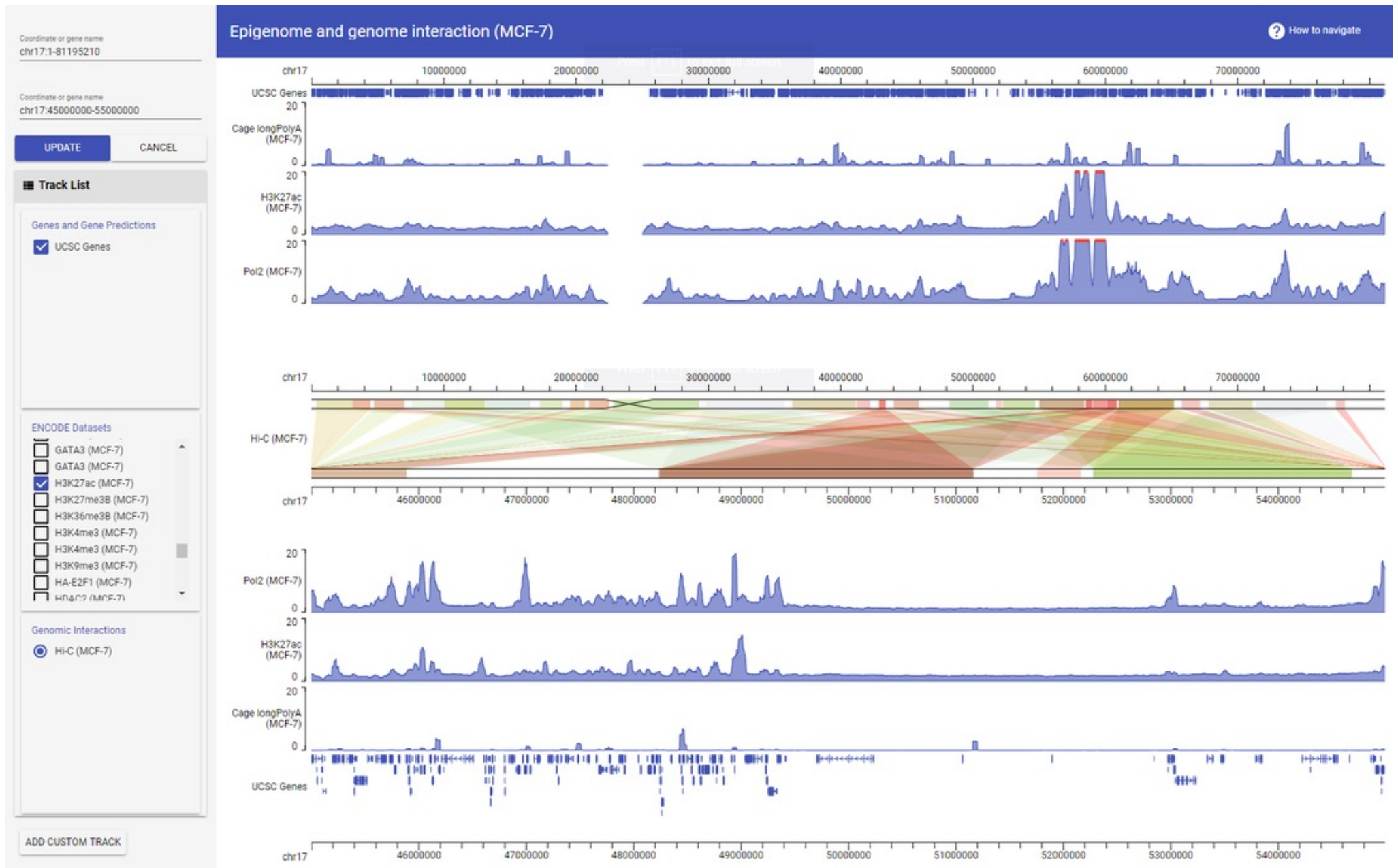


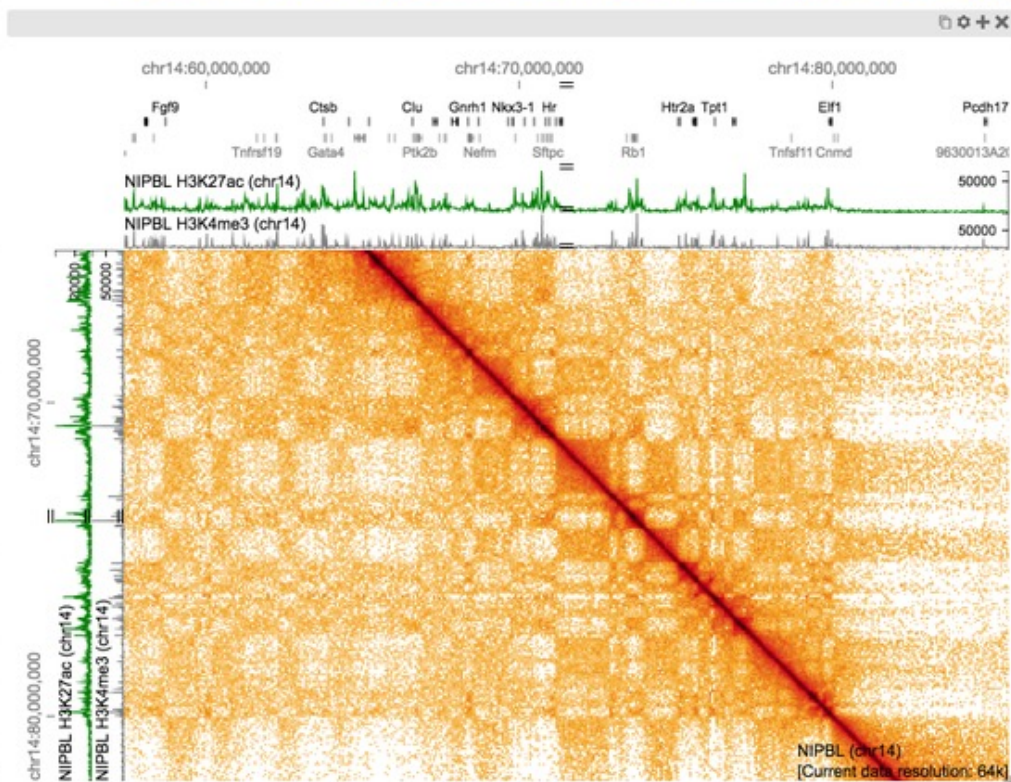
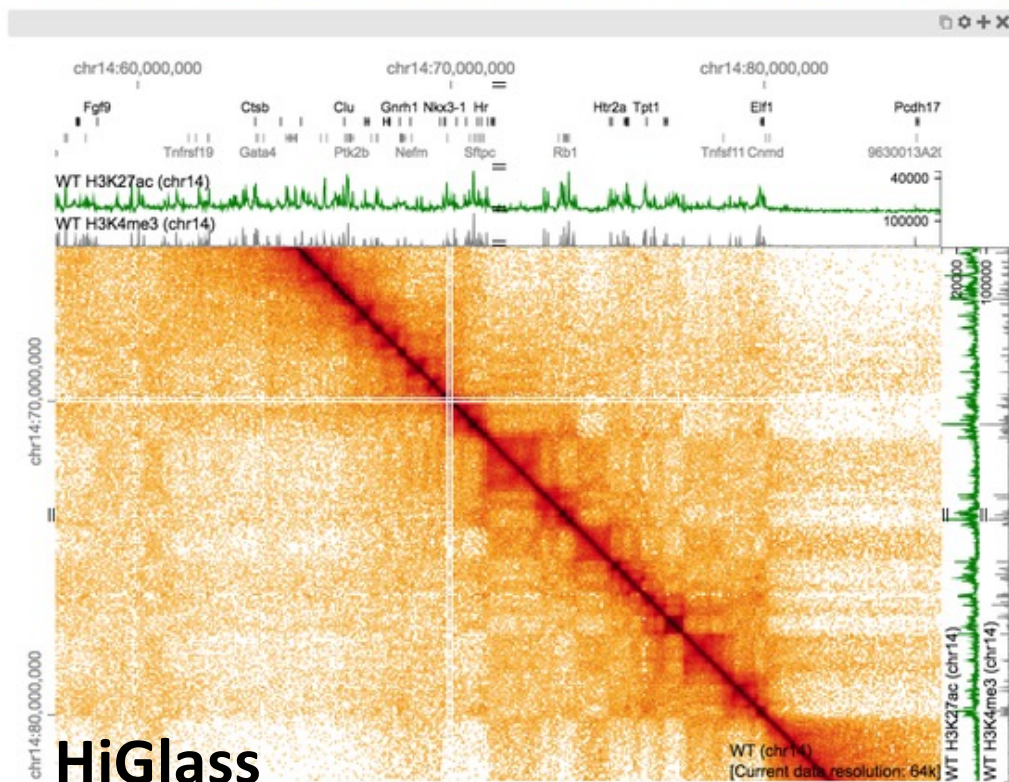
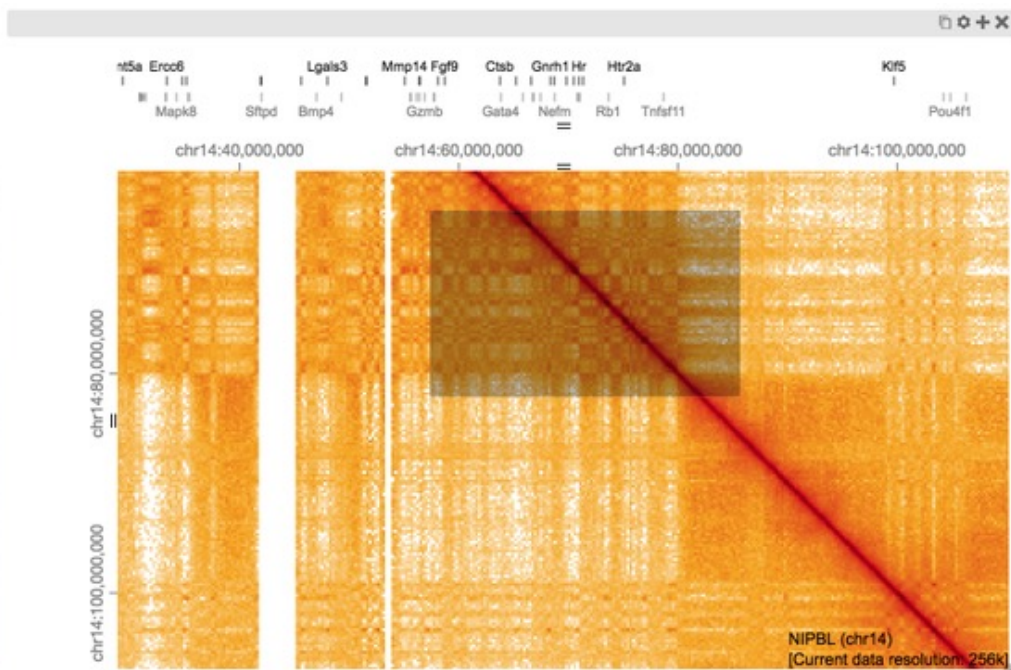
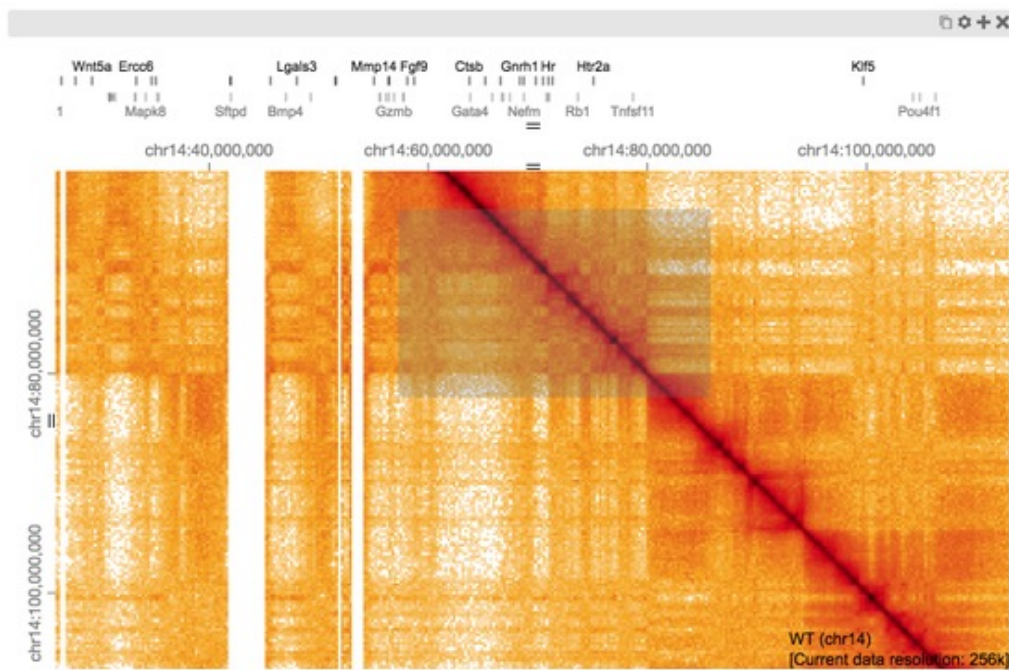
Nucleome Browser

START



3D genome browser





HiGlass

Software	Data Type	Description
3D Genome Browser	Hi-C, Virtual 4C	The 3D Genome Browser provides heatmap virtualizations of the Hi-C intra-chromosomal contact matrices. The browser allows the exploration of potential cis-regulatory elements and/or topologically associating domains (TADs) for the gene or genomic feature of interest.
3DIV	Hi-C	3DIV is a database which processed 315 Hi-C experiments from 80 human samples. 3DIV is an easy-to-use database providing epigenetic annotations and one-to-all chromatin interaction visualization with a variety of options.
4C-ker	4C	4C-ker is a method developed to analyze 4C-Seq (circularized chromosome conformation capture) data. 4C-ker is a Hidden-Markov Model based pipeline that identifies regions throughout the genome that interact with the 4C bait locus.
AutoChrom3D	3C, 4C, 5C, Hi-C	AutoChrom3D is an automatic pipeline to model chromatin regions with genomic size ranging from hundreds of kilobases to megabases at 8 kb resolution.
Centurion	Hi-C	A method that jointly calls all centromeres in a genome-wide Hi-C contact map.
ChIA-PET Browser	ChIA-PET	A software package for automatic processing of ChIA-PET sequence data, including linker filtering, mapping tags to reference genomes, identifying protein binding sites and chromatin interactions, and displaying the results on a graphical genome browser.
ChiaSig	ChIA-PET	ChiaSig finds significant interactions based on the NCHG model.
CHICAGO	Hi-C	A set of tools for calling significant interactions in Capture Hi-C data, such as Promoter Capture Hi-C.
CHROMATIX	Hi-C	CHROMATIX is 3-D chromatin folding software for deconvolving population Hi-C into single-cell ensembles and identifying significant higher-order chromatin interactions.
ChromContact	Hi-C	ChromContact is a web server for analyzing spatial contact of chromosomes from the publicly available Hi-C data.
chromoR	Hi-C	chromoR provides users with a statistical pipeline for analysing chromosomal interactions data (Hi-C data).
ChromSDE	Hi-C	A deterministic method which applies semi-definite programming techniques to find the best structure fitting the observed data and uses golden section search to find the correct parameter for converting the contact frequency to spatial distance.
ChromStruct	Chromosome conformation capture	ChromStruct is a Python code to estimate the 3D chromatin structure from chromosome conformation capture data are presented. The only input required is a text file containing a general real matrix of contact frequencies. The method is based on a multiresolution, modified-bead-chain chromatin model, evolved through quaternion operators in a Monte Carlo sampling.
CytoHiC	Hi-C	A Cytoscape plugin, which allow users to view and compare spatial maps of genomic landmarks, based on normalized Hi-C datasets.
diffHiC	Hi-C	A software package for the detection of differential interactions from Hi-C data. diffHiC provides methods for read pair alignment and processing, counting into bin pairs, filtering out low-abundance events and normalization of trended or CNV-driven biases.
EpiAlignment	Bigwig, Genome annotations	EpiAlignment is a dynamic programming algorithm for chromosomal similarity search, which is able to align two regions with both sequences and epigenomic information, such as histone modifications.
FastHiC	Hi-C	FastHiC is a fast peak caller for Hi-C data.
FisHiCal	Hi-C	FisHiCal integrates Hi-C and FISH data, offering a modular and easy-to-use tool for chromosomal spatial analysis.
Fit-Hi-C	Hi-C	Fit-Hi-C is a tool for assigning statistical confidence estimates to intra-chromosomal contact maps produced by genome-wide genome architecture assays such as Hi-C.

LACHESIS	Hi-C	A software tool to measure the thread of life. LACHESIS exploits contact probability map data (e.g. from Hi-C) for chromosome-scale de novo genome assembly.
Mango	ChIA-PET	A complete ChIA-PET data analysis pipeline that provides statistical confidence estimates for interactions and corrects for major sources of bias including differential peak enrichment and genomic proximity.
MAPS	PLAC-seq, HiChIP	MAPS is a pipeline to identify significant long-range chromatin interactions from PLAC-seq and HiChIP datasets.
MDM	ChIA-PET	MDM provides functions for the analysis of chromatin interactions using MC_DIST model, Two-Step model and One-Step model, based on ChIA-PET data.
MIA-Sig	ChIA-Drop	MIA-Sig is a python package for calling statistically significant multiplex chromatin complexes from non-enriched ChIA-Drop and protein-enriched ChIA-Drop datasets.
MICC	ChIA-PET	An easy-to-use R package to detect chromatin interactions from ChIA-PET sequencing data. By applying a Bayesian mixture model to systematically remove random ligation and random collision noise, MICC could identify chromatin interactions with a significantly higher sensitivity than existing methods at the same false discovery rate.
MOGEN	Hi-C	A 3D chromosome reconstruction method to make it capable of reconstructing 3D models of genomes from both intra- and inter-chromosomal Hi-C contact data.
NADfinder	NAD-seq	NADfinder is a Bioconductor package for bioinformatic analysis of the NAD-seq data, including normalization, smoothing, peak calling, peak trimming and annotation.
NuChart	Hi-C	An R Package to Study Gene Spatial Neighbourhoods with Multi-Omics Annotations.
Nucleome Browser	Multi-data	Nucleome Browser is a multimodal, interactive data visualization and exploration platform to integrate genomic, imaging, and 3D structure data.
PartSeg	3DFish	PartSeg allows super-resolution and 3DFish imaging data segmentation and analysis in the batch mode for hundreds/thousands of cells. This application is designed to help biologist with segmentation based on threshold and connected components.
PASTIS	Hi-C	Novel approach to infer a consensus three-dimensional structure of a genome from Hi-C data. The method incorporates a statistical model of the contact counts, assuming that the counts between two loci follow a Poisson distribution whose intensity decreases with the physical distances between the loci.
R3CPET	ChIA-PET	The package provides a method to infer the set of proteins that are more probably to work together to maintain chromatin interaction given a ChIA-PET experiment results.
SnapHiC	Hi-C	SnapHiC: Single nucleus analysis pipeline for Hi-C data is a tool to identifying chromatin loops from single cell Hi-C data.
TADbit	Hi-C	An algorithm for the identification of hierarchical topological domains in Hi-C data.
TADsplimer	Hi-C	TADsplimer is a computational tool to systematically detect topologically associating domain (TAD) splits and mergers across the genome between Hi-C samples.
TADtree	Hi-C	TADbit is a complete Python library to deal with all steps to analyze, model and explore 3C-based data.
THUNDER	Hi-C	THUNDER is a deconvolution method for bulk Hi-C data.
WashU Epigenome Browser	5C, Hi-C, ChIA-PET	The browser supports multiple types of long-range genome interaction data. This enables investigators to explore epigenomic data in the context of higher-order chromosomal domains and to generate multiple types of intuitive, publication-quality figures of interactions.
GMAP	Hi-C	GMAP (Gaussian Mixture model And Proportion test), is an algorithm to identify topologically associating domains and subdomains (Yu et al., Nature Communications, 2017). It provides a new and efficient tool to analyze simulated and experimental Hi-C data with increased statistical robustness and better consistency of the predictions compared to three state-of-the-art methods



4DN Software

[4DN Publications](#)[4DN Preprints](#)[Further Reading](#)

Here we compiled a list of tools available for data analysis and/or visualization of 4DN-related datasets

update this list, please send them to us via email:



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National Center for Biotechnology Information

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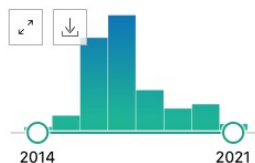
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RESULTS BY YEAR



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- ☐ Abstract
- ☐ Free full text
- ☐ Full text

ARTICLE ATTRIBUTE

- ☐ Associated data

ARTICLE TYPE

- ☐ Books and Documents
- ☐ Clinical Trial
- ☐ Meta-Analysis
- ☐ Randomized Controlled Trial
- ☐ Review

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204 results

- ☐ [SPIN reveals genome-wide landscape of nuclear compartmentalization.](#)
- 1 Wang Y, Zhang Y, Zhang R, van Schaik T, Zhang L, Sasaki T, Peric-Hupkes D, Chen Y, Gilbert DM, van Steensel B, Belmont AS, Ma J.
- Cite Genome Biol. 2021 Jan 14;22(1):36. doi: 10.1186/s13059-020-02253-3.
- Share PMID: 33446254 [Free PMC article.](#)
- ☐ [Close to the edge: Heterochromatin at the nucleolar and nuclear peripheries.](#)
- 2 Bizhanova A, Kaufman PD.
- Cite Biochim Biophys Acta Gene Regul Mech. 2021 Jan;1864(1):194666. doi: 10.1016/j.bbagr.2020.194666. Epub 2020 Dec 8.
- Share PMID: 33307247 [Review.](#)
- ☐ [Trans- and cis-acting effects of Firre on epigenetic features of the inactive X chromosome.](#)
- 3 Fang H, Bonora G, Lewandowski JP, Thakur J, Filippova GN, Henikoff S, Shendure J, Duan Z, Rinn JL, Deng X, Noble WS, Disteche CM.
- Cite Nat Commun. 2020 Nov 27;11(1):6053. doi: 10.1038/s41467-020-19879-3.
- Share PMID: 33247132 [Free PMC article.](#)
- ☐ [A map of cis-regulatory elements and 3D genome structures in zebrafish.](#)
- 4 Yang H, Luan Y, Liu T, Lee HJ, Fang L, Wang Y, Wang X, Zhang B, Jin Q, Ang KC, Xing X, Wang J, Xu J, Song F, Sriranga I, Khunsraksakul C, Salameh T, Li D, Choudhary MNK, Topczewski J, Wang K, Gerhard GS, Hardison RC, Wang T, Cheng KC, Yue F.
- Cite Nature. 2020 Dec;588(7837):337-343. doi: 10.1038/s41586-020-2962-9. Epub 2020 Nov 25.
- Share

[Launch Data Portal](#)

The 4D Nucleome Data Portal hosts data generated by the 4DN Network and other reference nucleomics data sets, and an expanding tool set for open data processing and visualization.

[Nucleome Browser](#)

Nucleome Browser is a multimodal, interactive data visualization and exploration platform to integrate genomic, imaging, and 3D structure



4DN pre-prints

See the [4DN bioRxiv channel](#).

Pre-print publications funded by NIH 4DN program			
Title	Authors	DOI	Date
Polymer physics and machine learning reveal a combinatorial code linking chromatin 3D architecture to 1D epigenetics	Esposito, A., Bianco, S., Chiariello, A. M., Abraham, A., Fiorillo, L., Conte, M., Campanile, R., Nicodemi, M.	10.1101/2021.03.01.433416	2021-03-01
Disruption of nuclear architecture as a cause of COVID-19 induced anosmia	Zazhytska, M., Kodra, A., Hoagland, D. A., Fullard, J., Shayya, H. J., Omer, A., Firestein, S., Gong, Q., Canoll, P., Goldman, J. E., Roussos, P., tenOever, B. R., Overdevest, J. B., Lomvardas, S.	10.1101/2021.02.09.430314	2021-02-09
Genome-wide variability in recombination activity is associated with meiotic chromatin organization	Jin, X., Fudenberg, G., Pollard, K. S.	10.1101/2021.01.06.425599	2021-01-07
Cohesin-mediated loop anchors confine the location of human replication origins	Emerson, D. J., Zhao, P. A., Klein, K., Ge, C., Zhou, L., Sasaki, T., Yang, L., Vennev, S. V., Gibcus, J. H., Dekker, J., Gilbert, D. M., Phillips-Cremins, J. E.	10.1101/2021.01.05.425437	2021-01-06
Systematic evaluation of chromosome conformation capture assays	Akgol Oksuz, B., Yang, L., Abraham, S., Venev, S. V., Krietenstein, N., Parsi, K. M., Ozadam, H., Oomen, M. E., Nand, A., Mao, H., Genga, R. M. J., Maehr, R., Rando, O., Mirny, L., Gibcus, J. H., Dekker, J.	10.1101/2020.12.26.424448	2020-12-27
Mechanical frustration of phase separation in the cell nucleus by chromatin	Zhang, Y., Lee, D. S. W., Meir, Y., Brangwynne, C. P., Wingreen, N. S.	10.1101/2020.12.24.424222	2020-12-24
Multi-omics analysis of chromatin accessibility and interactions with transcriptome by HiCAR	Wei, X., Xiang, Y., Shan, R., Peters, D. T., Sun, T., Lin, X., Li, W., Diao, Y.	10.1101/2020.11.02.366062	2020-12-19
Simple biochemical features underlie transcriptional activation domain diversity and dynamic, fuzzy binding to Mediator	Sanborn, A. L., Yeh, B. T., Feigertle, J. T., Hao, C. V., Townshend, R. J. L., Aiden, E. L., Dror, R. O., Kornberg, R. D.	10.1101/2020.12.18.423551	2020-12-18
SnapHiC: a computational pipeline to map chromatin contacts from single cell Hi-C data	Yu, M., Abnoui, A., Zhang, Y., Li, G., Lee, L., Chen, Z., Fang, R., Wen, J., Sun, Q., Li, Y., Ren, B., Hu, M.	10.1101/2020.12.13.422543	2020-12-15



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 External Program Consultants

Pre-print publications funded by NIH 4DN program			
Title	Authors	DOI	Date
Polymer physics and machine learning reveal a combinatorial code linking chromatin 3D architecture to 1D epigenetics	Esposito, A., Bianco, S., Chiariello, A. M., Abraham, A., Fiorillo, L., Conte, M., Campanile, R., Nicodemi, M.	10.1101/2021.03.01.433416	2021-03-01

Launch Data Portal

The 4D Nucleome Data Portal hosts data generated by the 4DN Network and other reference nucleomics data sets, and an expanding tool set for open data processing and visualization.

Nucleome Browser

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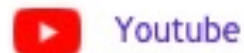
Nucleome Browser is a multimodal, interactive data visualization and exploration platform to integrate genomic, imaging, and 3D structure data.

Internal Wiki

The 4D Nucleome Network Wiki is an internal collaboration site, accessible to 4DN investigators and members only.

people.4dnucleome

people.4dnucleome provides a one-stop directory for all current 4DN members and allows logged in 4DN members to manage their and their team members' profiles, participate in the interval working groups, and more.

Multimedia Library**Stay Connected**

The achievement of 4DN

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Table 2 | Genomic technologies currently in use or in development in the 4DN Network

Assay Name	Assay Abbreviation	Key features
Chromosome Conformation Capture	3C ^{15,61} 4C ⁶² 5C ⁶³ Hi-C ¹⁷ Single-cell Hi-C ³⁶ Combinatorial single-cell Hi-C ⁶⁴ <i>In situ</i> Hi-C ⁶⁴ DNase Hi-C ⁶⁵ Micro-C ^{66,67} Capture Hi-C ^{68,69} TCC ⁷⁰ Distance Hi-C (UD) COLA ⁷¹	Interactions between specific genomic loci, including genome-wide studies Genome-wide interactions of a specific genomic locus Many loci against many loci Genome-wide map of all interactions in the nucleus Hi-C variant that enables mapping contacts within single cells Single-cell Hi-C variant using split-pool barcoding to map single cells Hi-C variant that performs digestion and ligation in intact nuclei Hi-C variant that digests chromatin using DNase Hi-C variant that digests chromatin using micrococcal nuclease Hi-C variant that incorporates selection of targeted genomic loci A variant of Hi-C using bead-coupling of complexes Variant of Hi-C using photo-activated crosslinkers to measure the distance between interacting DNA regions. A variant of Hi-C using frequent restriction cutters to map >2 simultaneous DNA interactions
Chromatin-interaction analysis by paired-end-tag sequencing	ChIA-PET ^{6,72}	Genome-wide map of interactions bound to a specific protein
Genome architecture mapping	GAM ³⁷	A cryosectioning method to map colocalized DNA regions in a ligation-independent manner
Split-pool barcoding of RNA and DNA	SPRITE	A ligation-independent method to barcode interacting RNA and DNA that enables mapping of higher-order contacts
RNA interaction with chromatin by paired-end-tag sequencing	RICh-PET (UD)	Genome-wide mapping for all ncRNA-chromatin interactions
Chromatin-immunoprecipitation aided RICh-PET	ChIP RICh-PET, UD	Genome-wide mapping for three-way interactions involving ncRNA, DNA loci and protein factors
DNA adenine methyltransferase identification	DamID and Single-Cell DamID ^{8,73}	Genome-wide mapping of molecular contact frequency of DNA loci to a Dam-methylase fusion protein: will be used to measure DNA proximity to different nuclear compartments
Tyramide signal amplification-sequencing	TSA-seq (UD)	Genome-wide mapping of estimated mean cytological distance (in μm) of a DNA locus in a nuclear compartment
Replicated DNA sequencing	Repli-seq ^{74,75}	Genome-wide mapping of the timing of DNA replication
Thousands of reporters integrated in parallel	TRIP ⁵¹	Genome-wide mapping of chromosome position effects on transcription and post-transcriptional processing
RNA antisense purification	RAP ^{49,50,76,77}	Mapping of DNA regions, RNA species and proteins that are in proximity to a specific ncRNA in the nucleus

UD, under development; SPRITE, split pool recognition of interactions by tag extension

Table 3 | Imaging technologies currently in use or in development in the 4DN Network

Application	Method (PMID)	Label/stain/dye	Feature
Visualizing DNA and RNA sequences	3D DNA and RNA FISH ⁷⁸ , seqFISH ⁷⁹ , MERFISH ^{80,81} , HiPMap ⁸²	Fl-labelled oligo and BAC probes Fl-labelled probes	Fixed cells, single- and multiple-target DNA and RNA detection High-throughput FISH and automated microscopy in fixed cells
	CASFISH ^{31,83} , CRISPRainbow ²⁹ , Cas live-cell imaging ^{30,38}	Fl-labelled dCas9 and guide RNAs (gRNAs) to label genomic loci; gRNAs can contain a stem loop bound by Fl-labelled coat proteins	Live and fixed cells—gRNAs against repetitive targets and/or collections of gRNAs that cover unique sequences
	Track first, identify later ⁸⁴	Barcoded Fl-labelled oligos and Fl-dCas9	Combination of SeqFISH and Cas9 live-cell imaging
	MS2, PP7, mSpinach, aptamers ^{85,86} , ParS-, LacO- or TetO-labelled genomic loci ^{87,88}	Fl-fusions that bind tagged RNA sequences Visualized by Fl-labelled ParB-, lac- or tet-repressor proteins	Live and fixed cells: transcription Live and fixed cells: genomic tags
High resolution ultrastructure and 3D organization	Transmission EM, SBEM, multiple-tit ⁸⁹ and serial-section EM tomography ⁹⁰	Heavy metals, colloidal nanogold, inorganic and organic probes and tags, ChromEM, Alexa 633, Fluoronanogold, Time-STAMP-YFP-MiniSOG	High resolution ultrastructure of macromolecules <i>in situ</i> ranging from 2D transverse projections through 70–250-nm sections to reconstruction of large 3D volumes (250 to >500nm thick)
	Multi-colour EM ⁹¹ , ChromEMT (UD) ⁴³	Orthogonal-correlated light and EM probes and DAB conjugates	Local ultrastructure and global 3D organization of chromatin as a continuum at nucleosome resolutions and megabase scales
	X-ray tomography ⁴¹	Quantitative LAC-generated contrast that reflects the bioorganic composition of unlabelled DNA, RNA and proteins	Mesoscale resolution (20–50 nm) in intact, unprocessed cells
	Correlated SIM and X-ray tomography ⁴²	Fl-labelled probes	Visualize ultrastructure and interactions of selectively labelled chromatin and protein simultaneously
Visualizing spatial and dynamic 3D nuclear organization	Wide-field, confocal/multiphoton	Fl probes: Fl protein, SNAP and Halo genomic tags, Alexa and Cy dyes, atto488, Suntag, SH2, Quantum dots, Janelia fluor dyes ¹⁰¹	Live and fixed cells: diffraction limited (>250 nm)
	Super resolution: 3D SIM, PALM, STORM ⁹² , STED, adaptive optics ⁹³ , tcPALM ²⁸ , SAIM ⁹⁴	Fl probes, photo-activatable and photo-switchable proteins; caged organic fluorophores ¹⁰¹ and organic fluorophores in thiol buffer coupled to SNAP, Halo, nanobodies or FISH oligos	Super-resolution imaging of nuclear organization in live and fixed cells (circa 10–20 nm)
	LLS, LLS-PAINT ^{95,96}	Fl-labelled probes	Live cell: 3D dynamics and nuclear organization
	Single-molecule imaging and tracking ^{97–100}	Fl-labelled and photo-activatable probes, ArrayG	Live cell: binding and search dynamics: residence time, search time, search mechanism

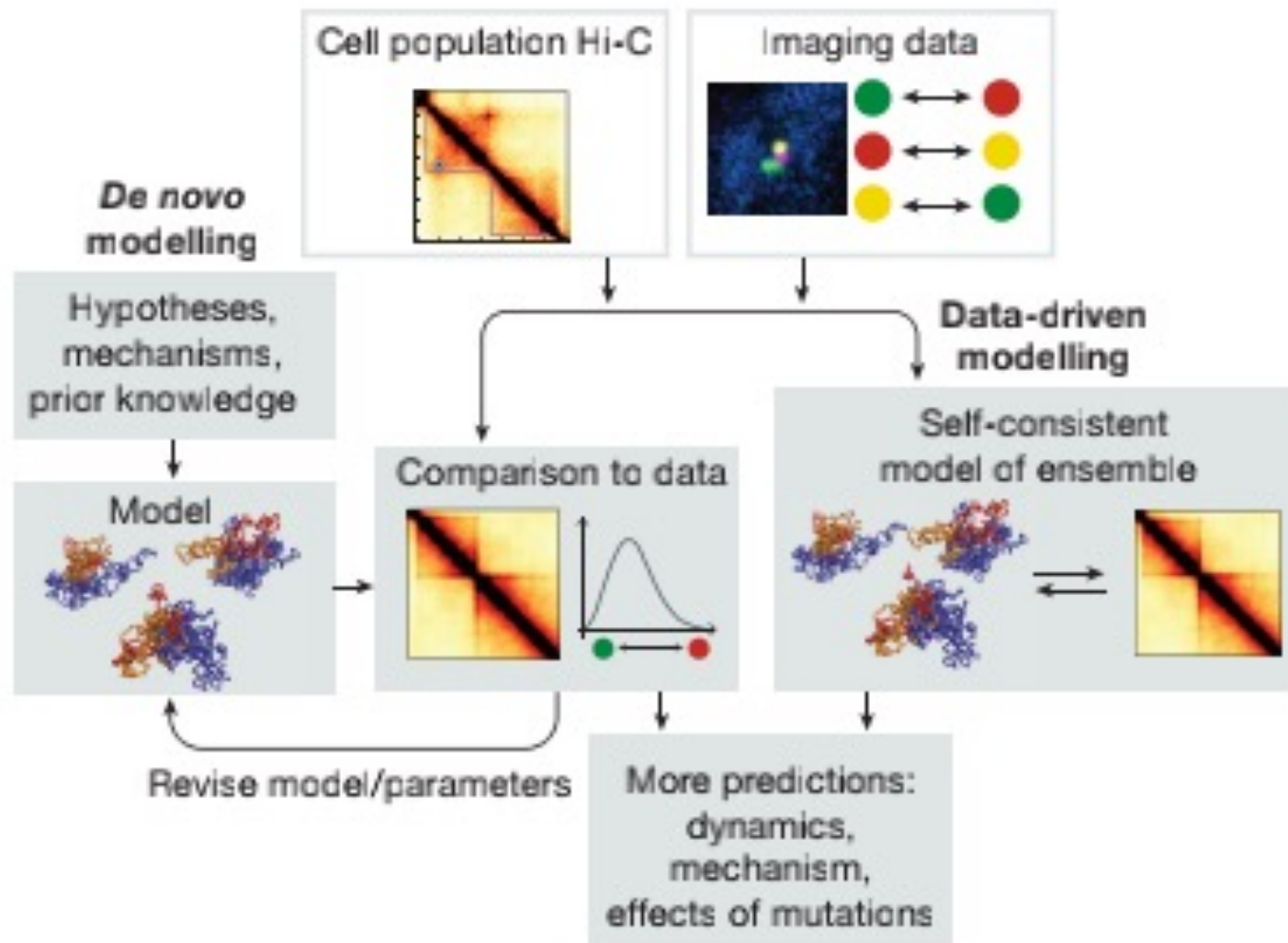
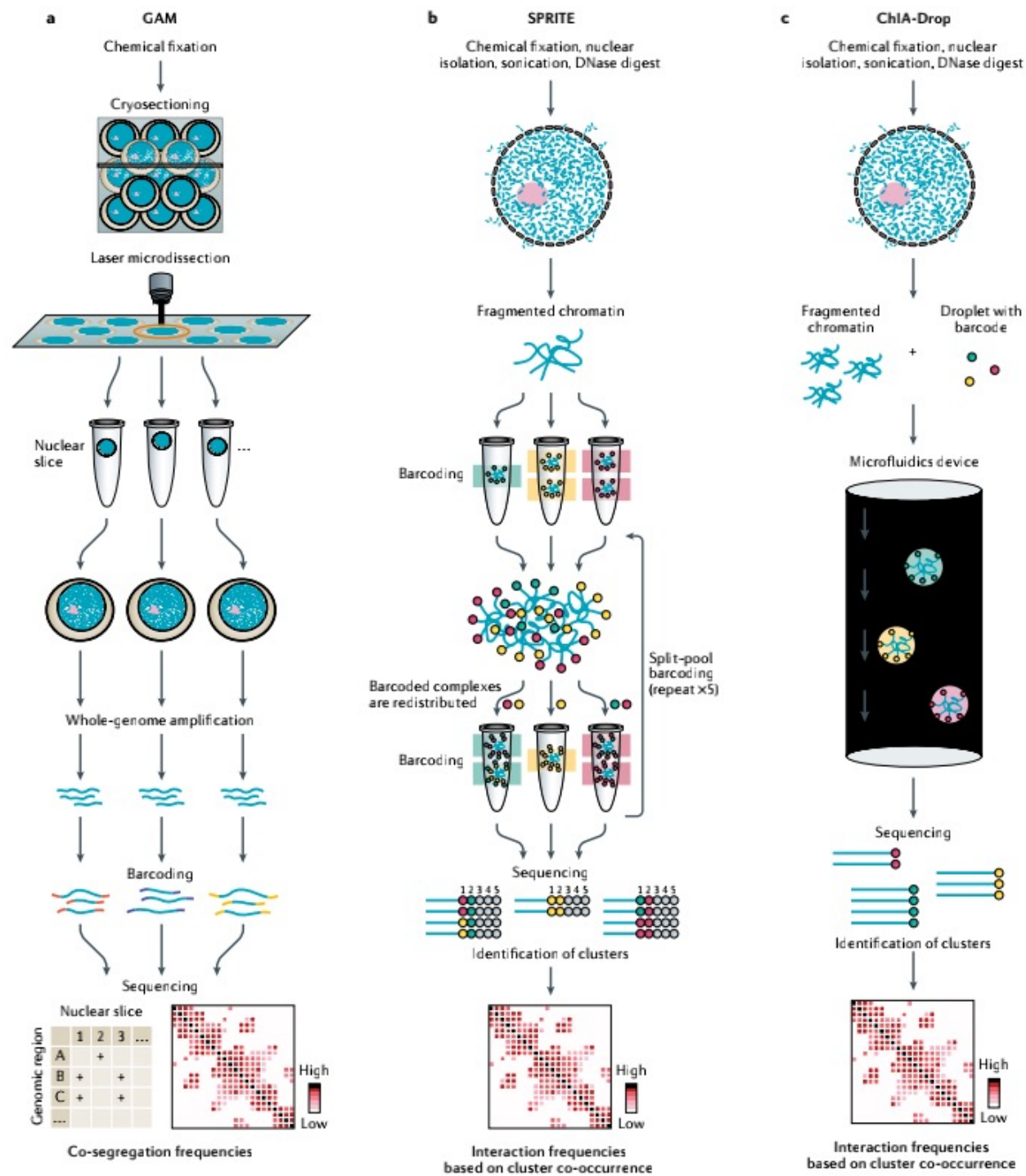


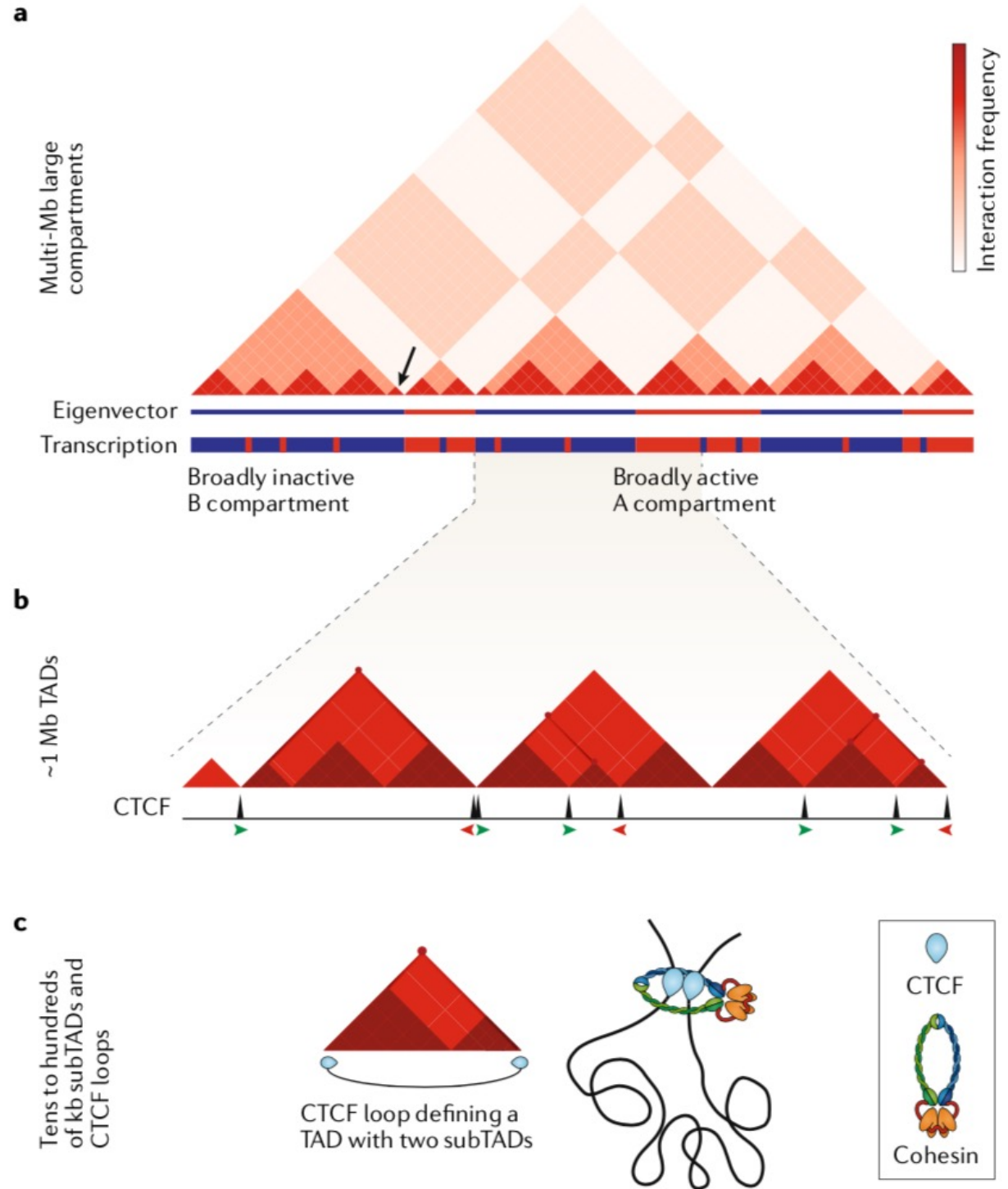
Table 1 | Comparison of methods used to detect chromatin contacts

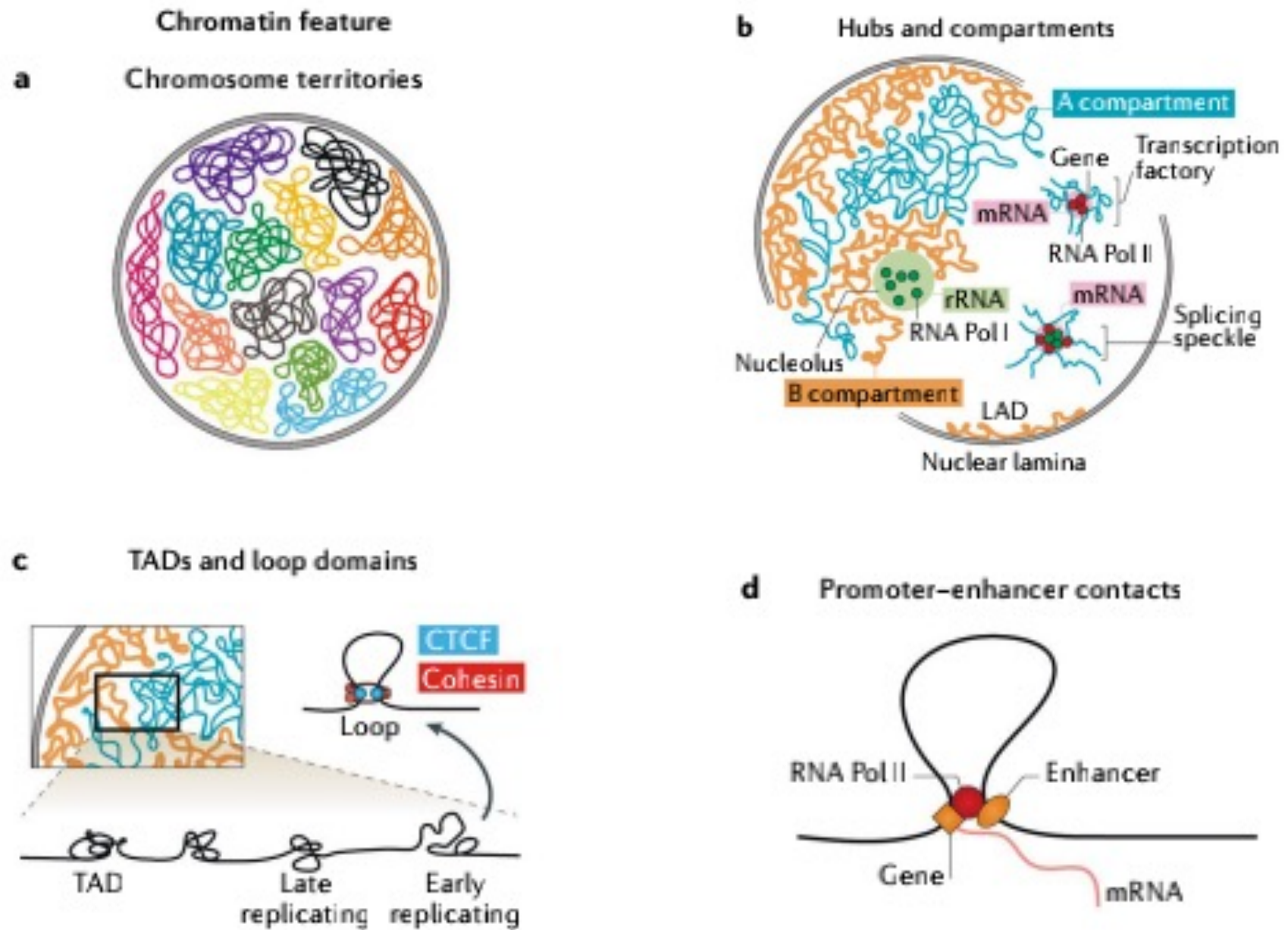
Assay	Description	Number of contacts per experiment	Multiplicity of contacts	Single-cell information	Number of cells	Detectable contacts	Protocol
3C-based methods							
3C	Proximity ligation and selection of target regions with primers, detection by quantitative PCR	One versus one	Pairwise	No	100 million ¹⁹¹	Protein-mediated	¹⁹¹
4C	Proximity ligation and enrichment for contacts with one bait region by inverse PCR, detection by sequencing	One versus all	Pairwise	No	Robust: 10 million ¹⁹³ , low input: 340,000 (REF. ¹⁹³)	Protein-mediated	¹⁹³
5C	Proximity ligation and enrichment for larger target region with primers, detection by sequencing	Many versus many	Pairwise	No	Robust: 50–70 million ¹⁹¹ , low input: 2 million ¹⁹⁶	Protein-mediated	^{191,196}
Hi-C	Proximity ligation and enrichment for all ligated contact pairs, detection by sequencing	All versus all	Pairwise	No	Robust: 2–5 million ⁹⁸ , low input: 100,000–500,000 (REFS ^{10,25,7})	Protein-mediated	^{64,191}
TCC	Tethered proximity ligation and enrichment for all ligated contact pairs, detection by sequencing	All versus all	Pairwise	No	25 million ⁵¹	Protein-mediated	⁵¹
PLAC-seq, ChIA-PET	Proximity ligation and pull-down of specific protein-mediated contacts, detection by sequencing	Many versus many	Pairwise	No	Robust: 100 million ¹⁹⁸ , low input: 500,000 (REF. ⁹³)	Protein-mediated (specific)	^{87,198}
Capture-C, C-HiC	Proximity ligation and target enrichment using probes for genomic regions of interest, detection by sequencing	Many versus all	Pairwise	No	Robust: 100,000 (REF. ¹⁹⁵), low input: 10,000–20,000 (REF. ¹⁷)	Protein-mediated	¹⁹⁵
Single-cell Hi-C	Proximity ligation and enrichment for all ligated contact pairs, detection by sequencing	All versus all	Pairwise	Yes	Hundreds	Protein-mediated	⁷¹
Imaging							
2D-FISH	Fixation to flatten cells, hybridization of fluorescent probes to target regions, measurement of 2D spatial distances	Between 2 and 52 regions*	Pairwise or more	Yes	Hundreds	All in spatial proximity	²⁰⁰
3D-FISH	Fixation of cells, hybridization of fluorescent probes for target regions, measurement of 3D spatial distances	Between 2 and 52 regions*	Pairwise or more	Yes	Hundreds	All in spatial proximity	²⁰¹
Cryo-FISH	Fixation of cells, cryosectioning, hybridization of fluorescent probes for target regions, measurement of 2D spatial distances	Between 2 and 52 regions*	Pairwise or more	Yes	Hundreds	All in spatial proximity	¹⁰¹
Live-cell imaging	Fluorescent labelling of genomic loci in living cells, measurement of spatial distances over time	Between 2 and 12 regions	Pairwise or more	Yes	Hundreds	All in spatial proximity	^{4,91,201}
Ligation-free methods							
GAM	Cryosectioning of fixed cells, DNA extraction from nuclear sections and sequencing, inferring spatial distances from co-segregation of genomic regions in nuclear sections	All versus all	Pairwise or more	Yes	Hundreds ¹⁰	All in spatial proximity	¹⁰
SPRITE	Fixation of cells, identification of crosslinked chromatin fragments by split-pool barcoding and sequencing	All versus all	Many	No	10 million ¹¹	Protein-mediated	¹¹
ChIA-Drop	Fixation of cells, identification of crosslinked chromatin fragments by droplet-based and barcode-linked sequencing	All versus all	Many	No	10 million ¹¹	Protein-mediated	¹¹

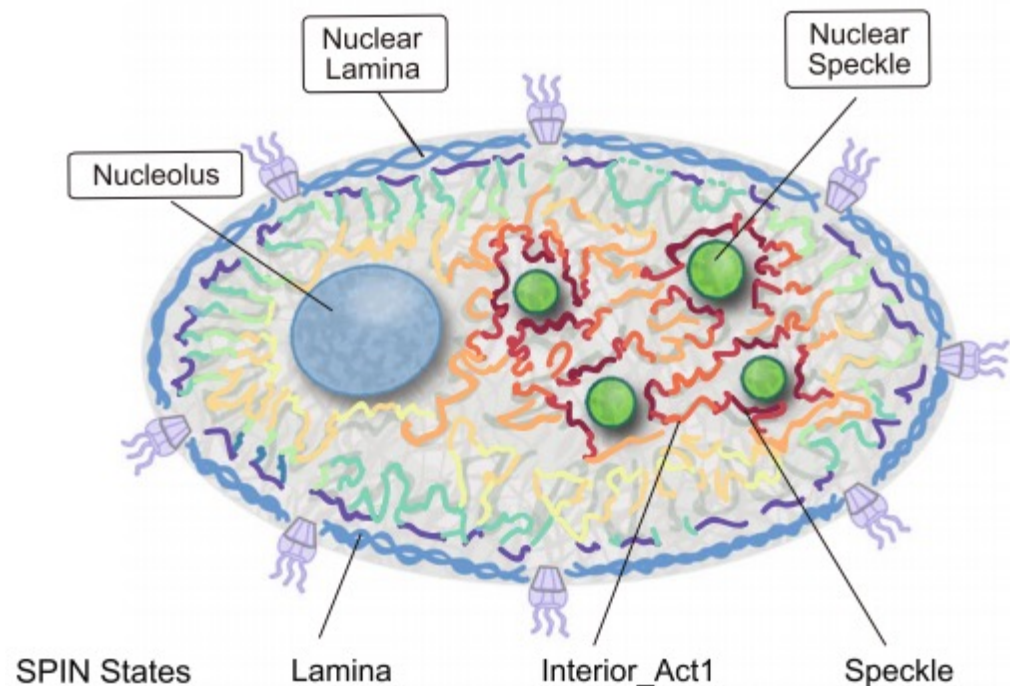
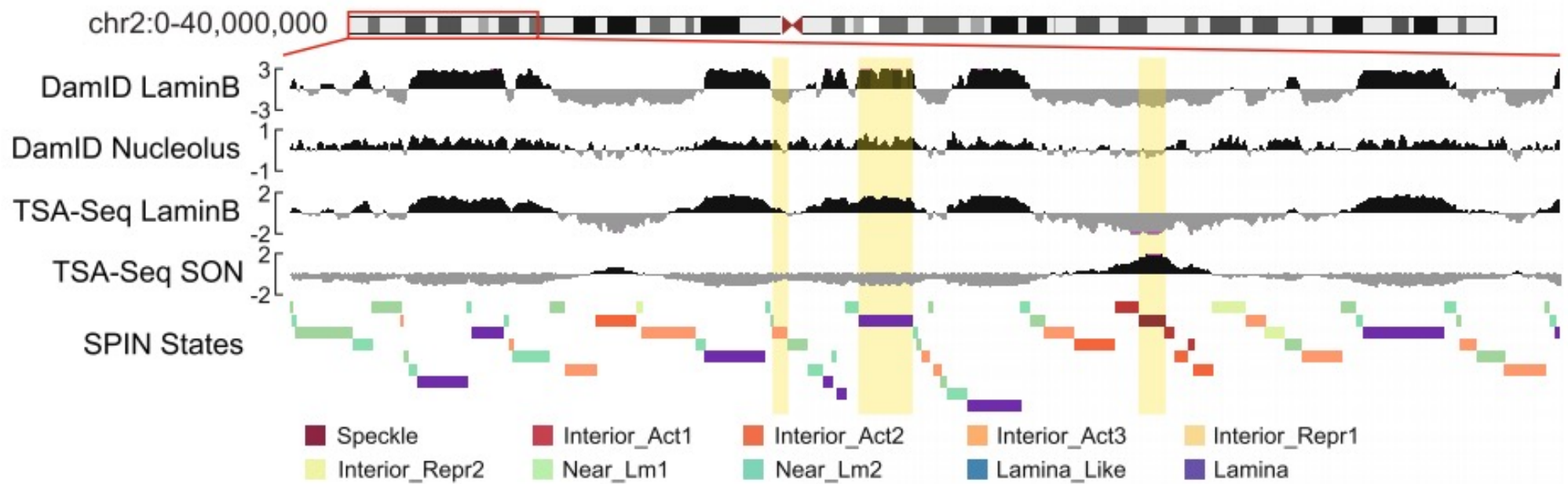
3C, chromosome conformation capture; 4C, circular chromosome conformation capture; 5C, chromosome conformation capture carbon copy; ChIA-Drop, chromatin-interaction analysis via droplet-based and barcode-linked sequencing; ChIA-PET, chromatin interaction analysis by paired-end tag sequencing; C-HiC, capture HiC; FISH, fluorescence in situ hybridization; GAM, genome architecture mapping; Hi-C, high-throughput chromosome conformation capture; PLAC-seq, proximity ligation-assisted chromatin immunoprecipitation sequencing; SPRITE, split-pool recognition of interactions by tag extension; TCC, tethered chromosome capture. *Classical FISH experiments rarely distinguish between more than 2–5 differentially labelled regions simultaneously²⁰⁴. Cycles of probe hybridization can increase this number up to 52 (REF.²⁰⁵).



Size

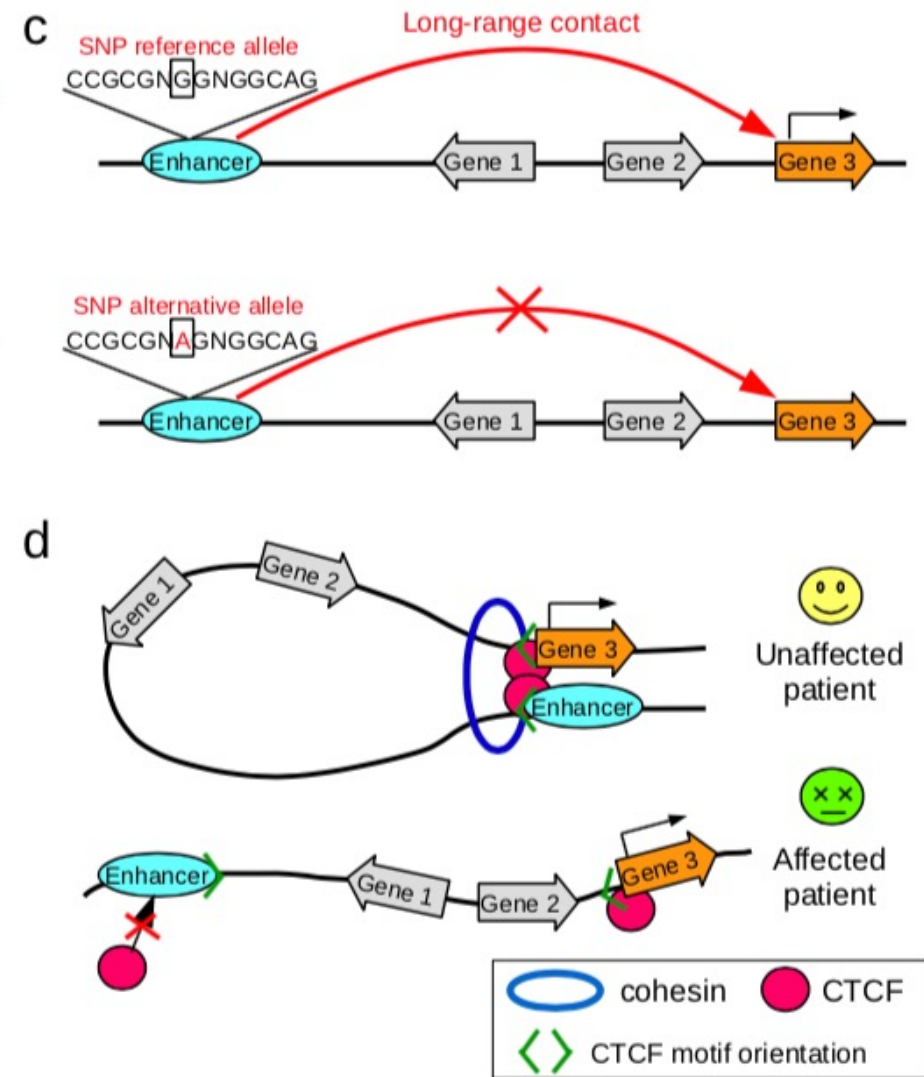
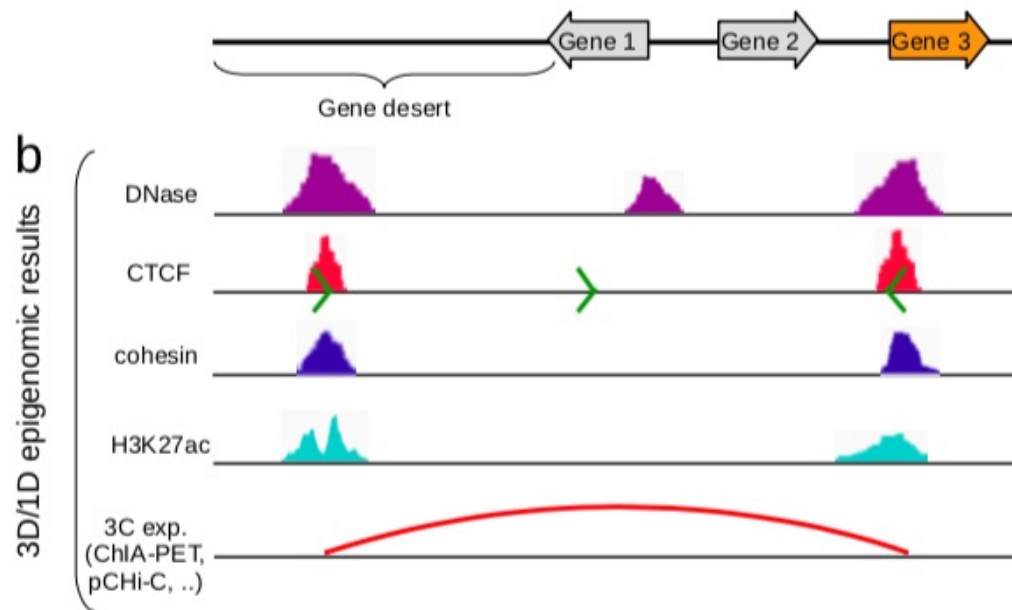
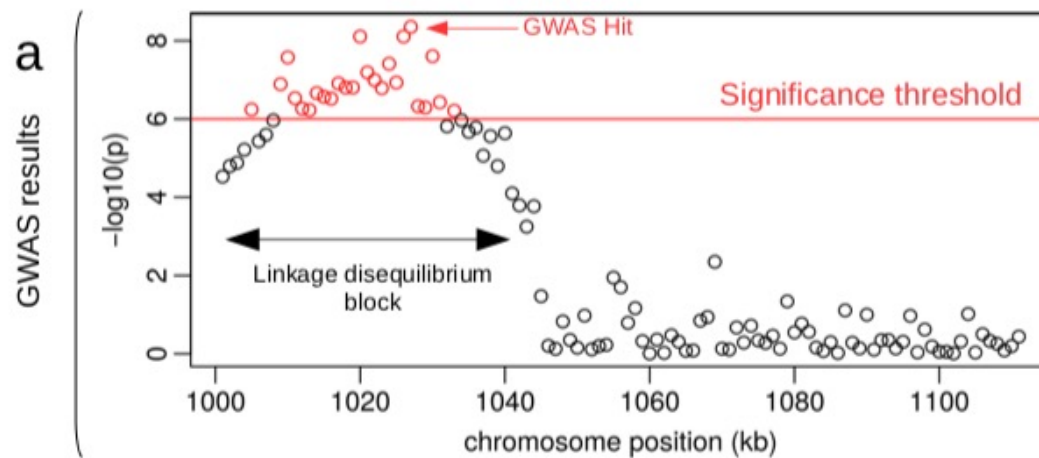




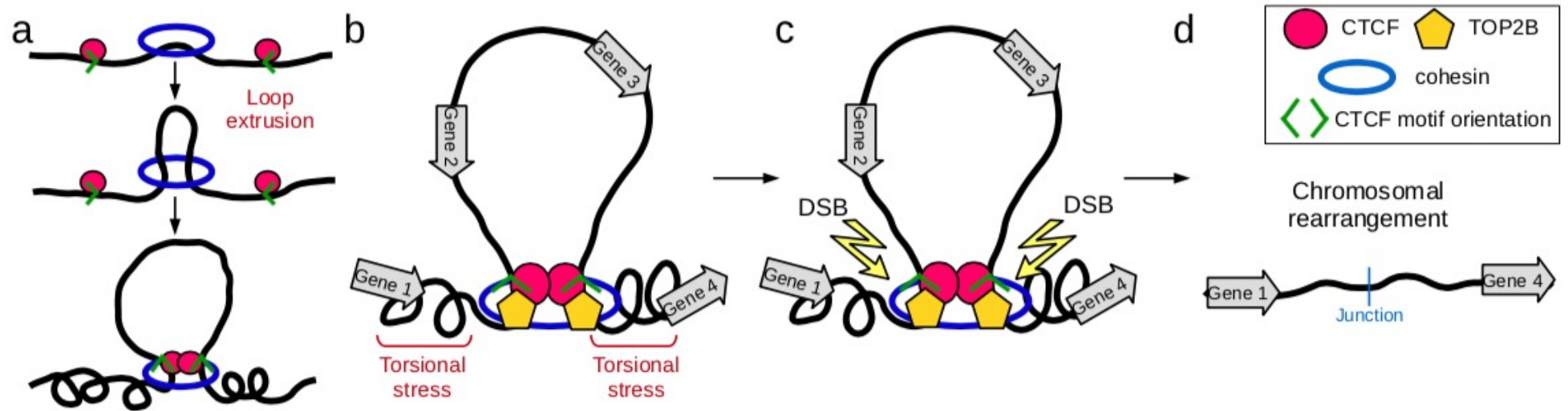


3D genome and disease

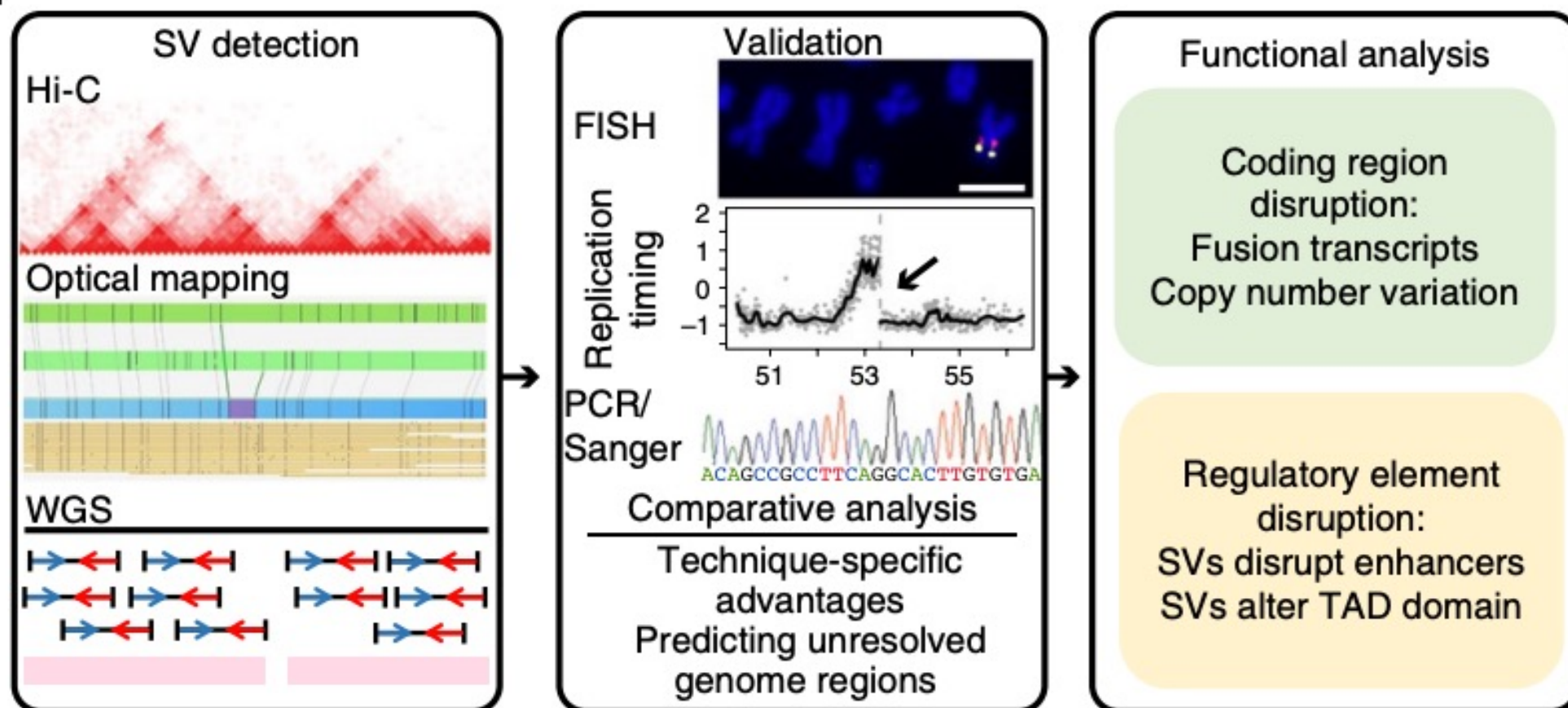
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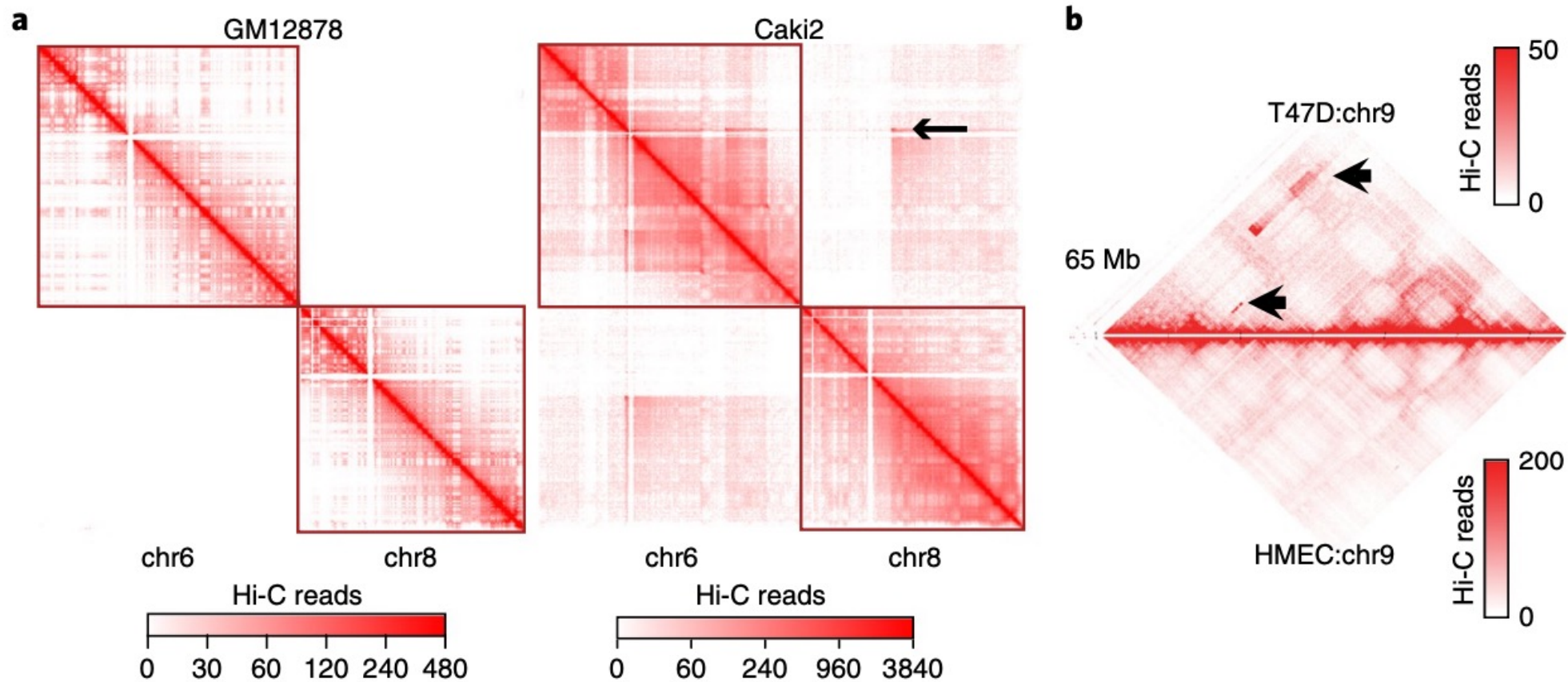


3D genome dynamic induces chromosomal rearrangements



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